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Modeling & Experiments

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To my beloved parents, Rafique and Salma

Preface

The recent advancements in the field of low power portable electronic devices and micro electromechanical systems (MEMS) technology have accelerated research in the field of energy harvesting. The main ambition is to make them self-sufficient by embedding an energy harvesting system within these devices, eliminating the requirement for periodic battery replacement or recharging. The embedded energy harvesting system can scavenge electrical energy by converting ambient energy sources such as solar, wind or mechanical motion energy. This book is focussed on the potential of converting mechanical motion or vibration energy into electrical energy and also investigates the effects of energy harvesting on the overall dynamics of the system.

Typically, there are three vibration-based energy harvesting techniques, (1) piezoelectric, (2) electrostatic and (3) electromagnetic. In piezoelectric vibration energy harvesting (PVEH), vibration energy is converted into useful electrical energy by using 'sensor effect' of piezoelectric materials. The work of this book is particularly presenting close-form, more accurate, experimentally validated mathematical modelling techniques. Additionally, MATLAB program codes are provided to allow researchers to more easily understand and apply complex PVEH system equations and design their own energy harvesting systems. The work presented in the book was mainly carried out during the doctorate and post-doctorate research of the author at renowned institutes.

The overall aims of this book are manifold: (1) a thorough theoretical and experimental analysis of a PVEH beam or assembly of beams; (2) an in-depth analytical and experimental investigation of a dual function piezoelectric vibration energy harvester beam/tuned vibration absorber (PVEH/TVA) or 'electromechanical TVA'; (3) example application of the dual function energy harvesting TVA to suppress the vibration of electronic box, and (4) ready-to-run MATLAB program codes to simulate PVEH mechanism.

Some of the distinct features of this book can be summarised as follows:

- An in-depth experimental validation of a PVEH beam model based on the analytical modal analysis method (AMAM), with the investigations conducted over a wider frequency range than previously tested.
- The precise identification of the electrical loads that harvest maximum power and that induce maximum electrical damping.
- A thorough investigation of the influence of mechanical damping on PVEH beams.
- A procedure for the exact modelling of PVEH beams, and assemblies of such beams, using the dynamic stiffness matrix (DSM) method.
- A procedure to enhance the power output from a PVEH beam through the application of a tip rotational restraint and the use of segmented electrodes.
- The theoretical basis for the concept of a dual function PVEH beam/TVA, and its realisation and experimental validation for a prototype device.
- A theoretical example illustrating the application of a dual function device to control the vibration of electronic box.
- An improved electrical circuit configuration is presented using the real time nonlinear electrical load.
- Easy to use programming codes to run simulations and observe the behaviour of dual function energy harvester tuned vibration absorbers.

Gavle, Sweden Sajid Rafique

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I would also like to express my gratitude to all the technicians who helped me in building the experimental set-up. I would also like to thank my friends and colleagues, who made this journey memorable and exciting.

Last but not the least, my parents and beloved family, I love you all and would not have done it without your support.

Contents

Abbreviations

AMAM	Analytical modal analysis method or modelling
DOF	Degree of freedom
DSM	Dynamic stiffness matrix/method
EH	Energy harvesting/Energy harvester
EM	Electromagnetic
IDE	Interdigitated electrodes
MFC	Micro-fibre composite
OC	Open circuit
PVEH	Piezoelectric vibration energy harvesting/harvester
PZT	Lead Zirconate Titanate
QP	Quick pack
RR	Rayleigh–Ritz
SC	Short circuit
TMD	Tuned mass damper
TVA	Tuned vibration absorber
VEH	Vibration energy harvesting

Nomenclature

A	Damping constant of a composite beam
A_a, A_b, A_c, A_d	Constants of Eq. (3.38), Chap. 3
a	A factor used for bimorph in series
B	Bending stiffness of a composite beam
$\tilde{B}$	Complex amplitude of a constant in Eq. (4.22)
b	Width of the piezoelectric and shim layer
C_1, C_2, C_3, C_4	Constants of displacement $\tilde{u}$, Chap. 4
C_p	Internal capacitance of the piezoelectric
c_a	Viscous damping coefficient
c_p	Strain rate damping coefficient
D_3	Charge density in '3' direction

D, D_e, D_m, D_g	Dynamic stiffness Matrix, of a beam, with tip mass and global dynamic stiffness matrix, Chap. 4
d_{ij}	Coupling constant in *ij* direction
E_3	Electric field in '3' direction
F	Real component of force
$\tilde{F}$	Complex amplitude of force
f	A constant, for piezoelectric layers connectivity
G	A coupling constant, used as coefficient of voltage
$G_{\ddot{u}_b v}(\omega)$	Cross-spectral density function relating signals $\ddot{u}_b$ and v
$G_{\ddot{u}_b \ddot{u}_b}(\omega)$	Power spectral density function of $\ddot{u}_b$
$H(x)$	Heavy side function
H	Henry, a unit for inductance
h_p	Height of piezoelectric layer
h_{sh}	Height of the shim layer
I	Moment of inertia
$i(t)$	Electrical current
j	$\sqrt{-1}$
k	Wave number in Chap. 4
k_a	Stiffness of absorber
L	Inductor
l	Length of energy harvesting beam
M	Internal bending moment, including '*A*' term
$\hat{M}$	Internal elastic bending moment of EH beam, Chap. 3
$M_A^{(s)}$	Modal mass of the target mode 's'
$\hat{M}_T, M_T$	Tip mass or central mass, Chap. 4
M_r	Mass normalisation constant
m_a	Mass of the piezoelectric absorber
m_b	Overhung mass of the piezoelectric beam
m_{eff}	Effective mass of the piezoelectric absorber
m_{red}	Redundant mass of the piezoelectric absorber
N	Number of modes included
N	Number of turns of the coil, Chap. 6
n	Mode number
n	A multiplication factor for external capacitances, Chaps. 5 and 6
P	Power
$\tilde{Q}$	Complex amplitude of shear force
q	Electric charge
R	Load resistance
$\bar{R}$	Effective mass percentage, Eq. 5.12
R_{ij}	Elements of receptance matrix with *ij* index
r	Mode number
s	Mode of the host structure, Chap. 5
$s_{1,2,\ldots6}$	Elements of dynamic stiffness matrix, Chap. 4
t_p, t_{sh}	Thickness of piezoelectric layer and shim layer

t	Thickness of host
$u, \tilde{u}$	Absolute vibration displacement of beam, real and complex
$u_b, \tilde{u}_b$	Base vibration displacement, real and complex
$u_{rel}, \tilde{u}_{rel}$	Relative displacement (real and complex) of beam with respect to u_b
$v, \tilde{v}$	Generated voltage, real and complex notation
x	x-coordinate, or direction in '1'
Y_p, Y_{sh}	Young's Modulus of piezoelectric and shim layers
y	y-coordinate, or direction in '2'
y_A	Amplitude of the attachment point, Chap. 5
y_a	Amplitude of the absorber
z	z-coordinate, or direction in '3'
z_c	Distance from centre of piezoelectric layer to the neutral axis of the bimorph, Chap. 3
Z	Impedance of the circuit
α	A constant, Chap. 4
α_r	An electromechanical coupling constant
$\beta(\omega)$	Relative tip displacement to base acceleration FRF
$\beta_{abs}(\omega)$	Absolute tip displacement to base acceleration FRF
$\beta, \tilde{\beta}$	An electromechanical coupling constant (real and complex), Chap. 4
$\gamma_r^u, \gamma_r^\theta$	Translation and rotational motion constant
$\bar{\gamma}_r^u, \bar{\gamma}_r^\theta$	Translational and rotational motion constants, Chap. 4
δ	Dirac delta
δ	Strain in the piezoelectric material
$\varepsilon_{33}^T, \varepsilon_{33}^S$	Permittivity at constant stress and constant strain in '3' direction
$\eta_r, \tilde{\eta}_r$	Modal coordinates, real and complex
θ_0	Rotation DOF at x = 0, Chap. 4
θ_l	Rotation DOF at $x = l$, Chap. 4
λ_r	Dimensionless frequency parameters for mode r
μ	Mass ratio
ζ_{opt}	Optimal damping ratio
ζ_r	Damping ratio of mode r
ρ_p	Density of a piezoelectric material
ρ_{sh}	Density of a shim material
σ_1	Stress in x-direction
σ_r	A constant for mode r
φ_r	Mass normalised eigenfunctions
ψ_r	Un-normalised mode shapes
χ_r	An electromechanical coupling constant
ω	Excitation frequency
ω_A, ω_B	Frequencies at half power points
ω_r	Resonance frequency of mode 'r'
ω_a	Resonance frequency of absorber
Γ_0, Γ_l	Rotation excitation at clamped and free end

Δ	A constant, used in Chap. 4
Ψ_r	An electromechanical coupling constant used in Chap. 5
Ω	A symbol for Ohm, use for units of resistance
$\Omega_A^{(s)}$	Target frequency of the host of mode

Chapter 1
Introduction

1.1 Background of PVEH

The word ‘piezo’ means pressure, while ‘electric’ refers to electricity. This means an electric field is generated when a piezoelectric material is stressed. Piezoelectric materials have the capability to produce electric voltage when deformed due to vibrations (direct effect), and on the other hand, they deform when subjected to an externally applied electric voltage (converse effect) [1]. For energy harvesting mechanism, it is the *direct* piezoelectric effect which allows the material to absorb mechanical vibration energy from its host structure or surroundings and convert it into electrical energy, and thus forms the basis of the field of vibration-based piezoelectric energy harvesting. It is important to comprehend that the *direct and converse* effects coexist in a piezoelectric material. Therefore, in a system where the direct piezoelectric effect is of particular interest (as in energy harvesting) overlooking the presence of the converse piezoelectric effect (or backward coupling) will lead to erroneous results [2].

Typically, a piezoelectric energy harvester is a cantilevered beam with one or more piezoelectric layers, attached to a non-piezo material that is referred to as a ‘shim’. The main purpose of the metallic ‘shim’ layer is to provide stiffness and strength to the piezoelectric energy harvester. The harvester is attached to a vibrating host structure (as shown in Fig. 1.1), and the dynamic strain induced in the piezoelectric layers generates an alternating voltage output across the electrodes covering the piezoelectric layers.

In return, the generated voltage modifies the dynamic properties of the harvester (i.e. damping, stiffness and resonance frequency). It is essential to note that the term PVEH system, used in this book, refers to a base excited piezoelectric energy harvesting beam, unless otherwise specified. Figure 1.1 depicts the block diagram of PVEH mechanism.

The foremost advantages of employing piezoelectric materials in vibration energy harvesting are their higher power densities and ease of implementation [2].

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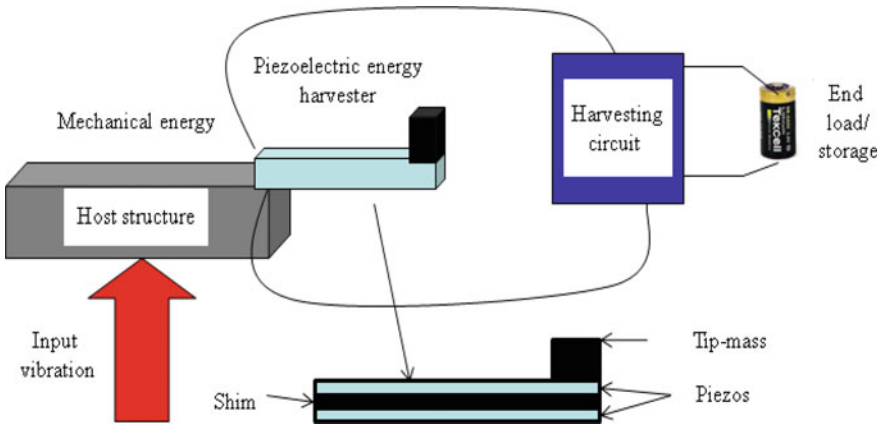

Fig. 1.1 Schematic diagram of a typical PVEH mechanism

Moreover, owing to the well-established thin-film and thick-film fabrication techniques, piezoelectric energy harvesters can be fabricated in both macro-scale and micro-scale easily [3]. It is also worth mentioning that the lifetime of a piezoelectric harvester can almost be unlimited if the applied force and working temperature are kept within the operational range of the material [4]. Furthermore, piezoelectric technology is sufficiently established, and a broad range of piezoelectric materials are conveniently available to use in energy harvesting.

PVEH is a multi-disciplinary field and involves in-depth understanding of mechanics, electrical circuitry and material engineering. For these reasons, there has been a constant effort to develop modelling techniques for PVEH systems by a wider research community belonging to different backgrounds. Therefore, several oversimplified and approximate mathematical modelling assumptions have been used in the literature to model the coupled PVEH mechanism. Erturk and Inman [5] identified some major modelling issues that had appeared in the literature during the preceding few years and showed that some of them might be misleading due to weak mathematical assumptions. The same authors later presented an accurate distributed parameter model of a PVEH beam based on the analytical modal analysis method (AMAM) [6].

This is important to note that the AMAM approach is primarily tractable to simple-cantilevered uniform-section beams. Moreover, owing to the modal representation of the vibration, an adequate number of modes must be included in the solution to ascertain accuracy. Moreover, the existing research in PVEH lacks the insight to investigate the *energy harvester-host structure* interactions which can have potential applications in vibration control. The limitations and gaps in the existing PVEH modelling techniques provided the primary motivation to the work presented in this book. The first part of the research mainly concerns a theoretical and experimental investigation of a PVEH system that is more in-depth than some of the previous work [7]. The theoretical investigation in this part of the book is based on the AMAM. The experimental study used a double-cantilevered configuration which cancelled out the bending moment on either side of the clamp reducing the rotation effects, as present in the previous studies [7–9]. In the second part of this book, a modelling technique based on dynamic stiffness method (DSM) is presented [10]. The existing AMAM is also reformulated such that its formulae are condensed to encompass all previously analysed systems. Moreover, the electric load is generalised to arbitrary linear impedance. Damping-related assumptions are thoroughly investigated in Chaps. 3 and 4. The accuracy of AMAM is verified against DSM, and the limitations of AMAM are discussed. The DSM is also used to inspect the effect of the piezoelectric energy harvester on the dynamics of its host structure. This leads to the next important part of this research, which presents the application of energy harvesting to vibration control. The concept of a dual function EH beam/TVA or 'electromechanical Tuned Vibration Absorber' is introduced and analysed theoretically in Chap. 5. A prototype electromechanical TVA is then built, and the theory is experimentally validated for different electrical circuit configurations. It is worthwhile to note that the work presented in Chaps. 5 and 6, combines the classical mechanical TMD [11, 12] and its electrical analogue (i.e. the piezoelectric shunt circuit) [13, 14], overcoming their relative disadvantages while retaining their advantages, as detailed in Sect. 5.1.1. Chapters 5 and 6 illustrates the vibration attenuation capabilities of a PVEH system on its host structure. However, an in-depth investigation of energy harvesting/storage capabilities of the proposed EH/TMD device involves suitable modifications in the proposed model to include nonlinear elements used in AC-DC rectifications. Chapter 7 presents the analytical study of the application of a dual function device to optimally attenuate the vibration of an electronic box mounted on a vibrating host structure. The final chapter presents summary of the book and the future trends along with potential application areas of vibration energy harvesting devices. Appendix-C presents details about the equipment and hardware used in the experimental study of this book. Appendix-A presents MATLAB programs of working code of complex electromechanical equations of the mathematical models presented in this book. These working codes shall provide a convenient launch pad to see through the effects of electromechanical and geometrical parameters on the PVEH system.

1.2 Aims and Objectives of the Book

Piezoelectric vibration energy harvesting is a multi-disciplinary field and, hence, involves an in-depth knowledge of mechanics, material engineering, electrical circuitry and charge storage devices. The book effectively investigates the relevant knowledge of all these areas and critically evaluates the problems and issues persistent, hindering the development of effective energy harvesting systems. Therefore, the overall aims of this book are manifold:

1. *To present a thorough theoretical and experimental analysis of a piezoelectric energy harvesting beam (system) or assembly of beams, with the aim of presenting a deeper intuition of the dynamics of piezoelectric vibration energy harvester (PVEH).* This aim primarily has following objectives.

   - To present specifically a thorough experimental validation of a distributed parameter analytical modal analysis model of a cantilever energy harvester without a lumped/tip mass in order to investigate the implementation of the AMAM at *higher* resonance frequencies (120–130 Hz).
   - To carry out a comprehensive comparative study into the identification of (mechanical) *modal damping* of the harvester, using different techniques, based on experimental voltage and tip frequency responses.
   - To discuss electromechanical *coupling* effects of the energy harvesting mechanism on its dynamics, by probing the shifts in the resonance frequency, mechanical response, resonant voltage, resonant current and resonant power outputs of the system against different loads, '*R*'.
   - To present a mathematical modelling method (based on the dynamic stiffness method) which can offer an exact close-form modelling technique, by eliminating the modal transformation and modal approximation as required by AMAM, either for a single piezoelectric EH beam or assembly of beams, having the same or different cross sections and with any arbitrary boundary conditions.
   - To carry out a meticulous comparative study of a piezoelectric bimorph using DSM and AMAM to verify the accuracy of the AMAM.
   - To show the applicability of DSM to the analysis of systems those are more complex than the typical clamped-free uniform-section PVEH beam.
   - To present an analytical study to investigate the electromechanical effects of piezoelectric *energy harvester-host structure* interaction on the dynamic response of the host structure, using DSM.

2. *To show an in-depth analytical and experimental study of the idea of a dual function piezoelectric vibration energy harvester beam/tuned vibration absorber (TVA) (to be referred to as an"electromechanical TVA") and its prospective application to vibration control*. This aim has following objectives:

   - To present a theoretical model, using AMAM and DSM, to analyse the concept of the electromechanical TVA, when applied as a tuned mass damper (TMD) to attenuate a vibration mode of a generic host structure.
   - To show an in-depth comparative analysis to evaluate the implementation of AMAM and DSM to the proposed electromechanical TMD for diverse electrical circuits.

- To present the details of the complete test set-up which was used in the experimental study.
- To present a test example of implementation of the theory of dual function vibration energy harvesting tuned vibration absorbing device to suppress the vibration of a symbolic electronic box.

1.3 Major Contributions of the Book

The research carried out in this book is intended to provide the reader with an in-depth insight into the electromechanical behaviour of PVEH systems and its impending application in vibration control. The contributions of this book are outlined in the following list:

1. The first part of the research presented in this book uses the closed-form distributed parameter AMAM as its theoretical basis but considerably contributes towards the knowledge of PVEH through a theoretical and experimental analysis that is more in-depth. A salient feature of this phase is that the previously published notion that the electrical load that generate maximum power is the same as that which creates maximum vibration attenuation (of the vibrating beam) [6, 15] is not correct. It is shown that the load that generates maximal resonant power is much higher than the load that yields a minimum response of the tip of a base-excited PVEH cantilever. Moreover, the study presents, graphs demonstrating the theoretical and experimental variation with electrical load of the resonance frequency, resonant voltage amplitude, resonant power and resonant deflection amplitude. These graphs provide a deeper insight into the electromechanical interaction within the harvester. Previous studies, e.g. [6, 7], have only shown the variation with load of the voltage and power at two fixed excitation frequencies (equal to the short- and open-circuit resonances, respectively). Nyquist plots allow for a more thorough validation of the theory than FRF magnitude plots used in preceding studies since they depict both the magnitude and phase of the FRFs. The centre of the Nyquist circle is seen to shift in a more noticeable way with the electrical load than in the FRF magnitude plots, illustrating the significance of electromechanical coupling in a more noticeable manner. Moreover, such Nyquist plots are also useful for the identification of the mechanical modal damping of the PVEH device.
2. In the second part of the research, a mathematical modelling technique based on the dynamic stiffness matrix (DSM) method is developed to model piezoelectric beams. DSM requires less elements than the finite element method for an assembly of uniform-section beams [16], offering a more accurate solution for high-frequency applications. In contrast to AMAM, the DSM readily lends itself to the modelling of beams with different boundary conditions or assemblies of beams of different cross sections.

3. Analytical investigations performed in this book reveal that AMAM converges to DSM if a sufficiency of modes is used in the AMAM.
4. A thorough investigation of damping, and the damping related assumptions, in PVEH beam analysis is presented in the first two parts of the research.
5. Analytical investigations using DSM revealed that a significant increase in the power output from a base-excited PVEH cantilever could be achieved through the application of a tip rotational restraint and the use of segmented electrodes.
6. The analytical investigations using DSM revealed the neutralising effects of a tuned harvester beam on the vibration at its base for different electrical loads. The findings suggest the use of a piezoelectric beam shunted by variable capacitance for the dual function of adaptive vibration neutralisation/energy harvesting. The vibration neutraliser is one type of tuned vibration absorber that is designed to suppress harmonic vibration at a particular excitation frequency.
7. The third part of the research extends the above concept to the other type of tuned vibration absorber—the tuned mass damper (TMD)—which suppresses a particular vibration mode of a generic host structure over a broad band of excitation frequencies. In-depth theoretical and experimental investigations are presented to validate the concept of the dual EH/TMD beam device or 'electromechanical' TMD. This device comprises a pair of bimorphs shunted by resistor-capacitor-inductor circuitry. The optimal damping required by this TMD is generated by the PVEH effect of the bimorphs. The results demonstrate that the ideal degree of vibration attenuation can be achieved by the proposed device through appropriate tuning of the circuitry. It is revealed that vibration reduction factors of 10 or more are achievable by an EH/TMD beam device whose effective mass is less than 2% of the equivalent modal mass of the host structure. The EH effect provides a far easier way of controlling/adjusting the TMD damping compared to conventionally damped TMDs. The proposed dual EH/TMD beam device combines the advantages of the classical (mechanical) TMD and the electrical vibration absorber, presenting the prospect of a functionally more readily adaptable class of 'electromechanical' tuned vibration absorbers.

1.4 Overall Structure of the Book

Chapter 1—*Introduction*: This chapter presents the overview, outline of the research conducted along with objectives and contributions to knowledge. The chapter concludes with a summary of the book organisation.

Chapter 2—*Energy Harvesting Overview*: This chapter gives a comprehensive background of the subject and also provides a critical review of the already published work in the core areas relevant to this book.

Chapter 3—*Distributed parameter modelling and experimental validation of a piezoelectric bimorph EH beam*: This chapter presents a theoretical and experimental study of a PVEH system that is more in-depth than available in the literature.

Chapter 4—*Mathematical modelling of Energy Harvesting beams using DSM and AMAM methods*: In this chapter, modelling of energy harvesting beams using dynamic stiffness method (DSM) is presented, and the accuracy of AMAM is verified with DSM for the case of a bimorph. The AMAM is also reformulated into a condensed form that encompasses all previously analysed systems in the literature. The applicability of DSM to the study of more complex PVEH systems is demonstrated.

Chapter 5—*A Theoretical study into an electromechanical beam Tuned Mass Damper*: This chapter investigates theoretically the concept of a dual EH/TMD beam, in which a suitably shunted energy harvesting beam is used as a TMD on an arbitrary structure.

Chapter 6—*Experimental validation of "electromechanical beam" tuned mass damper model*: In this chapter, a prototype electromechanical TMD formed from a pair of suitably shunted bimorphs is tested to validate the concept and theory introduced in the previous chapter.

Chapter 7—Example application of v*ibration suppression of an electronic box using dual functionEH/TVA*: In this chapter, an analytical study illustrating the application of the theory developed in Chap. 5 is presented. The response of electronic box at its first resonance frequency is optimally suppressed by the implementation of the dual function EH/TVA.

Chapter 8—*Summary and future research*: *The chapter illustrates summary of the work presented in the book along with highlighting topics for future* in this field.

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Chapter 2
Overview of Vibration Energy Harvesting

2.1 Background of Energy Harvesting

The idea of extracting energy from ambient sources and accumulating and storing it for a useful purpose is called energy harvesting (EH). The idea of harvesting energy from ambient sources is not fresh, and its history dates back to the windmill and the waterwheel [1]. For many decades, researchers have been establishing techniques to harvest energy from heat and other ambient sources. However, owing to the low energy conversion efficiency and a higher power requirement by many electronic applications, the field of EH did not attract enough attention in the past.

With a global desire for harvesting ‘green energy’ from ambient sources and recent advances in low-powered portable wireless electronic devices, the area of EH has attained greater attention over the past few years. The drastic reduction in size and power utilisation of modern electronic devices has buoyant researchers and industry to discover schemes to implant an endless power supply means within these systems, which can harvest energy from their surroundings for their entire lifespan.

Conventionally, these low-powered systems are designed to function on limited electrochemical batteries, which need intermittent replacement (e.g. alkaline battery AA) or recharging (e.g. Nickel-Zinc, Nickel-Cadmium, Lithium-ion batteries). However, the advancement in conventional battery technology has not been rapid enough to fulfil the long-lasting power needs of these portable wireless electronic systems [1, 2]. Therefore, a periodic battery-recharging or substitution is an indispensable requirement for these electronic applications. In many applications, this is not a practical choice, e.g. battery-recharging where a standard power supply is unavailable in distant areas, or a battery-replacement in sensors implanted in giant structures or in implanted medical devices. Figure 2.1 illustrates advancements in the performance of laptop computers (a mobile technology example) on a logarithmic scale relative to a laptop from 1990 [1]. It can be seen from the graphs that the progress in battery energy express the slowest tendency in mobile

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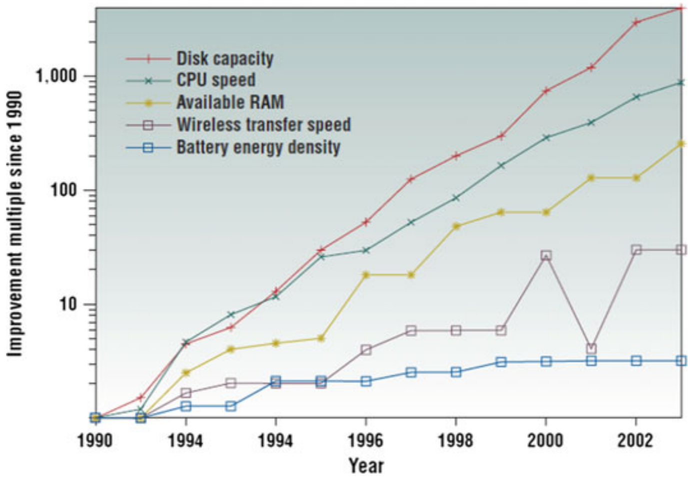


Fig. 2.1 Relative advancements in processing and computing technology from 1990 to 2003 [1]

computing [1]. For these reasons, conventional batteries are no longer a rational energy resource in most modern electronic devices owing to the issues of recharging, replacement or disposal.

Tables 2.1 and 2.2 below present a short survey of specific power production capabilities of some common ambient energy sources together with the power consumption needs by some modern electronic devices. The power values in both tables are extracted from the published literature and are shown here for illustrative purposes only, as these values can be altered in different operating conditions. Thus, care should be taken in making comparative studies.

Table 2.1 Electrical Power generation capacities of some ambient sources

Energy source	Conversion mechanism	Energy level	Reference
Vibration	Piezoelectric	100 mW/cm^3	[2, 3]
	Electromagnetic	0.5–8 mW/cm^3	[4]
	Electrostatic	8nW–42.9 μW/cm^3	[5]
Light	Photovoltaic (solar)	100 mW/cm^2	[6]
	Photovoltaic (indoor)	100 μW/cm^2	
Ambient radiation	Radio frequency	$\leq$1 mW/cm^2	[1]
Wind	Turbine	200–800 μW/cm^2	[5]
Thermal	Thermoelectric, thermionic, thermo-tunnelling	60 μW/cm^2	[1]

Table 2.2 Power utilisation by some portable wireless electronic devices

Electronic device	Power requirement	Reference
Electronic watch or calculator	1 μW	[6]
Implanted medical devices	10 μW	[7]
HTC Touch Pro phone (active mode) without GPS	29.1 μW	[8]
Hearing aids	100 μW	[6]
Hearing aids	1 mW	[7]
ET thermistor	3.5 mW	[9]
HTC Touch Pro phone (active mode) without GPS	24.8 mW	[8]
Bluetooth transceiver	45 mW	[6]
Palm MP3	100 mW	[6]
Phototransistor filter	150 mW	[10]

It can be observed from Tables 2.1 and 2.2 that the majority of the wireless electronic devices have the prospects of being powered by an implanted energy harvesting system in the near future.

2.2 Energy Harvesting from Vibrations

Williams and Yates [11] described three major vibration energy harvesting mechanisms: (1) electromagnetic, (2) electrostatic and (3) piezoelectric harvesting. In their research [11], Williams and Yates investigated the case of a lumped parameter base-excited model to analyse the electrical power generation for electromagnetic energy harvesting.

Electromagnetic energy harvesters are designed on the theory of Faraday's law of electromagnetic induction which describes that when a conductor (wire) is moved in a magnetic field, or vice versa, a voltage is generated around the conductor. The quantity of voltage produced by the electromagnetic induction is proportional to the rate of change of magnetic flux around the conductor. A classic electromagnetic energy harvester consists of a permanent magnet, a vibrating cantilever and a conducting coil as shown in Fig. 2.2 [12]. This technique does not require any outside power source to kick-start the energy harvesting process. However, the amount of energy harvested is proportional to the cube of the frequency of vibration [11], reducing its usefulness for low-frequency conditions. Moreover, the highest voltage generated by this transduction mechanism is usually quite low [13].

Electrostatic vibration energy conversion works on the rule of variable capacitance. The capacitance of a pre-charged capacitor changes as the distance between the two plates of a capacitor is varied. The charge on a capacitor is associated with the capacitance as: $Q = CV$, where Q is the charge on either plate, V is the voltage difference between the plates and C is the capacitance. The working principle of an

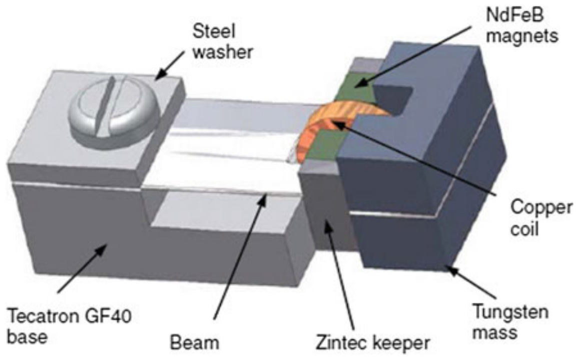


Fig. 2.2 Schematic representation of a micro electromagnetic EH [12]

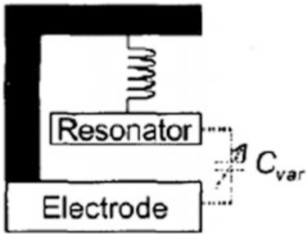


Fig. 2.3 Working principle of an electrostatic vibration-based EH [14]

electrostatic energy harvester is shown in Fig. 2.3 [14]. By keeping either Q or V constant across an initially charged capacitor, any variation in C can cause variation in Q or V, which results in electrical charge production.

It is worth mentioning that the electrostatic vibration EH technique is not self-driven and requires an external power source to initially charge the capacitor and, thus, switch on the energy harvesting process. Moreover, the space available, within a capacitor itself, to allow the relative motion of the two capacitor plates is very narrow, which imposes durability hazards for the components.

Further details, elucidations and applications of electrostatic and electromagnetic vibration energy harvesting can be studied in references [11, 12, 14–18]. Table 2.1 illustrates that PVEH has a notably higher energy density level than both electromagnetic and electrostatic vibration EH techniques. For this basis and the other motives mentioned in Chap. 1, Sect. 1.1, this book is focused only on PVEH.

2.3 Piezoelectric Vibration Energy Harvesting (PVEH)

Previous research work broadly investigated the use of piezoelectric materials in vibration-to-electricity energy harvesters [2, 19–21]. To explore the ability of the piezoelectric energy harvesting technique from vibrations, researchers not only published different mathematical models but also provided experimental results

[22–25] to validate their proposed models. More generally, the research work in PVEH is mainly focused on:

- To develop mathematical techniques [26–34] to model the PVEH mechanism;
- To improve the geometrical and physical composition of harvesters [23, 35, 36] for maximising output;
- Enhancing the ability of electrical circuitry to extract more power [37–39];
- Studying the backward-coupling effects of piezoelectric energy harvesting on the dynamics of the system [27, 40–44];
- Improving the power storage medium [45, 46].

2.3.1 Improvements in Modelling Techniques

The modelling and analysis of the base-excited piezoelectric energy harvesting beams portrayed in Fig. 2.4 have fascinated many researchers with the aim of predicting the electrical output for given base motion input. In reference [22], it is illustrated that highly different modelling techniques had published in the literature during the preceding few years, and some of them might be ambiguous due to weak mathematical assumptions. For example, the early modelling attempt in reference [23] was based on a lumped parameter single degree of freedom (SDOF) model. Erturk and Inman [26, 28, 29] later showed that SDOF models for the distributed parameter systems in Fig. 2.4 may produce highly imprecise predictions. It was shown that, in the vicinity of the first natural frequency and without a tip mass, the errors in relative motion transmissibility functions when using a SDOF model can be greater than 35% irrespective of the mechanical damping [28]. Moreover, it was also presented that the error of using a SDOF model decreases as the ratio of tip mass to distributed mass of the beam is increased [28]. However, this technique (SDOF) continues to be used in theoretical studies, e.g. in reference [47].

A more exact approximation to the distributed parameter system has been acquired via the Rayleigh–Ritz type discrete formulation, e.g. [32]. This method uses a transformation of the displacements using appropriately selected basis functions. Hamilton's Principle is then used to obtain discrete mass, stiffness and damping matrices in the transformed space. Analytical solutions that are limited to a single vibration mode expression have also been presented [48, 49]. Furthermore, these latter works have either ignored the essential effect of the piezoelectric coupling on the system dynamics [48] or oversimplified it as viscous damping [49]. Erturk and Inman [22] identify a more recent work [30] that is basically inconsistent since the piezoelectric effect is introduced on the basis of a static tip-force/deflection correlation.

A significant improvement in analytical modelling was made by Erturk and Inman [40], who applied an analytical modal analysis method (AMAM) to an Euler–Bernoulli model of a clamp-free uniform-section unimorph without tip mass. The piezoelectric coupling was precisely included through the piezoelectric

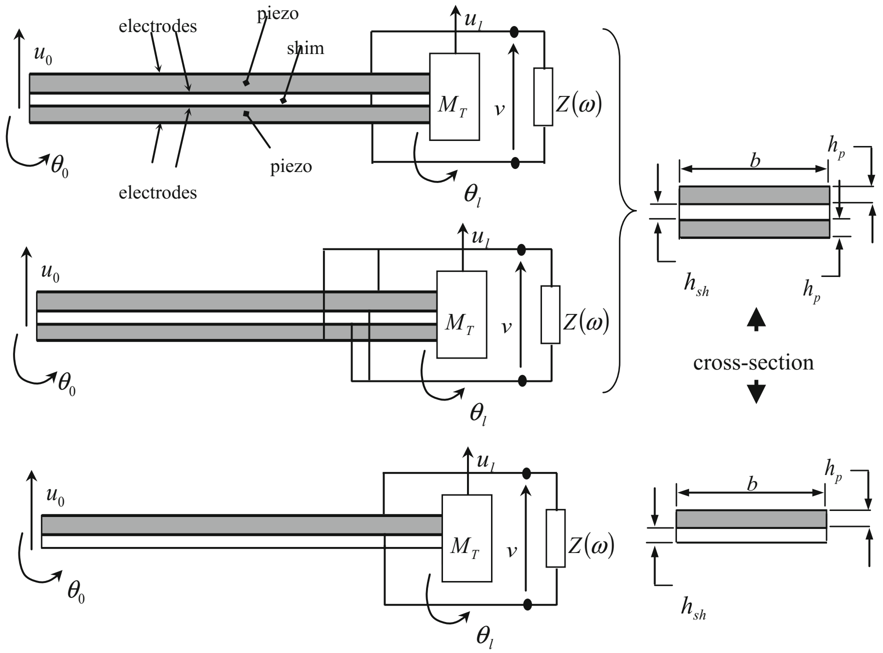


Fig. 2.4 Base-excited piezoelectric EH beams (series-connected bimorph, upper figure; parallel-connected bimorph, middle figure; unimorph beam, lower figure)

constitutive relations. Terms accounting for material ('Kelvin-Voigt' or 'strain-rate') damping were included directly into the wave equation, in addition to ambient damping. The resultant wave equation with electromechanical coupling was transformed into modal space using the clamped-free bending modes of the undamped, electrically uncoupled beam. An analogous technique was later used in [22] to model a bimorph with a tip mass. Results for the frequency response functions (FRFs) relating the voltage and tip motion to the base excitation were illustrated in [22] over the first resonance region and experimentally validated. The presence of a tip mass in [22] reduced the influence of the distributed inertia of the beam and limited effective operation to low frequencies (e.g. 45–50 Hz resonance in [22]). A more rigorous experimental validation, covering the higher end of the frequency range of application, that most harvesters are designed for actual vibration conditions is necessary. This gap in the research is addressed in Chap. 3 of this book, which has been published in reference [25], and constitutes contribution no. 1 in Sect. 1.3.

None of the works in [22, 25, 40] validate their modelling over a wide frequency range against an alternative technique. However, Elvin and Elvin [50] presented that their numerical Rayleigh–Ritz (RR) solution converges to the AMAM solution [40] provided that appropriate basis functions are chosen for the RR method, and enough number of modes/basis functions are used in both methods. It is obvious from the literature that PVEH beam modelling lacks an exact method that does not require approximations via modes/basis functions. This gap in the research is addressed in Chap. 4 of this book through the novel application of the dynamic stiffness method to PVEH beams [51–55]. The work in Chap. 4 has been published in reference [27] and contains contribution no. 2 in Sect. 1.3.

It is also worth mentioning that most of the distributed parameter modelling techniques, e.g. [22, 25, 29, 40] are based on a simple resistive impedance. Elvin and Elvin [50] stated that their model can entertain complex circuitry connected to the piezo but very little detail is presented in this regard. The work in this book addresses this gap in the research by generalising the analysis to generic resistor-capacitor-inductor circuitry.

2.3.2 Progress in Geometric Configuration of Piezoelectric Harvester

Previous research extensively investigated different piezoelectric energy harvesting configurations with the aim of increasing the power harvesting ability. The configuration of the PVEH device can be changed through diverse ways such as tuning the resonant frequency of the harvester; using multi-layers of the piezoelectric material to maximise the active volume; studying performance comparisons of various piezoelectric materials; changing the electrode pattern; changing the poling and stress direction; altering the coupling mode of the harvester; and optimising the geometrical shape of the device.

2.3.2.1 Resonant Tuning the Harvester

Most vibration-based piezoelectric energy harvesters produce peak power when the resonant frequency of the harvester matches with the frequency of the input ambient vibration. Any difference between these two frequencies can result in a considerable lessening in generated power. This is a basic disadvantage of piezoelectric resonant vibration energy harvesters which confines their power generation capability in real applications [47]. The aim of this section is to abridge the previous research investigating the resonance functionality of the harvester and its effects on power output. It is important to note that the contribution nos. 1, 2, 4 and 5 listed in Sect. 1.3, offer in-depth study of the power output of the PVEH system over a broad range of frequencies around the resonance.

The work [33] used an auxiliary structure which was tuned to the resonance frequency of the most dominant mode of vibration of three-storey host structure and connected it to the main structure. The auxiliary structure consisted of a mechanical fixture, a cantilever beam and a PZT element bonded on it as shown in Fig. 2.5.

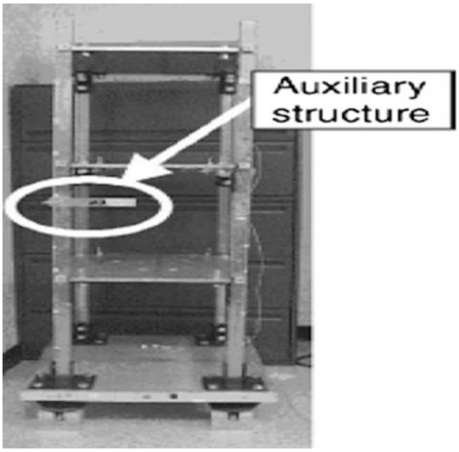


Fig. 2.5 Resonant Tuned auxiliary structure attached to a vibrating host structure [33]

A PZT bonded directly to the three-storey host structure generated 0.057 V, the PZT attached to the mistuned auxiliary structure produced 0.133 V and the PZT attached to the tuned auxiliary structure produced 0.335 V [33]. It is important to note that the PZT attached to a mistuned auxiliary structure produced twice more voltage output than the conventional technique of attaching it directly to the main host structure. The work in [33] only focussed on analysing the power outputs with and without the tuned auxiliary structure and did not study the alterations in the mechanical response of the host structure at the point of attachment.

The work [56] investigated the concept of active self-tuning of a piezoelectric generator to deal with the problem of mistuning of the harvester. The resonance frequency of the piezoelectric generator was harmonised with the resonance frequency of the ambient source by varying the effective stiffness or mass. Comparing to the extra power generated by adaptive tuning of the harvester with the power consumed by the tuning mechanism, the idea of self-active tuning did not prove efficiently in this approach [56].

A more effective method of achieving active self-tuning is through the addition of a microcontroller [57]. The microcontroller gets its power from the upper piezoelectric element of the bimorph, whereas the energy harvesting system is attached to the lower piezoelectric element of a bimorph as shown in Fig. 2.6.

The natural frequency of the power harvesting device was matched with the excitation frequency of the ambient vibration by changing the stiffness of the harvester beam [57], and a 30% average increase in harvested power was observed. The work stated that the major gains in the efficiency are achieved by reducing the structural stiffness and mechanical damping, and increasing the effective mass [57]. However, an inexact SDOF model was used to obtain the voltage equation, which is not an accurate selection as explained in Sect. 2.3.1. While the oversimplified model may be sufficient for preliminary power output study, it is not enough to study the effects of EH and its backward-coupling effects on the dynamics of the system.

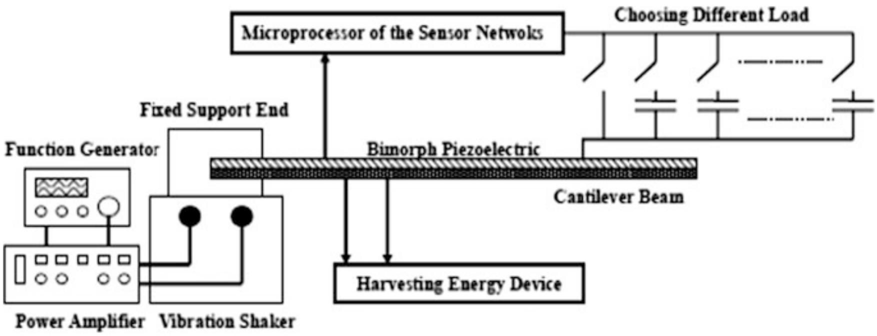


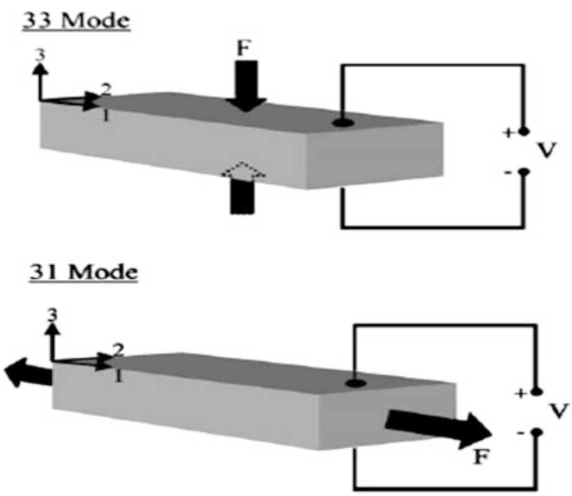
Fig. 2.6 Block diagram of experimental set-up of EH system [57]

2.3.2.2 Coupling Mode Effects on Power Output

The capability of a piezoelectric material to transform mechanical energy into electrical energy depends also on the piezoelectric coupling mode used in the energy harvesting system. Two practical piezoelectric coupling modes of operation exist, which are the −31 mode and the −33 mode as shown in Fig. 2.7 [23]. In −31 mode, force is applied perpendicular to the poling direction, whereas in the −33 mode the force is applied along the poling direction as shown in Fig. 2.7.

It is important to have proper consideration of piezoelectric mode of operation (−31 or −33) when designing a PVEH system since the power output and the dynamic properties of PVEH system fully depend on it. The −31 mode has a lower coupling coefficient than the −33 mode. The aim of this section is to investigate the performance of PVEH models in −31 and −33 modes as it is the main design feature with regard to the electromechanical conversion capacity of the harvester.

Fig. 2.7 Two piezoelectric coupling modes, −33 mode and −31 mode, loading operations [23]



The work [58] investigated a piezoelectric stack configuration in the −33 mode, and the results were compared to the results from an equal volume of a cantilever beam operating in −31 mode. The stiffer stack configuration was not an appropriate choice to create higher strain in the harvester for low-intensity ambient vibrations. Therefore, a lower electrical power output was achieved, even though the loading was applied in higher coupling mode (−33 mode) [58]. On the other hand, the cantilever beam operating in −31 mode (having lower coupling coefficient) produced twice as much power (compared to the stack configuration under the same loading conditions) [58]. This shows that in low-intensity vibration environment, a cantilever EH operating in −31 mode is more efficient since it produces greater strain. For high-intensity vibration conditions, namely vibrations at the bases of manufacturing machinery, a stack configuration operating in −33 mode is more useful and robust, and generates more power [58]. For these reasons, piezoelectric energy harvesters have been more yielding in −31 coupling mode since more strains can be generated with smaller input forces [58]. Moreover, the resonant frequency of the system is much lesser in −31 mode because of the reduced stiffness, thereby increasing the effectiveness of the EH system to operate at low ambient vibrations [23, 58]. Therefore, a clamp-free beam, with a tip mass at the free end as shown in Fig. 2.8, has been a useful PVEH configuration investigated by majority of researchers.

The rectangular clamp-free configuration is the most rigorously researched design in the PVEH literature. However, other designs have also been researched. For example, the works in [59, 60] examined analytically and experimentally the effectiveness of a membrane type geometry [60]. A lumped parameter model was used to study the effects of design and process parameters, such as residual stress, substrate thickness, piezoelectric thickness and electrode coverage, on the electromechanical *coupling coefficient* [60]. The work showed that the coupling coefficient and thereby the power output of the harvester enhanced considerably by the appropriate selection of residual stress, substrate thickness and the covered area of the electrodes in membrane harvester. It was observed that for a substrate of thickness 2 μm, altering the thickness of piezoelectric layer from 1 to 3 μm, increased the coupling coefficient 4 times. The experimental work presented that, for a membrane having an initial residual stress of 80 MPa, the coupling coefficient was improved by 150% [59]. It should be noted that the work used a lumped

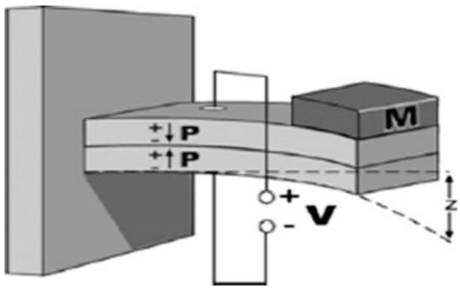


Fig. 2.8 Functioning of a piezoelectric bimorph in a −31 mode

parameter model to study the distributed parameter system which may yield highly inaccurate predictions as explained in Sect. 2.3.1.

2.3.2.3 Comparison of Various Piezo Materials and Shapes

A range of piezoelectric materials with a variety of electromechanical properties are easily available. The work in [35] explores the energy harvesting competences of three piezoelectric materials (Macro-fibre composites (MFC), quick pack (QP) and quick pack interdigitated electrodes (QP IDE)), attached to the same aluminium beam as shown in Fig. 2.9. In this method, all of the materials under consideration experienced the same input vibration excitation. The MFC contained piezoelectric fibres embedded in epoxy matrix that permitted greater flexibility and strain, and it utilised interdigitated electrode pattern which permitted the electric field to be applied along the length of the fibre, enabling it to function in a higher d_{33} coupling mode [35]. The work included the first 12 modes of the beam, and the amount of power harvested was recorded for all the materials.

The results illustrated that the quick pack showed to be more capable, extracting more power compared to MFC and quick pack IDE. It is important to note that the interdigitated electrode pattern of the MFC and quick pack decreased the capability of the devices to accumulate more charge and thereby reduced the net harvested power output. Furthermore, it was found in [35] that a high electromechanical coupling and permittivity values increases a material's ability to transfer more mechanical energy into electrical energy under given conditions. Moreover, the quantity of power produced by a brittle piezoelectric element may be smaller than the one produced by a flexible piezoelectric material. This may be due to the reason that the brittle materials can absorb less strain and hence produce a reduced power output. The work in [35] only compared the experimental power outputs of different

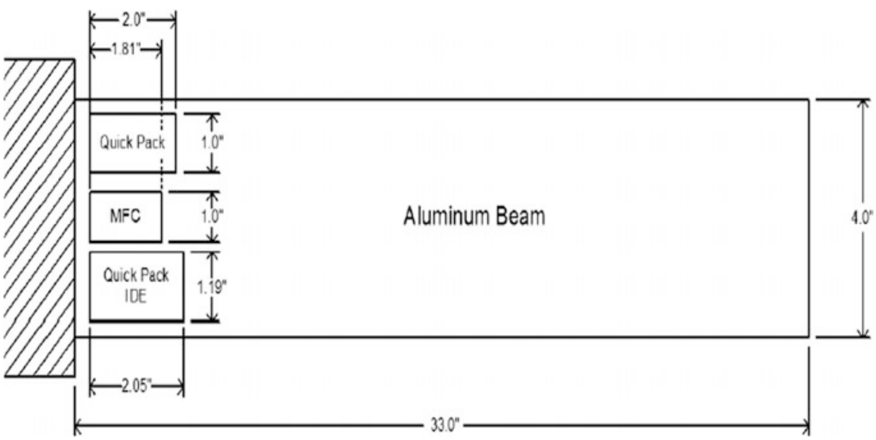


Fig. 2.9 QP, QP IDE, MFC patches bonded to same aluminium beam [35]

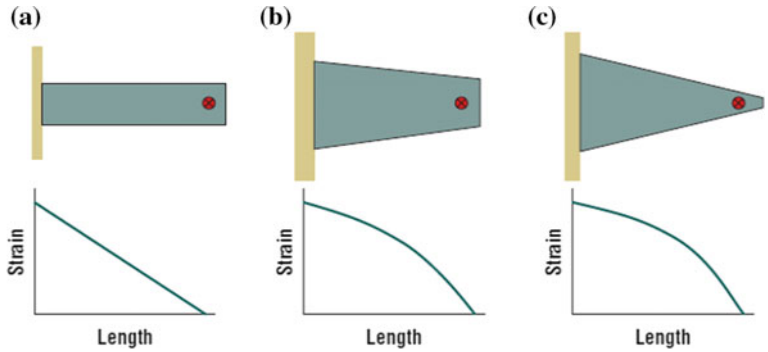


Fig. 2.10 Relative bending energies and strain profiles for different beam configurations. Red circle indicates load point [3]

piezoelectric materials. The work neither used any mathematical model nor any other alternative theoretical method to evaluate and confirm the experimental results.

Typically, a rectangular cantilevered geometry has been a popular (e.g. unimorphs or bimorphs) option among the researchers working in PVEH. However, the work in reference [3] demonstrated that, with an increasingly trapezoidal-shaped cantilever, the strain distribution is consistently distributed all over the structure in comparison with a rectangular beam that contains a non-uniform strain distribution (Fig. 2.10). It was also shown that, for the same volume of PZT energy harvesters, a trapezoidal cantilever can generate more power than a rectangular beam [3].

In [61], the performance of various triangular- and rectangular-shaped beams, as shown in Fig. 2.11, was studied. It was demonstrated that the triangular-shaped beams performed better than the rectangular ones in terms of tolerable excitation amplitude and maximum power output [61].

In another study [62], different design configurations were studied to increase the specific power of the harvester by optimising its geometric shape. The power output for two trapezoidal-shaped harvesters (the wider side clamped or free) was compared with that of a conventional rectangular harvester. It was shown that the proposed optimised trapezoidal geometries produced more power output than the rectangular geometry, as can be seen in Fig. 2.12 [62].

To date, there are diverse piezoelectric materials and design configurations that are published in the literature with the definitive aim of maximising the output and thereby the efficiency of the energy harvesting system. Figure 2.13 shows examples of some generally available configurations of piezoelectric materials [19]. It is important to note that the work [3, 19, 61, 62] efficiently examines EH abilities of diverse harvester geometries to attain higher power but ignored the effects of energy harvester-host structure interaction in their models. These gaps in knowledge are addressed in this book (i.e. Chapter 4, 5, 6) through contributions nos. 6 and 7 listed in Sect. 1.3.

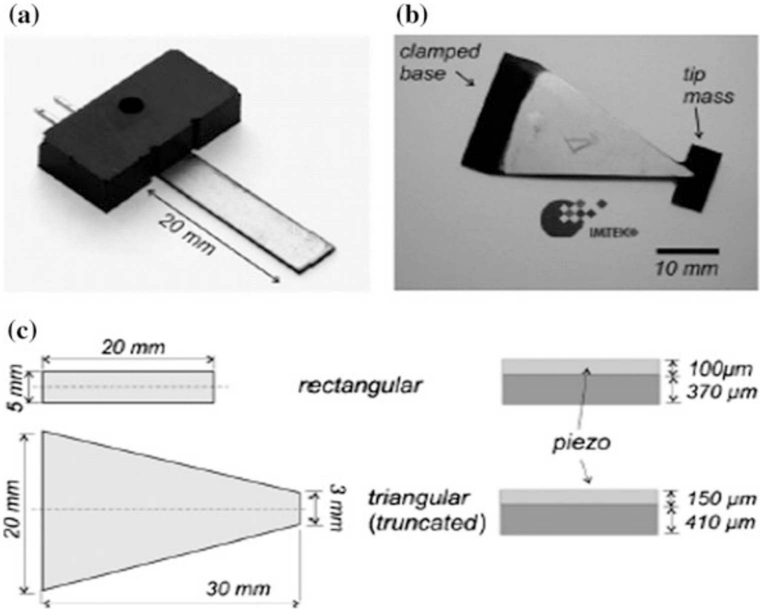
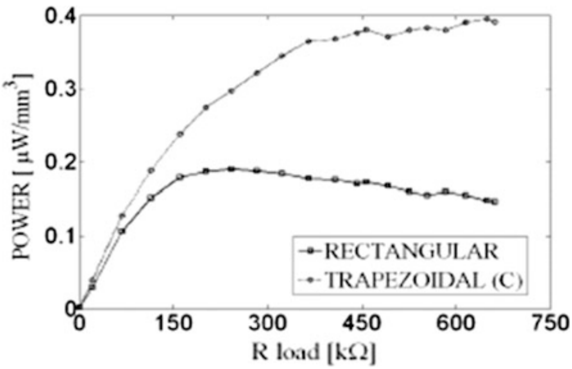


Fig. 2.11 Examples of different piezoelectric EH beam shapes [61]

Fig. 2.12 Comparison of power outputs of rectangular- and trapezoidal-shaped EH [62]



2.3.3 Application of PVEH in Vibration Control

Prior to considering the application of PVEH to vibration control, it is best to give a more general background to vibration control research, mainly tuned vibration absorbers. A tuned vibration absorber (TVA), in its most basic form, is an auxiliary system whose parameters can be tuned to suppress the vibration of a host structure. As discussed by von Flotow et al. [42], the auxiliary system is generally a structure that is equivalent to a spring-mass-damper system, in which case the TVA is

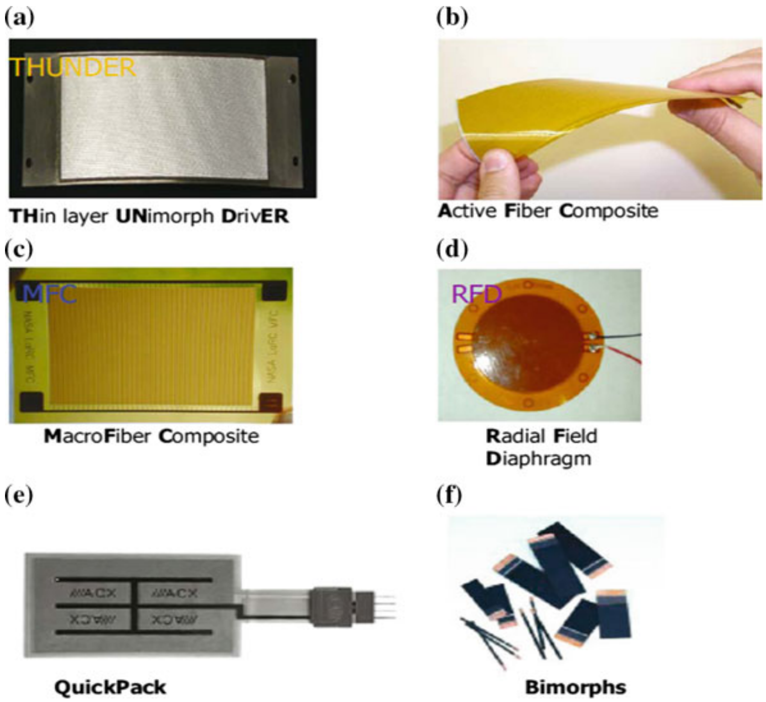


Fig. 2.13 Examples of some piezoelectric energy harvester configurations [19]

referred to here as a 'mechanical' TVA. The tuned frequency ω_a of the TVA is defined as its lowest undamped resonance frequency with its base (point of attachment) fixed. The mechanical TVA suppresses the vibration at its point of attachment to the host structure through the application of an interface force [63].

The TVA can be used in two distinct ways, resulting in different optimal tuning criteria and design requirements:

(a) It can be tuned to suppress the modal contribution from a specific troublesome natural frequency Ω_s of the host structure over a wide band of excitation frequencies.
(b) It can be tuned to suppress the vibration at a specific troublesome excitation frequency, in which case it acts as a notch filter.

When used in application (b), a mechanical TVA is referred to as a tuned vibration neutraliser (or undamped TVA). In this case, the TVA is tuned to the excitation frequency, i.e. the condition $\omega_a = \omega$ defines optimal tuning. In this condition, the neutraliser plants an anti-resonance at its point of attachment to the host structure, and the host vibration is suppressed over a very narrow bandwidth centred at ω_a. In the absence of damping in the neutraliser, there is total attenuation of the vibration. The attenuation degrades with increasing absorber damping. In

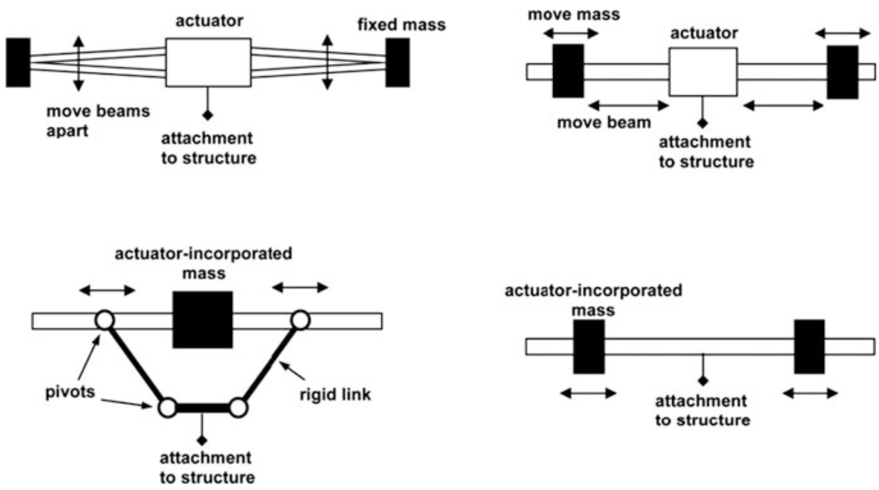


Fig. 2.14 Examples of adaptable beam type TVAs [64]

practice, a neutraliser can be conveniently implemented in the form of a simple beam-like structure [64], as shown in Fig. 2.14 Such beam-like configurations allow retuning of the device (adaptation) under variable conditions through the adjustment of the effective beam cross section or beam span. The method for the derivation of the equivalent two-degree-of-freedom model (minus the damping) for the beam structures in Fig. 2.14 was presented in [64].

When used in the application (a), a mechanical TVA is referred to as a tuned mass damper (TMD). In this case, ω_a is tuned to a frequency that is slightly lower than that of the targeted vibration mode Ω_s. A TMD needs the exact amount of optimal amount of damping in order to suppress the contribution of the targeted mode to the vibration frequency response at the point of attachment over a wide band of excitation frequencies. The practical implementation of the exact amount of damping in conventionally damped TMDs poses a design challenge. Furthermore, once implemented, such damping may be hard to adjust in response to varying system parameters. Moreover, the prerequisite for damping means that simple beam-like designs (Fig. 2.14) are complex to realise for TMDs.

As mentioned by von Flotow [42], a TVA can also be realised with other physics. The most relevant in the present context is the ‘electrical’ TVA [41], which is normally used for application (a) discussed above (i.e. analogous to the TMD). In such a device, the auxiliary system is a piezoelectric shunt circuit. A piezoelectric patch is attached directly to the host structure (typically a cantilever beam) and connected across an external inductor-resistor circuit, as shown in Fig. 2.15 [41].

The piezoelectric patch is used to transfer the vibration energy of the host structure into electrical energy and added a capacitor effect into the circuit, turning it into an R-L-C circuit. The electrical energy is then dissipated most efficiently as Joule heating through the resistor when the electrical resonance produced by the

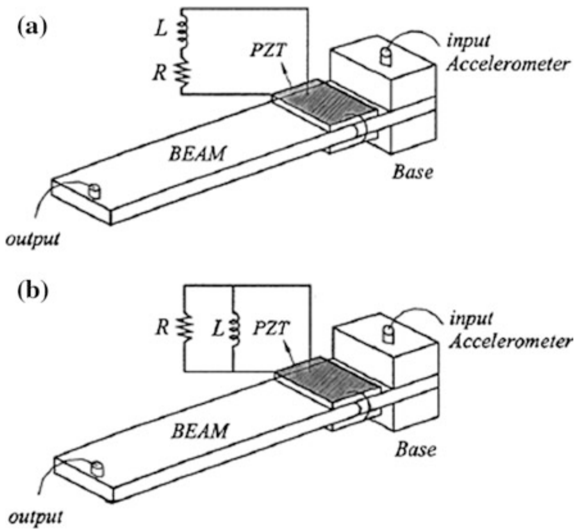


Fig. 2.15 **a** An 'electrical' TVA attached to host structure (cantilever beam) with series R-L and **b** with parallel R-L [41]

L-C components is close to the frequency of the targeted mode. An optimal resistance value provides that the contribution of the targeted mode to the vibration frequency response at a chosen location is suppressed over a wide band of excitation frequencies.

The work in [65] investigated the use of such an electrical TVA (i.e. shunted piezo patch bonded to a host structure whose vibration is to be controlled) for various types of electrical circuits: (1) resistive shunt which dissipates energy and works as a damper, (2) an inductive shunt which results in resonant L-C circuit, (3) a capacitive shunt that changes the stiffness of the piezoelectric element and (4) a switched shunt that controls the energy transfer [65]. In [44, 65–67], the electrical damping effects added by the piezo patch to the system to which it is bonded are analysed. To estimate the electrical damping, the resonance frequency and the damping of the fundamental vibration mode were measured at open circuit, short circuit and when the energy harvesting system was in operation [44]. For the case studied, the mechanical coupling coefficient of the system and the mechanical loss factor were calculated as 0.264 and 2.3%, respectively [44]. In [66], a model was developed and the amount of electrical energy generated by vibrating piezoelectric patches was predicted. The work in [66] presented a relatively simple approach of visualising the electrical damping effects through the comparison of the impulse response of a cantilever beam for very low load, an optimum load (yielding maximum power), and a very high load. The comparison of the three graphs indicated a considerable rise in damping of the system which can be noted from the settling time shown in Fig. 2.16. Similar illustrations and studies investigating the electrical shunt damping effects on the structural response are discussed in references [68–72].

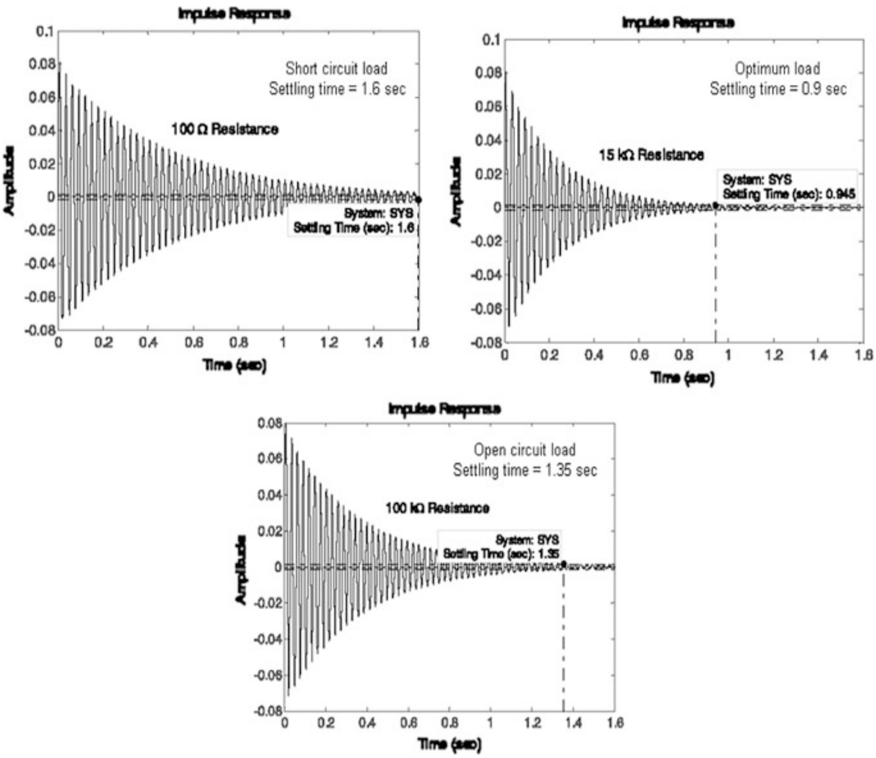


Fig. 2.16 Comparison of damping at different loads [66]

Relative to conventional viscoelastic material damping treatments that can be applied to beams, the electrical TVA has the following advantages: (a) robustness and compactness; (b) relatively less temperature dependence; and (c) ease of controlling the damping level for the desired vibration suppression [41, 68, 72, 73]. These advantages also apply when comparing the electrical TVA to the conventional TMD, which is also limited by space constraints due to its larger size. However, the analysis required to predict the optimal parameters of the electrical TVA are not tractable to complex generic host structures. Such analysis involves the setting up of the coupled electromechanical equations of the host structure with the shunted piezo patch attached and their transformation to modal space, in much the same way as the AMAM analysis (Chap. 3 of this book). The transfer function governing the modal vibration at a specified location could then be extracted and optimised [41]. For this reason, the electrical TVA has only been analysed and presented for the simplest of host structures, namely the cantilever beams in Fig. 2.15 [41]. In contrast, the conventional theory of the classic TMD is readily applicable to any arbitrary host structure since the only host structure data needs are the frequency Ω_s and mass $M_A^{(s)}$ of the targeted mode. The relative shortcomings of the mechanical TVA and electrical TVA are overcome through contribution no. 7 listed in Sect. 1.3 and presented in Chaps. 5 and 6, wherein a dual EH/TVA beam or 'electromechanical' TVA is introduced.

2.3.4 Application of PVEH in Nanogenerators

The PVEH systems can also have enormous applications in nanotechnology devices [74]. It is found from current advancements in the fabrication of PZT nanofibers that such nanofibers may have an even superior piezoelectric voltage constant, higher bending flexibility and larger mechanical strength than bulk PZT which would support its use in nanogenerators [74]. Owing to these properties of PZT nanofibers, it would likely to generate higher voltages and power outputs than other types of piezoelectric geometric configurations for a given volume and same input vibration energy source. Therefore, the properties of PZT nanofibers have introduced considerable interest in taking benefit of these nanofibers in the development of nanogenerators. It is worth mentioning that, although, the concept of harvesting vibration energy by application of nanofiber piezo materials have shown noteworthy improvements, still there is enormous prospects exist for further research to facilitate the development of nanowire-based generators for future miniature applications [74].

2.4 Chapter Summary

This chapter has presented a broad review on PVEH as well as its application to vibration control. It started by giving a brief background of energy harvesting and its necessity for potential applications. This was then followed by a critical review of vibration-based energy harvesting with focus on PVEH modelling techniques. Previous research relating to the improvements in the piezoelectric energy harvester configurations was then reviewed in detail. Finally, research into the electromechanical coupling effects of PVEH on the dynamics of the system was discussed. The shortcomings and gaps in the research were highlighted at different stages in this review in order to explain the drive for the various novel contributions listed in Sect. 1.3.

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Chapter 3
Distributed Parameter Modelling and Experimental Validation

3.1 Background

Typically, a piezoelectric vibration energy harvester (PVEH) is a cantilevered beam having one or two piezoelectric layers (unimorph or bimorph, respectively). The PVEH beam is attached to a vibrating host structure at its base. The terminals of the piezoelectric beam are connected to a sophisticated electrical circuit, consisting of ac-to-dc current conversion elements to facilitate charging of a battery or storage capacitor. However, majority of researchers used a simple resistive load connected across the harvester to derive simplified mathematical models to predict the electrical output for a given base motion input [1–4]. The mathematical modelling techniques employed in energy harvesting literature include oversimplified single degree of freedom (SDOF) models [3, 5], the Rayleigh–Ritz discrete parameter approach [6] and the distributed parameter modelling method [1, 2]. In a discrete parameter model, the motion of a structure is represented by a finite number of coordinates. The SDOF model is a special case of a discrete parameter model in which the motion of the whole structure is represented by a single coordinate. In a distributed parameter model, the inertia, as well as the elasticity of the structure, is considered as being continuously distributed. Although the SDOF approach provides initial insight into an energy harvesting system, it is oversimplified and ignores several key features such as the dynamic mode shapes and the accuracy of the strain distribution along the bender [4]. Therefore, a more detailed distributed parameter modelling approach is required to correctly model the behaviour of the structure at any point along its length. Erturk and Inman [1] presented a closed-form continuous (distributed parameter) modelling method for a base-excited cantilever unimorph beam without a tip mass. In a later paper [2], the same authors presented experimental results of a base-excited distributed parameter bimorph having a lumped tip mass.

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3.2 Modelling of a Bimorph Using Distributed Parameters

In this section, a closed-form mathematical model using distributed or continuous parameters is presented. In order to validate the method, the vibrating piezoelectric beam is only having its own distributed inertia without any lumped or tip mass. A bimorph beam as shown in Fig. 3.1 consists of two piezoelectric layers bonded on top and bottom of a thin middle metallic plate. The main purpose of this middle metallic layer also known as 'shim' is to provide strength and stiffness to the piezoelectric beam. Therefore, a bimorph can be electrically connected either in series or in parallel depending on the poling directions of the bottom and top piezoelectric layers [4]. When the poling direction in the top and bottom piezoelectric layers are in opposite direction, the EH is said to be connected in series, and as Fig. 3.1 shows, the poling directions of top and bottom layers are opposite as indicated by their respective arrows. On the other hand, if the poling of both layers of a bimorph is in same direction, then the harvester is said to be connected in parallel. In series connectivity, higher voltage and low current is achieved whereas higher current and low voltage is attained in parallel connection configuration [4]. However, the overall power output of the harvester remains the same whether the harvester is connected in series or in parallel since net power is the product of output current and voltage. In this work, a series-connected bimorph is used as illustrated in Fig. 3.1. In following sections, a step-by-step derivation is presented to calculate the relations for electromechanically coupled voltage, current, power and tip displacement FRFs for the case of harmonic base excitation. The derivation follows that in [2]; however, no lumped mass is attached at the tip in the present study. The incremental mathematical derivation is presented here for completeness and also to facilitate readers to comprehend the method. Another distinct aspect of this work is that the applicability of the FRFs to random base excitation is investigated theoretically and experimentally which was not considered in [2].

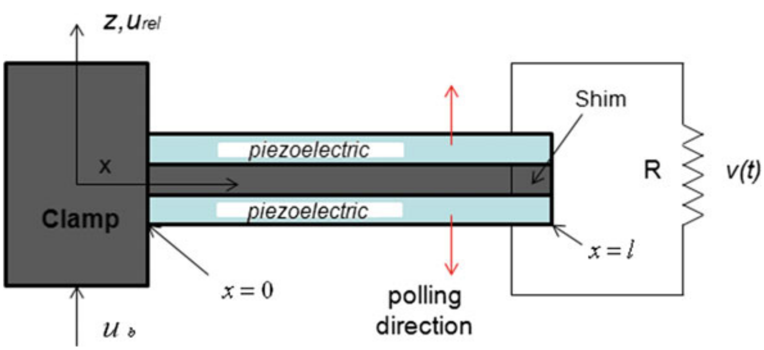


Fig. 3.1 A bimorph, internally connected in series, excited by base motion

3.2.1 Mechanical Model with Electrical Coupling Effects

In this section, dynamic equations of motion of a piezoelectric bimorph are derived having backward electrical coupling effects. Let the bending stiffness of the composite beam is B, distributed mass per unit length is m, the externally connected electrical load is R, and the voltage appearing across the resistor is $v(t)$ [4]. Ignoring the rotation of the clamp itself, the absolute transverse displacement of any point, located at a distance 'x' from the clamp, along the length of the beam can be represented as [7],

$$u\,(x,t) = u_b(x,t) + u_{rel}(x,t) \tag{3.1}$$

where $u_b(x,t)$ is representing the absolute transverse displacement of the clamp, and $u_{rel}(x,t)$ is describing the transverse displacement at distance x relative to the moving base (i.e. the flexural displacement) [4].

The Euler–Bernoulli beam equation representing the motion of the bimorph can be written as [1]

$$B\frac{\partial^4 u}{\partial x^4} + A\frac{\partial^5 u}{\partial x^4 \partial t} + c_a\frac{\partial u}{\partial t} + m\frac{\partial^2 u}{\partial t^2} = 0 \tag{3.2}$$

where

$$B = b\left[\frac{Y_{sh}h_{sh}^3}{12} + Y_p\left\{\frac{h_p h_{sh}^2}{2} + h_{sh}h_p^2 + \frac{2}{3}h_p^3\right\}\right] \tag{3.3}$$

and

$$A = c_p I_p + c_s I_{sh} \tag{3.4}$$

Combining Eq. (3.1) and (3.2) together will yield:

$$\begin{aligned} &B\frac{\partial^4 u_{rel}(x,t)}{\partial x^4} + A\frac{\partial^5 u_{rel}(x,t)}{\partial x^4 \partial t} + c_a\frac{\partial u_{rel}(x,t)}{\partial t} + m\frac{\partial^2 u_{rel}(x,t)}{\partial t^2} \\ &= -c_a\frac{\partial_b(x,t)}{\partial t} - m\frac{\partial^2 u_b(x,t)}{\partial t^2} \end{aligned} \tag{3.5}$$

Equation (3.5) can also be rewritten as

$$\begin{aligned} &\frac{\partial^2 \widehat{M}}{\partial x^2} + A\frac{\partial^5 u_{rel}(x,t)}{\partial x^4 \partial t} + c_a\frac{\partial u_{rel}(x,t)}{\partial t} + m\frac{\partial^2 u_{rel}(x,t)}{\partial t^2} \\ &= -c_a\frac{\partial u_b(x,t)}{\partial t} - m\frac{\partial^2 u_b(x,t)}{\partial t^2} \end{aligned} \tag{3.6}$$

where internal elastic moment is denoted by $\widehat{M}$, the internal structural damping of the composite beam is represented by A, the bending stiffness of the beam is denoted by B, mass per unit length is m and c_a represents viscous damping coefficient per unit length at ambient conditions. It should be noted that both viscoelastic and ambient damping terms satisfy proportional damping criteria so it is mathematically suitable to utilise them in the modal analysis solution [4].

The term of relative transverse displacement $u_{rel}(x,t)$, with respect to the vibrating base, can be represented in modal coordinates by using the modal expansion theorem [8] as shown below,

$$u_{rel}(x,t) = \sum_{r=1}^{\infty} \phi_r(x)\eta_r(t) \tag{3.7}$$

where $\phi_r(x)$ are the mass normalised eigenfunctions of mode r of a clamp-free beam, and $\eta_r(t)$ are the corresponding modal coordinates of the mode r. It is important to note that these eigenfunctions $\phi_r(x)$ pertain to electrically uncoupled and undamped conditions, and can be written as [9]:

$$\phi_r(x) = \sqrt{\frac{1}{ml}}\left[cosh\frac{\lambda_r}{l}x - cos\frac{\lambda_r}{l}x - \sigma_r\left(sinh\frac{\lambda_r}{l}x - sin\frac{\lambda_r}{l}x\right)\right] \tag{3.8}$$

σ_r can be written as (at $x = l$)

$$\sigma_r = \frac{(sinh\lambda_r - \sin\lambda_r)}{(\cosh\lambda_r + \cos\lambda_r)} \tag{3.9}$$

where l represents the overhung length of the piezoelectric beam, and λ_r's are dimensionless frequency parameters that may be calculated by the following transcendental characteristic relation of a cantilever beam.

$$1 + \cos\lambda_r \cosh\lambda_r = 0 \tag{3.10}$$

The undamped resonance frequency of the rth mode (r = 1, 2, 3, …) of a continuous clamp-free beam can be determined by the following relation.

$$\omega_r = \left(\frac{\lambda_r}{l}\right)^2\sqrt{\frac{B}{m}} \tag{3.11}$$

In Eq. (3.6), the internal moment term $\widehat{M}$ can be calculated by using the well-known constitutive equations of piezoelectric materials [10]. Figure 3.2 shows cross section of a bimorph beam having a metallic shim sandwiched between top and bottom piezoelectric layers [4].

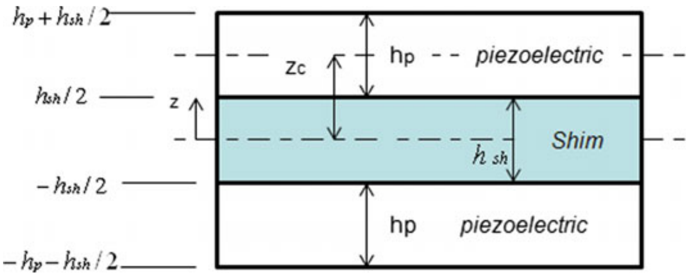


Fig. 3.2 Cross section of a bimorph beam; from neutral axis, upwards distances is +ve

Bending of the piezolayers produces a stress in the '1' direction which, in turn, creates a voltage in the '3' direction, and so the constitutive equations of the piezoelectric layers defining the electromechanical relation can be written as [4]

$$D_3 = d_{31}\sigma_p + \varepsilon_{33}^T E_3 \tag{3.12}$$

$$\delta_p = \frac{\sigma_p}{Y_p} + d_{31}E_{31} \tag{3.13}$$

where Y_p is the Young's Modulus of piezoelectric material, D_3 is the electrical displacement (charge density) in electric poling direction, ε_{33}^T is the permittivity at constant stress, σ_p and δ_p are the stress and strain in the piezoelectric layer, E_3 is the electric field, and d_{31} is the piezoelectric strain coefficient [4]. The stress in the middle metallic shim can be calculated using Hooke's law as $\sigma_{sh} = Y_{sh}\delta_{sh}$ where σ_{sh}, δ_{sh} represent axial stress and strain, respectively, and Y_{sh} is the Young's modulus of elasticity of the metallic shim.

At a given section x, the bending of the beam produces opposite stresses in the top and bottom piezoelectric layers, the top layer is in tension, and the bottom layer is in compression and vice versa. It is for this basis that, in the present case of series-connected piezolayers, the piezolayers have to be oppositely poled [4]. The opposite stresses in the top and bottom layers produce a bending moment in the composite piezobeam which can be written as

$$\widehat{M}(x,t) = \int_{\frac{h_{sh}}{2}}^{\frac{h_{sh}}{2}+h_p} Y_p(\delta + d_{31}E_3)bzdz + \int_{\frac{-h_{sh}}{2}}^{\frac{h_{sh}}{2}} Y_s\delta bzdz + \int_{\frac{-h_{sh}}{2}-h_p}^{\frac{-h_{sh}}{2}} Y_p(\delta - d_{31}E_3)bzdz \tag{3.14}$$

where h_{sh} and h_p are thicknesses of middle shim and the piezoelectric layers, respectively. δ represents the generic strain in the bimorph beam at a distance z from the neutral axis. Substituting for the strain $\delta = -z\partial^2 u_{rel}/\partial x^2$ and the electric field $E_3 = -v(t)/(2h_p)$ and performing the integrations in Eq. (3.14) will yield:

$$\widehat{M}(x,t) = B\frac{\partial^2 u_{rel}(x,t)}{\partial^2 x} + \vartheta\, v(t) \tag{3.15}$$

In Eq. (3.15), the electrical coupling effects present in the internal bending moment (mechanical domain) can be represented by a constant term, ϑ, and can be written as:

$$\vartheta = -\frac{d_{31}Y_p b\left(h_p + h_{sh}\right)}{2} \tag{3.16}$$

In Eq. (3.16), $\vartheta v(t)$, the piezoelectric coupling term in mechanical domain is a function of time only, so in order to substitute it into equation of motion (Eq. (3.6)), it should be multiplied by $[H(x) - H(x-l)]$, where $H(x)$ is a Heaviside function [2]. Subsequently, the resulting equation of motion can be transformed into modal coordinates from physical parameters by substituting for u_{rel} from Eq. (3.7) and using the conditions of orthogonality of the eigenfunctions [1] will yield:

$$\frac{d^2\eta_r(t)}{dt^2} + 2\zeta_r\omega_r\frac{d\eta_r(t)}{dt} + \omega_r^2\eta_r(t) + \chi_r v(t) = -m\frac{\partial^2 u_b(x,t)}{\partial t^2}\int\limits_{x=0}^{l}\phi_r(x)dx \tag{3.17}$$

where χ_r consists of mechanical and electrical terms and can be represented as:

$$\chi_r = \vartheta\frac{d\phi_r(x)}{dx}\bigg|_{x=l} \tag{3.18}$$

It can be noted in Eq. (3.18) that the excitation provided due to ambient damping term, as shown by the first term on the right-hand side of Eq. (3.5) or (3.6), has been ignored, as in [1, 3, 4].

3.2.2 *Electrical Circuitry Equation with Backward Mechanical Coupling*

In this section, the electrical circuit equation having mechanical backward-coupling term will be obtained using the piezoelectric constitutive relations given in Eqs. (3.12) and (3.13) [4]. The electrical charge accumulated at the electrodes can be calculated by integrating Eq. (3.12) over the whole surface area:

$$q = \int\limits_{x=0}^{l}\left(d_{31}Y_p\overline{\delta}_p + \varepsilon_{33}^S E_3\right)bdx \tag{3.19}$$

where $\bar{\delta}_p$ is the bending strain along the middle surface of the piezolayer, and b represents the width of the piezoelectric layer. Applying Ohm's law, the current through a load resistor R can be calculated by taking the time derivative of the charge, Eq. (3.19):

$$i(t) = \frac{v(t)}{R} = \frac{d}{dt}\left[\int_{x=0}^{l} \left(d_{31} Y_p \bar{\delta}_p + \varepsilon_{33}^S E_3\right) b dx\right] \tag{3.20}$$

Substituting for the electric field $E_3 = -v(t)/(2h_p)$ and the strain $\bar{\delta}_p = -z_c \partial^2 u_{rel}/\partial x^2$, where $z_c = (h_p + h_s)/2$ (as in Fig. 3.2) in Eq. (3.20) will yield:

$$\left(\frac{\varepsilon_{33}^S bl}{2h_p}\right)\frac{dv(t)}{dt} + \frac{v(t)}{R} = -Y_p d_{31} b z_c \int_{x=0}^{l} \frac{\partial^3 u_{rel}(x,t)}{\partial x^2 \partial t} dx \tag{3.21}$$

Substituting the value of $u_{rel}(x,t)$ from Eq. (3.7) into Eq. (3.21) then gives

$$\frac{C_p dv(t)}{2\,dt} + \frac{v(t)}{R} = \sum_{r=1}^{\infty} \alpha_r \frac{d\eta_r(t)}{dt} \tag{3.22}$$

where α_r is a constant and can be written as

$$\alpha_r = -Y_p d_{31} b z_c \int_{x=0}^{l} \frac{d^2\phi_r(x)}{dx^2} dx = -Y_p d_{31} b z_c \frac{d\phi_r(x)}{dx}\bigg|_{x=l} \tag{3.23}$$

The capacitance term C_p, as in Fig. (3.22), of the piezoelectric patch can be defined as

$$C_p = \varepsilon_{33}^S bl/h_p \tag{3.24}$$

where permittivity at constant strain is ε_{33}^S and can be written as $\varepsilon_{33}^S = \varepsilon_{33}^T - d_{31}^2 Y_p$. It is important to note that a piezoelectric element can be represented as a current source in parallel with its internal capacitance. The components of the electrical circuit include externally connected load resistor R, the internal capacitance of the piezoelectric layers and the current source $i_p(t)$ [4].

3.2.3 Derivation of FRFs

Assuming base excitation $u_p(t)$ is harmonic and can be represented as:

$$u_b = R_e\left\{\tilde{u}_b e^{j\omega t}\right\} \tag{3.25a}$$

Thus, all other time-varying quantities are also harmonic and can be expressed as:

$$\begin{aligned} &v(t) = R_e\left\{\tilde{v}e^{j\omega t}\right\}, u(x,t) = R_e\left\{\tilde{u}(x)e^{j\omega t}\right\}, u_{rel}(x,t) = R_e\left\{\tilde{u}_{rel}(x)e^{j\omega t}\right\} \\ &\eta_r(t) = R_e\left\{\tilde{\eta}_r e^{j\omega t}\right\} \end{aligned} \tag{3.25b–e}$$

where $\tilde{u}_b$, $\tilde{v}$, $\tilde{u}$, $\tilde{u}_{rel}$ and $\tilde{\eta}_r$ are the complex amplitudes of the respective quantities, and ω is the excitation frequency in rad/s. Substituting for $u_p(t)$, $v(t)$ and $\eta_r(t)$ from Eqs. (3.25a, 3.25b–e) into Eq. (3.17) and rearranging give [4]:

$$\tilde{\eta}_r = \frac{(F_r - \chi_r \tilde{v})}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{3.26}$$

where

$$F_r = m\omega^2\gamma_r^u\tilde{u}_b, \quad \gamma_r^u = \int_{x=0}^{l} \phi_r(x)dx = \frac{2\sigma_r}{\lambda_r}\sqrt{\frac{l}{m}} \tag{3.27a, b}$$

Substituting from Eqs. (3.25a, e) for $v(t)$ and $\eta_r(t)$ into Eq. (3.22) and then substituting for $\tilde{\eta}_r$ from Eq. (3.26) and rearranging yield the relation for voltage FRF $\tilde{v}(\omega)$ as below:

$$\tilde{v}(\omega) = \frac{\tilde{v}}{-\omega^2\tilde{u}_b} = \frac{\sum_{r=1}^{\infty}\left(\frac{-jm\omega\gamma_r^u\alpha_r}{\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega}\right)}{\left(\frac{1}{R}+j\omega\frac{C_p}{2}\right)+\sum_{r=1}^{\infty}\left(\frac{j\omega\alpha_r\chi_r}{\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega}\right)} \tag{3.28}$$

The base acceleration-normalised voltage FRF $\tilde{v}(\omega)$ is seen to be the complex amplitude of the voltage divided by the complex amplitude of the base acceleration $\ddot{\tilde{u}}_b$, and the 'current FRF' is the complex amplitude of the current divided by the complex amplitude of $\ddot{\tilde{u}}_b$ [4]. Hence, the relation for current FRF can simply be obtained by dividing voltage FRF by the resistance R. Similarly, the instantaneous power can be calculated from the relation v^2/R, and the peak power is therefore $\left|\tilde{v}^2\right|/R$ [2]. Hence, the amount of electrical power generated is obtained by taking the square of the modulus of the voltage FRF and dividing it by R, and this quantity is defined as the 'power FRF' [4]. In this book, peak or instantaneous power is used, in line with most of the previous research in PVEH. However, the FRF of the average power can be obtained by dividing the peak power by 2.

Substituting for $\tilde{v}^2$ from Eq. (3.28) into Eq. (3.26) and using Eq. (3.7) will yield the relative tip response FRF as:

$$\beta(\omega) = \frac{\tilde{u}_{rel}(l)}{-\omega^2 \tilde{u}_b} = -\sum_{r=1}^{\infty} \left\{ \gamma_r^u - \chi_r \frac{\sum_{r=1}^{\infty} \left(\frac{j\omega\gamma_r^u \alpha_r}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \right)}{\left(\frac{1}{R} + j\omega\frac{C_p}{2} \right) + \sum_{r=1}^{\infty} \left(\frac{j\omega\alpha_r\chi_r}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \right)} \right\} \times \left(\frac{m\phi_r(l)}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \right) \tag{3.29}$$

As mentioned above and also in Eq. (3.25d), $\tilde{u}_{rel}(l)$ is the complex amplitude of the relative tip displacement. In the testing, the tip vibration is measured with a laser sensor which records absolute tip displacement or velocity. From Eq. (3.1), the absolute and relative tip displacement FRFs are related by the following equation [4]:

$$\beta_{abs}(\omega) = -\frac{1}{\omega^2} + \beta(\omega) \tag{3.30}$$

3.2.4 Reduced Expressions of FRFs for a Single Mode

Primarily in PVEH research, only the first mode of the harvester is considered as mostly it has dominant modal factor. The FRFs obtained in the last section include contributions of any number of modes; however, these FRFs can be reduced to a single mode (first mode or for any particular mode r) [4]. The single-mode FRFs can be obtained by solving Eq. (3.28) and (3.29) for one mode only, namely rth mode, and can be reduced to the following relation:

$$\tilde{v}(\omega)|_r = \frac{-j2\omega m R\gamma_r^u \alpha_r}{(2 + j\omega C_p R)(\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega) + (j2\omega R\alpha_r\chi_r)} \tag{3.31}$$

$$\beta(\omega)|_r = \frac{-(2 + j\omega C_p R) m\gamma_r^u \phi_r(l)}{(2 + j\omega C_p R)(\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega) + (j2\omega R\alpha_r\chi_r)} \tag{3.32}$$

In Eqs. (3.31, 3.32), the subscripts 'r' represents that the FRFs are single-mode approximations centred around the frequency ω_r.

3.2.5 Application to Non-Harmonic Base Excitation

The FRF expressions derived above are applicable for input harmonic excitations which relate complex output amplitude of the beam (tip displacement or voltage) to the complex amplitude of the input base excitation. However, considering the system is linear and following standard signal processing theory [11], the same FRF expressions as presented in Eqs. (3.28, 3.29, 3.31, 3.32) should also be valid for non-harmonic excitation [4]. For a deterministic signal u_b, the FRF expressions would simply be defined by the ratio of the Fourier Transforms of the output and input signals [11]. In the case of a non-deterministic or random signal u_b, the FRFs would define the ratio of the cross-spectral density to the input power spectral density [11]. For example, according to this theory, the voltage FRF expression on the right-hand side of Eq. (3.28) or (3.31) would be equal to [4]:

$$G_{\ddot{u}_b v}(\omega)/G_{\ddot{u}_b \ddot{u}_b}(\omega) \tag{3.33}$$

where $G_{\ddot{u}_b v}(\omega)$ represents the cross-spectral density function relating to signals $\ddot{u}_b$ and v, and $G_{\ddot{u}_b \ddot{u}_b}(\omega)$ is the power spectral density function of $\ddot{u}_b$.

3.3 Experimental Validation of the Model

The distributed parameter model presented above was experimentally validated, and the experimental results of the voltage, current, power and tip displacement FRFs were compared to the simulation results of the model. Figure 3.3b shows the experimental set-up used in this study. A bimorph connected to a resistor load was clamped from one end and was mounted to an electrodynamic shaker as shown in Fig. 3.3a. Rotational effects at the clamp, causing unbalance, were minimised by attaching an identical dummy bimorph at the other side of the clamp, as shown in Fig. 3.3a [4]. If a single cantilever is used, as in previous energy harvesting work [2], then the dynamic bending moment at the base of the cantilever inclined to rotate the clamp [4]. This tendency has to be resisted by an identical and opposite external moment applied by the shaker on the clamp [4]. It can be noted that the moment on the shaker was reduced due to the use of two symmetric cantilevers since the dynamic bending moments cancel each other out at the roots of the two cantilevers. It is handy practice to implement a base-excited, with no-base-rotation cantilever configuration as a symmetric double cantilever, as can be seen from references [4, 12, 13].

The bimorphs used in the study were manufactured by Piezo Systems Inc, and each of those was made up of two PZT-5H4E layers bonded on the top and bottom surfaces of a thin aluminium shim. As mentioned in the previous section, each bimorph was series-connected (Fig. 3.3a), and its material, electromechanical and geometric properties, as provided by the manufacturer, are given in Table 3.1 [4].

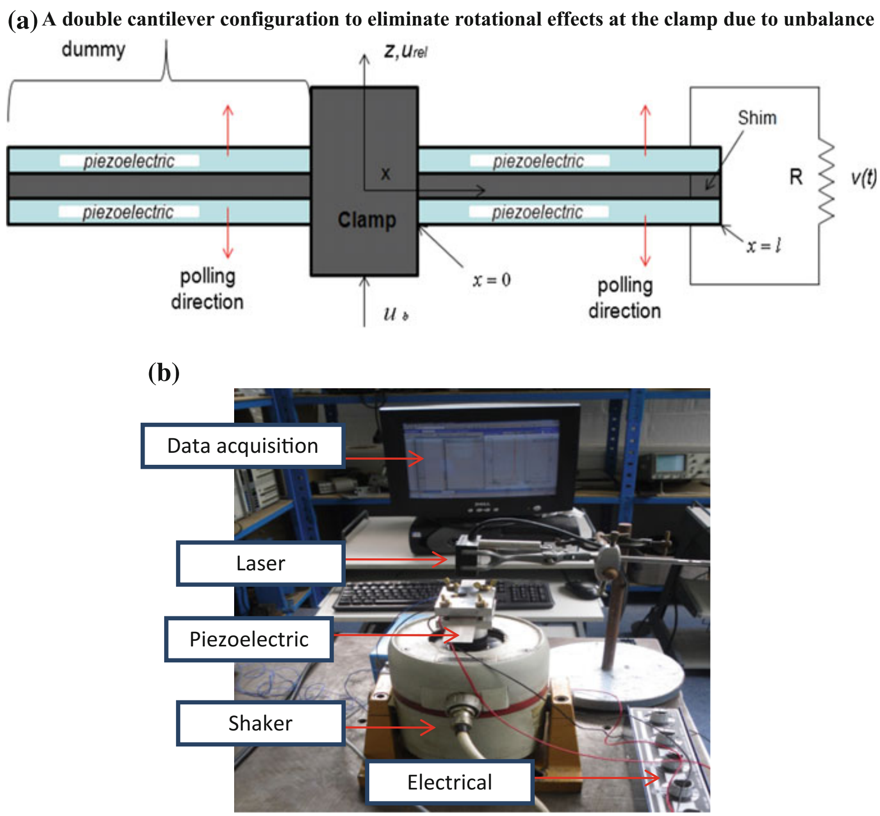


Fig. 3.3 **a** Schematic of Double cantilever bimorph beams, **b** Experimental set-up

Table 3.1 Characteristics of the piezoelectric energy harvesting bimorph beam [4]

Property	Units	Value
Length of the beam & substrate	mm	60
Width of the beam & substrate, b	mm	25
Thickness of each piezoelectric, h_p (upper & lower layers)	mm	0.267
Thickness of the substrate, h_{sh}	mm	0.3
Young's modulus of the piezoelectric, Y_p	GPa	62
Young's modulus of the substrate, Y_{sh}	GPa	72
Density of the aluminium shim substrate	Kg/m^3	2700
Density of the piezoelectric	Kg/m^3	7800
Relative dielectric constant (at constant stress)		3800
Piezoelectric constant, d_{31}	pm/volt	−320

A laser sensor (model MEL M5L/4-10B24NK, with resolution 0.0005 mm and sensitivity of 0.54 V/mm) was used to measure the absolute tip displacement, and a PCB 352C22 accelerometer (resolution 0.002 g rms and sensitivity 9.08 mV/g) was used to measure the input base (at clamp) acceleration.

The external electrical load used was purely resistive and was controlled manually through a variable resistor box. A pc-controlled data acquisition system, LMS Scadas 5 with LMS Test.Lab Rev 7A software, was used to generate an excitation signal to the shaker and to manage the signals and computing the FRFs [4].

A band-limited white noise random signal was used to excite the test device with a frequency spectrum of 0–320 Hz (i.e. including the electrically uncoupled first undamped natural frequency ω_1). The FRFs were computed by using the data acquisition software by calculating the cross-spectral densities and power spectral densities of the acquired signals, as discussed previously in Sect. 3.2.5. Ten different electrical loads of values 100 Ω, 1 kΩ, 5 kΩ, 10 kΩ, 15 kΩ, 25 kΩ, 50 kΩ, 100 kΩ, 500 kΩ and 1000 kΩ were used [4]. The natural frequency of the first mode of the bimorph was measured at 121.7 Hz for a small value of $R = 100\ \Omega$, and there was no noticeable shift in this frequency when R was further raised to the next higher load of 1 kΩ [4]. Hence, the electrically uncoupled resonance frequency of the first mode (ω_1) at short-circuit conditions was measured as 121.7 Hz. It is important to note that this value closely matches to the theoretical value, calculated by Eq. (3.11), for ω_1 (121.1 Hz). It can be observed in Fig. 3.4a that the maximum value of the input random signal $\ddot{u}_b$ did not exceed 1 g, for all values of the load resistors. Furthermore, the operation of the EH system in the linear range was ascertained by monitoring the coherence functions relating v with $\ddot{u}_b$ and $u(l,t)$ with $\ddot{u}_b$ [4]. For a linear system and in the absence of measurement noise, the coherence functions should be close to unity for the frequency range of interest [11]. It can be seen from Fig. 3.4c that the coherence functions are close to unity, around the first resonance frequency, for the corresponding experimental voltage FRF and tip velocity FRF as shown in Fig. 3.4b at a load of 1 kΩ.

3.3.1 Mechanical Damping Estimation

For a small value of R, the single-mode FRF Eqs. (3.31, 3.32) can be approximately expressed as:

$$\tilde{v}(\omega)|_r \approx \frac{-j\omega m R \gamma_r^u \alpha_r}{\omega_r^2 - \omega^2 + j2\left\{\zeta_r + \frac{R\alpha_r\chi_r}{2\omega_r}\right\}\omega_r\omega} \tag{3.34}$$

$$j\omega\beta(\omega)|_r \approx \frac{-j\omega m \gamma_r^u \phi_r(l)}{\omega_r^2 - \omega^2 + \mathrm{j}2\left\{\zeta_r + \frac{R\alpha_r\chi_r}{2\omega_r}\right\}\omega_r\omega} \tag{3.35}$$

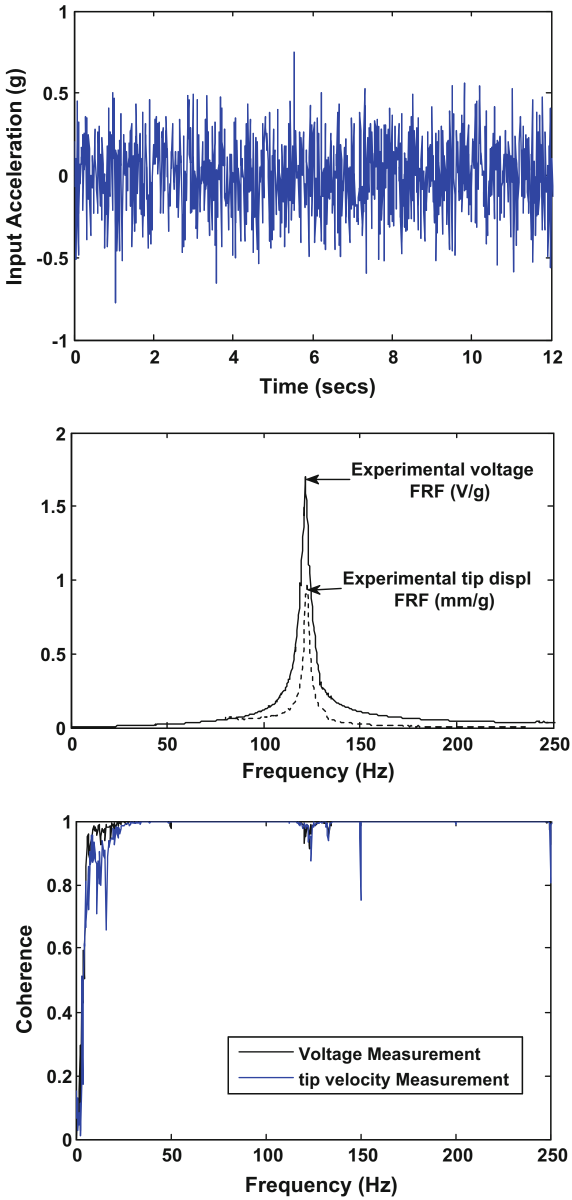


Fig. 3.4 (top) Input acceleration at base $\ddot{u}_b$; (middle) Experimental tip velocity and voltage FRFs at 1 kΩ and (bottom) their corresponding coherence functions [4]

It is noted that the approximate expression for $\beta(\omega)|_r$ has been multiplied by $j\omega$ to get an expression that is of similar form to $\tilde{v}(\omega)|_r$, $j\omega\beta(\omega)|_r$ is the single-mode expression for the relative tip velocity FRF per unit base acceleration amplitude [4]. From modal testing theory [14], the expressions for $\tilde{v}(\omega)|_r$, $j\omega\beta(\omega)|_r$ in Eqs. (3.34, 3.35) produce a circle passing through the origin with a diameter along the

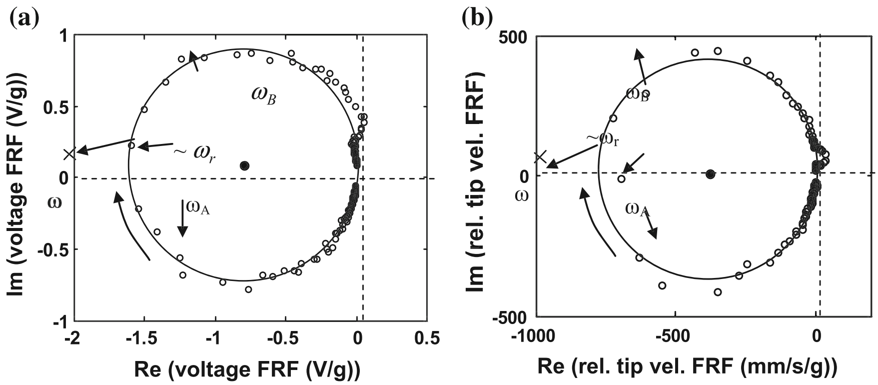


Fig. 3.5 Nyquist plots for 1 kΩ of **a** voltage FRF and **b** relative tip velocity FRF (using circle-fit technique on experimental data) [4]

horizontal axis when plotted real part of the FRF versus the imaginary part. This type of graph is known as Nyquist plot. In Fig. 3.5, this observation is approximately validated by experimental Nyquist plots of the voltage FRF and the relative tip velocity FRF for 1 kΩ over the frequency range 80–160 Hz, where the circles shown were fitted through the experimental data points using a least-squares fit [4]. It is noted that the Laser sensor measured absolute tip displacement, and the relative tip velocity FRF $j\omega\beta(\omega)$ was obtained from the measured *absolute* tip displacement FRF $\beta(\omega)$ using Eq. (3.30).

From Eqs. (3.34, 3.35), the *equivalent electromechanical* damping ratio $\hat{\zeta}_r$ of the mode r is given by

$$\hat{\zeta}_r = \zeta_r + \frac{R\alpha_r\chi_r}{2\omega_r} \tag{3.36}$$

From Eq. (3.34) or (3.35) and following the method similar to that in [14], the damping ratio $\hat{\zeta}_r$ can be computed by the half power point formula written below:

$$\hat{\zeta}_r = \frac{\omega_B - \omega_A}{2\omega_r} \tag{3.37}$$

According to Eqs. (3.34, 3.35), the condition $\omega = \omega_r$ reveals the intersection of the frequency point with the real axis. Modal testing theory [14] explains specifically that in the FRFs of the form of Eqs. (3.34, 3.35), the condition $\omega = \omega_r$ approximately relates to the location of maximal spacing between the two consecutive data points of the experimental Nyquist plot [4]. These conditions are effectively validated in Fig. 3.5. Furthermore, the locations of the half power point frequencies $\omega_{A,B}$ are located 90° on either side of the position of ω_r. The frequencies $\omega_{A,B}$ as required in Eq. (3.37) can be spotted, in the Nyquist plot of Fig. 3.5, by using the circle fit. As an alternative to the circle-fit (Nyquist plot)

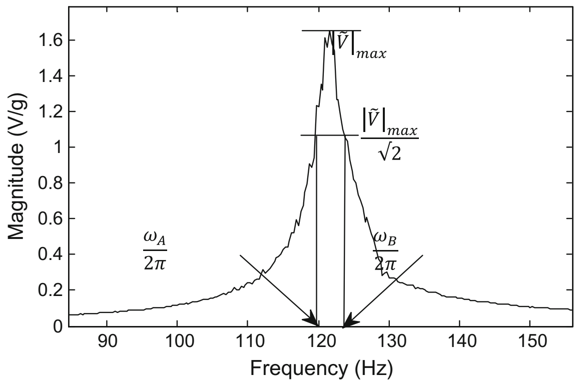


Fig. 3.6 Calculating damping by 'Peak-amplitude method' using voltage FRF at 1 kΩ

method, the frequencies $\omega_{A,B}$ can also be located using the magnitude versus frequency plots of the flexural tip velocity FRF or voltage FRF using 'peak-amplitude method' [14] technique, as shown in Fig. 3.6 [4].

Table 3.2a shows the estimates for $\hat{\zeta}_r$ obtained by applying both methods (peak amplitude, circle fit), as explained above for each type of FRF (flexural tip velocity FRF and voltage FRF). It can be noted that all of the damping estimates were found to be closely matched with each other. However, in general, one expects the circle-fit method to be more trustworthy since the peak-amplitude method is greatly sensitive to the frequency resolution of the data points [4]. It was also noted that, for the values of *R* significantly smaller than 1 kΩ, considerable noise contamination in the voltage FRF was observed. This is according to the fact that the numerator of Eq. (3.34) is directly proportional to load *R,* and, conversely, the tip velocity FRF yields better results under short-circuit conditions, as can be noted in Eq. (3.35) [4]. However, this FRF was potentially vulnerable to errors induced by the vibration of the laser head. Having determined the equivalent electromechanical modal damping ratio $\hat{\zeta}_r$, the mechanical modal damping ratio ζ_r could be determined from Eq. (3.36), since all other parameters in the second term of this equation were quantifiable [4]. Table 3.2b shows the estimates for ζ_r. The average value of the damping ratios, shown in Table 3.2b, was used for computing the theoretical FRFs.

Table 3.2 Modal damping ratio estimates [4]

(a) Equivalent electromechanical modal damping ratio $\hat{\zeta}_r$			
FRF	Load *R* (kΩ)	Circle fit $\hat{\zeta}_r$ (%)	Peak amplitude $\hat{\zeta}_r$ (%)
Relative tip velocity FRF	1	1.3	1.33
Voltage FRF	1	1.45	1.48
(b) Mechanical modal damping ratio ζ_r			
FRF	Load *R* (kΩ)	Circle fit ζ_r (%)	Peak amplitude ζ_r (%)
Relative tip velocity FRF	1	1.0	1.03
Voltage FRF	1	1.15	1.18

It is noted that in [2], ω_r was determined as the natural frequency at short-circuit conditions, and then, ζ_r was found by trial and error by fitting the theoretical magnitude–frequency plots of Eq. (3.31) or (3.32) to the measured data. It is evident that the Nyquist plot method, presented here, not only avoids trial and error, but also provides a deeper understanding and is itself a means for validating the theory [4].

3.3.2 Comparison of Experimental and Theoretical FRFs

In this section, voltage, current, power and tip velocity FRFs are presented, and critical comparison is made between the experimental and the theoretical results. The FRFs are obtained using different resistor loads ranging between 10^2 Ω ('short' circuit) and 10^6 Ω ('open' circuit). It is important to note that the resonance frequency of the energy harvester depends on the value of the resistor load and can be located at a point where FRF yields maximum magnitude. For a given resistance, the peak magnitudes of all electrical and mechanical FRFs occurred at virtually the same frequency [4]. This phenomenon coincides with the simulation results of the mathematical model presented in this chapter. The resonance frequency of 121.7 Hz was noted for a very low resistance of 1 kΩ. This may be due to the fact that, for small nonzero resistance, the electrical effect is equivalent to a viscous damper proportional to R, as evident from Eq. (3.35) [4]. On the other hand, the resonance frequency of the harvester increases for higher resistance values (up to 500 kΩ). The simulation results presented in the following sections are obtained for single-mode case, as in Eqs. (3.31, 3.32) with $r = 1$, ignoring the effects of higher modes. For symmetry, the simulation results obtained from Eqs. (3.31, 3.32) were multiplied by g (9.81 m/sec^2) since the experimental base acceleration readings from the measurement channel were in g's of acceleration.

3.3.2.1 Magnitude Plots of Voltage FRFs

The comparison of experimental and the theoretical voltage FRFs for different values of load resistors are presented in Fig. 3.7. It can be noted from magnitude–frequency plots of the measured and predicted voltage FRFs that the resonance frequency increases from short-circuit to open-circuit conditions. For this case, the measured resonance frequency at 500 kΩ was 128.6, and it was almost unaffected by increasing the resistance to 1 M Ω [4]. Hence, this frequency can be termed as the 'open-circuit' frequency. The theoretical open-circuit frequency of 128.4 Hz matches remarkably with the corresponding measured value of 128.6 Hz.

The theoretical variation of the resonance frequency with resistance can be acquired by taking the modulus of $\tilde{v}(\omega)|_r$ in Eq. (3.31), differentiating this with

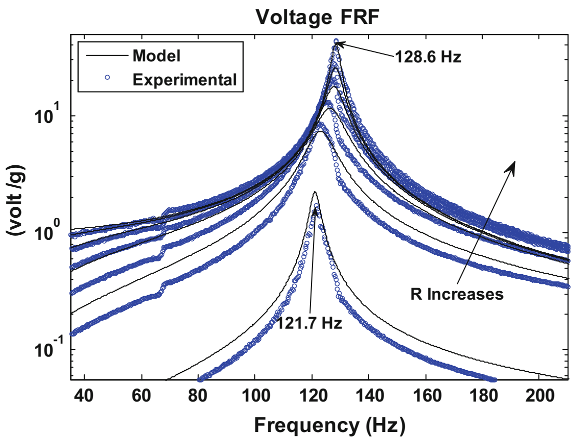


Fig. 3.7 FRFs of generated voltage at 6 different loads (1 kΩ, 10 kΩ, 25 kΩ, 50 kΩ, 100 kΩ, 500 kΩ) [4]

respect to ω, and equating the result to 0 [4]. The resulting equation will be the cubic in ω^2 as below:

$$A_a(\omega^2)^3 + A_b(\omega^2)^2 + A_c\omega^2 + A_d = 0 \tag{3.38}$$

where

$$A_a = 2K_a^2, A_b = 4 + K_a^2K_b^2 - 2\omega_r^2K_a^2 - 2K_aK_c, A_c = 0, A_d = -4\omega_r^4 \tag{3.39a–c}$$

$$K_a = RC_p, K_b = 2\zeta_r\omega_r, K_c = 2R\alpha_r\chi_r \tag{3.40a–c}$$

The resonance frequency at given R is then given by the positive real value of ω that satisfies the above equation, and solving for r = 1 (fundamental mode) over a range of R gives the curve in Fig. 3.8 [4]. At low resistances, it can be noted that the slope of the curve is horizontal. This is in line with the fact that the rate of change of resonance frequency with R is negligible for small R due to the fact that, at such low

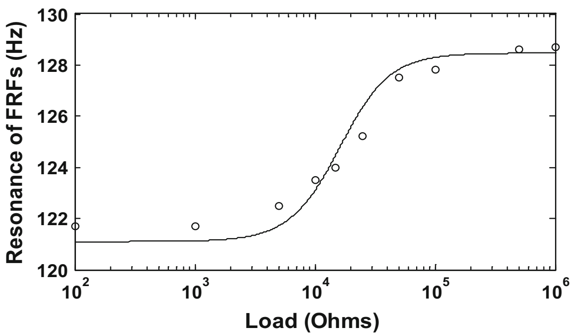


Fig. 3.8 Change in resonance frequency with electrical load (experimental data points with theoretical black line)

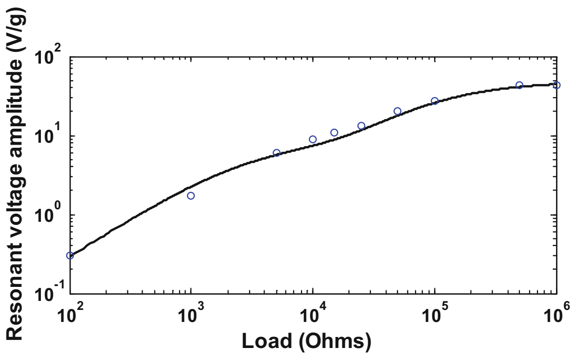


Fig. 3.9 Change in resonant voltage amplitude with electrical load (experimental data points with theoretical curve)

resistances, the electrical effect is only equivalent to viscous damping (Eq. (3.34)) [4].

The magnitude of the voltage FRF at corresponding resonance frequency increases monotonically with the resistor load as shown in Fig. 3.7. These observations are demonstrated in Fig. 3.9, where the magnitude of the voltage FRF at resonance also called resonant voltage amplitude is plotted as a function of load [4]. The theoretical curve is acquired by calculating the modulus of $\tilde{v}(\omega)|_r$ at the theoretical resonance frequencies of Fig. 3.8 [4].

Figure 3.7 shows that the magnitude of the voltage FRF at a *fixed* frequency also increases monotonically with the electrical load, and this can be further observed in Fig. 3.10, which shows magnitude of the voltage FRF, at the short-circuit and open-circuit frequencies, at different electrical loads [4].

Figure 3.10 above demonstrates that the value of voltage output at these two frequencies increases at similar way, but at low loads, the output is higher for the short-circuit frequency since the system is close to short-circuit conditions [1, 4].

For the similar reason, the voltage at the open-circuit frequency or at higher loads becomes higher than that at the short-circuit frequency since the system shifts nearer to open-circuit conditions.

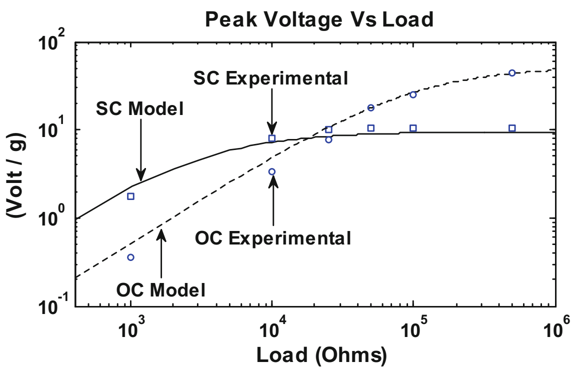


Fig. 3.10 Amplitude of voltage at open-circuit and short-circuit frequencies

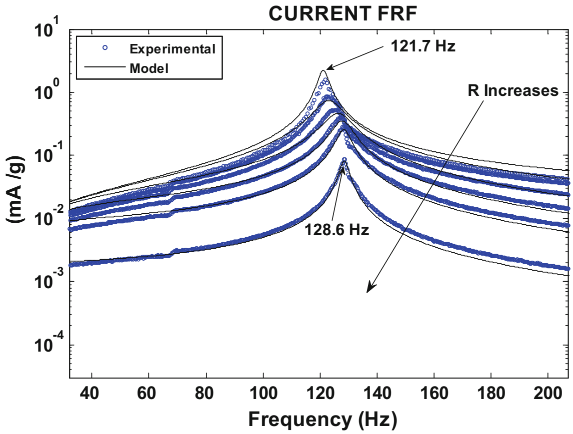


Fig. 3.11 Current FRFs at different loads (1 kΩ, 10 kΩ, 25 kΩ, 50 kΩ, 100 kΩ, 500 kΩ) [4]

3.3.2.2 Magnitude Plots of Current FRFs

The current FRFs also showed similar monotonic trend with load but in a reverse manner to the voltage FRFs. In current FRF, the magnitude of the generated current decreases as the load was increased. Figure 3.11 shows comparison of current FRFs, experimental and theoretical, for various resistor loads ranging from 1 kΩ to 500 kΩ.

Figure 3.12 demonstrates the magnitude of the current FRF at resonance frequency plotted as a function of corresponding resistor load. It is important to note that the behaviour with increasing the load is found to be the reverse of that of Fig. 3.9.

Figure 3.13 illustrates changes in value of the magnitude of the current FRF, by varying the electrical load, at the short-circuit and open-circuit frequencies, and it is found that the behaviour with increasing the electrical load is again seen to be the reverse of that in Fig. 3.10 [4].

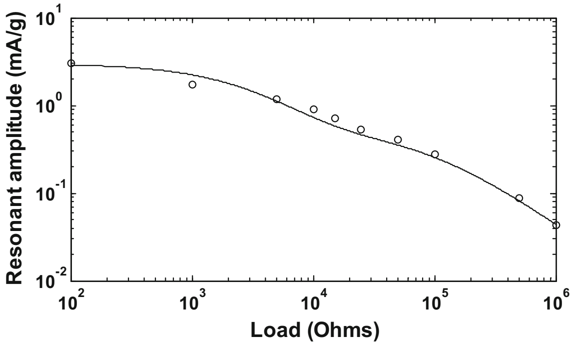


Fig. 3.12 Changes in resonant current amplitude with electrical load (experimental data points with theoretical curve)

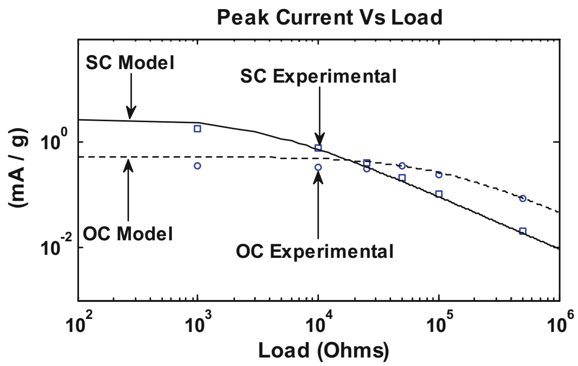


Fig. 3.13 Amplitude of current at open-circuit and short-circuit frequencies

3.3.2.3 Power Output FRFs

The power FRF can be defined by the established relation of electrical circuitry as $\left|\tilde{v}^2(\omega)\right|/R$, or can also be expressed as the product of the moduli of the voltage and current FRFs. Since the voltage and current FRFs have reverse monotonic variations with load, the variation of the power FRF with load will not be monotonic [4]. This fact can be noted in Fig. 3.14 which presents the comparison of theoretical and experimental power FRFs for six different loads. It is important to note that the maxima of the power FRFs in Fig. 3.14 happen at the resonance frequencies depicted in Fig. 3.8, and the values at these maxima are referred to here as 'resonant power' [4]. Figure 3.15 illustrates the variation of the resonant power with electrical load. The theoretical graph demonstrates that the resonant power has turning points at loads of 3 kΩ, 16.2 kΩ, 85 kΩ, with the middle load giving a local minimum point and the other two providing practically equal maxima [4]. It shall be noted that the horizontal axis is logarithmic. Hence, the first peak which occurs in the low-resistance side where the horizontal subdivisions represent smaller increments would appear much sharper (i.e. thinner) when plotted on a linear horizontal (resistance) scale instead of logarithmic scale. It can be noted from the experimental data points that the optimal resonant load exists in between 50 and 100 kΩ, which also validates the optimal resonant load of 85 kΩ as determined by the theoretical curve.

The resonant power curve in Fig. 3.15 can be determined by squaring the resonant voltage FRF values in Fig. 3.9 and dividing by the corresponding resistance values [4]. As a result of squaring the voltage FRF, the discrepancies between theory and experiment are further amplified. However, another main reason why the discrepancies appear large in Fig. 3.15 is because of the range of the vertical axis is much narrower than that in Fig. 3.9 (the limits are 1 to 10 in Fig. 3.15 and 0.1 to 100 in Fig. 3.9) [4].

Figure 3.16 shows the variation in power FRFs, evaluated at different loads, when excited at two fixed frequencies, namely the open- and short-circuit resonances. It can be noted that these curves have only one turning point and that the

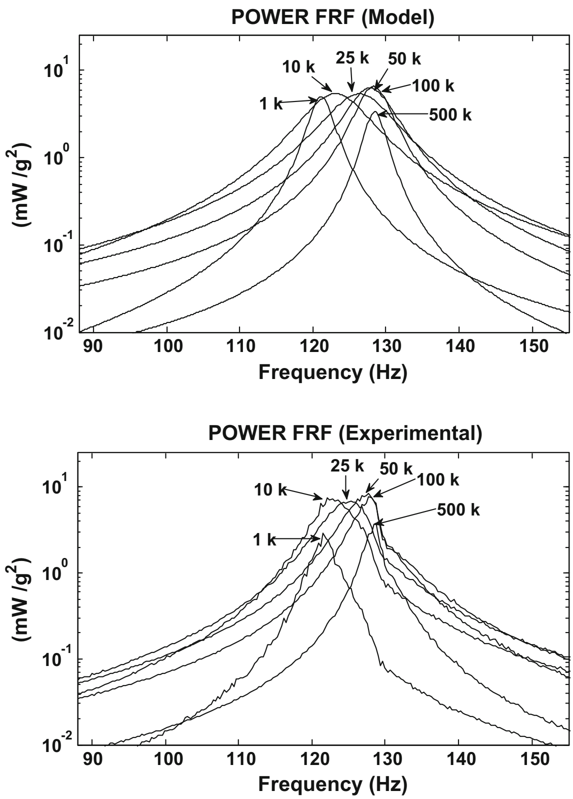


Fig. 3.14 Power FRFs comparison: Model and experimental plots [4]

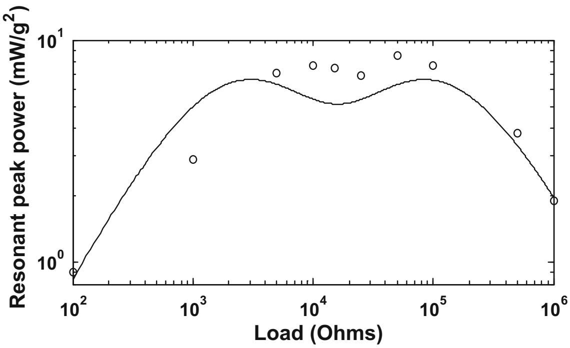


Fig. 3.15 Change in resonant power FRF with load (theoretical with experimental data points)

maximum value of power is almost the same in short-circuit and open-circuit conditions [4]. It can be seen from Fig. 3.16 that the optimum load is smaller when the system is excited at the short-circuit frequency.

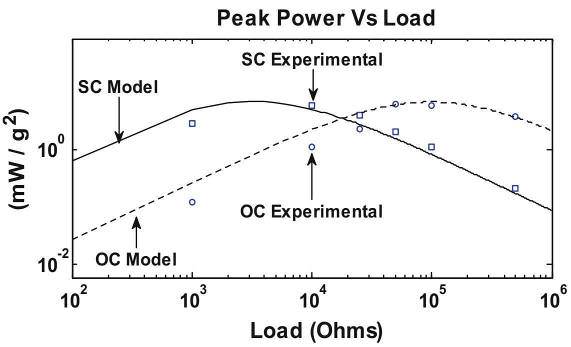


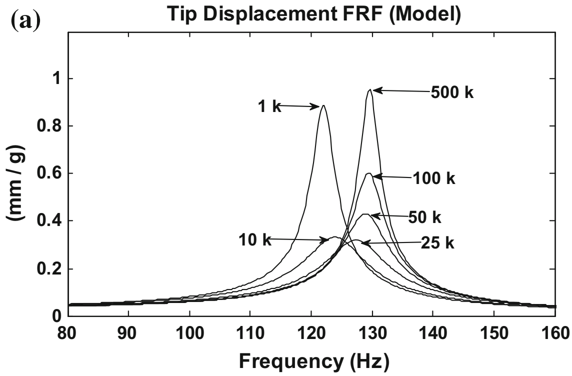
Fig. 3.16 Power FRF at open- and short-circuit resonance frequencies

3.3.2.4 FRFs of Tip Response

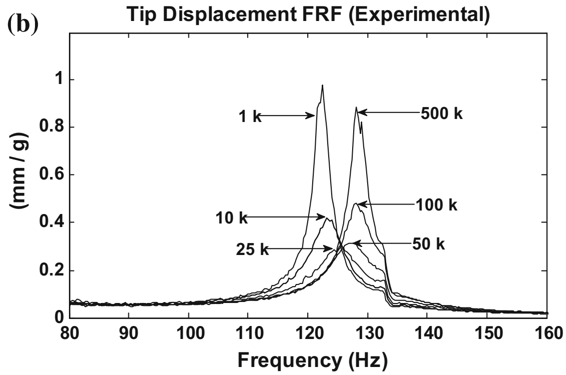
The experimental and the theoretical results for the absolute tip displacement FRF $\beta_{abs}(\omega)$ are presented in Fig. 3.17 for six different electrical loads. As previously explained, the laser recorded $\beta_{abs}(\omega)$, so the theoretical results were obtained by solving Eqs. (3.32) and (3.30). It can be seen that both measured and simulated values of the resonant tip response amplitudes show similar behaviour, of moving upwards and downwards, for different electrical loads. It is noteworthy that any alteration in the electrical load not only affects the amplitude of the tip but also shifts the resonance frequency of the system in the same way as it was observed in voltage, current and power FRFs [4]. For this study, the resonance occurs in excess of 100 (i.e. 121–128 Hz), the effect of the term, $1/\omega^2$, present in Eq. (3.30) is almost negligible, and hence, the resonances in $\beta_{abs}(\omega)$ and $\beta(\omega)$ are coincident. Hence, in order to locate the theoretical resonance frequency at a given resistance, one can differentiate the modulus of the relative displacement FRF in Eq. (3.32) and set the equation to zero [4]. Appendix B presents the details of the resulting polynomial equation in this regard. The resonances computed from this equation for different electrical loads produced a similar graph as was presented in Fig. 3.8. The theoretical and experimental variation of the resonant value of $\beta_{abs}(\omega)$ with electrical load is presented in Fig. 3.18. It can be seen that at low-load or short-circuit condition, the electrical power generated is negligible but the resonant tip response is high. The resonant tip response then begins to reduce as the electrical load is increased up to a certain value and beyond this value of electrical load, the resonant tip response starts to increase again as open-circuit conditions are reached, for which the electrical power generated is negligible [4]. It is due to the fact that additional damping is added to the system when mechanical energy is converted into electrical energy at certain range of electrical load. The induced additional damping also called as ‘electrical damping’ reduces the amplitude of the resonant tip response. The fine resolution of the theoretical graph in Fig. 3.18 shows that the resonant tip response is lowest for a theoretical load of 17 kΩ [4]. It is noteworthy that, although the electrical power generation results in supplementary damping, the theoretical load that provides the lowest resonant tip response in Fig. 3.18 (17 kΩ) does not correspond

Fig. 3.17 Amplitude of absolute tip response FRFs comparison: **a** model; **b** experimental [4]

(a)



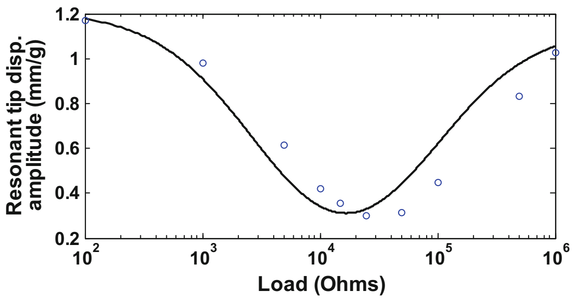
(b)



to the theoretical load that offers the maximum resonant power output in Fig. 3.15 (85 kΩ) [4]. In fact, the theoretical load where the tip response is minimum as in Fig. 3.18 is matched closely to the theoretical load that yields the local minimum point, shown in the middle region, of Fig. 3.15 (i.e. 16 kΩ). This occurrence verifies the fact that lowering the tip response of the piezoelectric cantilever energy harvester will reduce the generated electrical power.

This aspect is verified by the experimental data in Figs. 3.18 and 3.15: the load of 25 kΩ provided the lowest measured resonant tip response and a local minimum

Fig. 3.18 Change in resonant value of absolute (or relative) tip response FRF with resistor load (theoretical with experimental data points)



of the measured resonant power. It is important to note that this feature was not observed in the theoretical analysis of [1] and the theoretical and experimental analyses of [2] since resistance values were adjusted in larger steps than in the present case [4].

3.3.2.5 Evolution of Nyquist Plots with Load

Previous research [2] has only considered the magnitude of the FRFs to show the output of energy harvesting system. As FRFs are complex-valued functions that contain both magnitude (amplitude ratio) and phase information, hence, a more thorough validation of the derived FRFs in Eqs. (3.31, 3.32) is achieved by studying the Nyquist plots [4]. The Nyquist plots of voltage and flexural velocity FRFs against the electrical load are presented in Figs. 3.19 and 3.20, respectively. It is noted that the plots of both types of FRF remain almost circular, and the orientation of the circle of the voltage FRF relative to the origin is seen to be significantly affected by changing the resistance value, unlike the circle of the flexural tip velocity FRF (Fig. 3.20) [4]. This may be due to the fact that the voltage FRF is directly proportional to the value of R as shown in Eq. (3.31), whereas the resistance term is small in the numerator of the tip deflection FRF (Eq. (3.32)) compared to the other term (i.e. 2). Nyquist plots of the theoretical simulation and the experimental data demonstrate the same pattern and, with increasing electrical load, provide a powerful validation of the model.

It is observed that, despite the display of reasonably good correlation between theory and prediction, the experimental data points show deviations from the main circular pattern at some higher frequencies as shown in Figs. 3.19 and 3.20 [4]. These deviations can be seen as a minor circular outgrowth from the main circular pattern. They correspond to the knot (shoulder) around 133 Hz, more significantly, in the experimental tip response FRF magnitude plots in Fig. 3.17b and, to lesser extent, in the experimental voltage FRF plots in Fig. 3.7. It is likely that this may be due to the contribution of an unwanted torsional mode that was inadvertently excited in the experiments.

3.3.2.6 Limitations of the Theory

Overall, the measurement results show reasonable match with the theory despite some deviations. This is considered that measurement error alone is not enough to justify discrepancies between theory and experiment [4]. Following assumptions made in the mathematical modelling of PVEH system also contribute to the discrepancies:

- The electrodes that cover the upper and lower surfaces of each piezolayer were assumed to be completely conductive so that a single potential could be assumed for each electrode surface

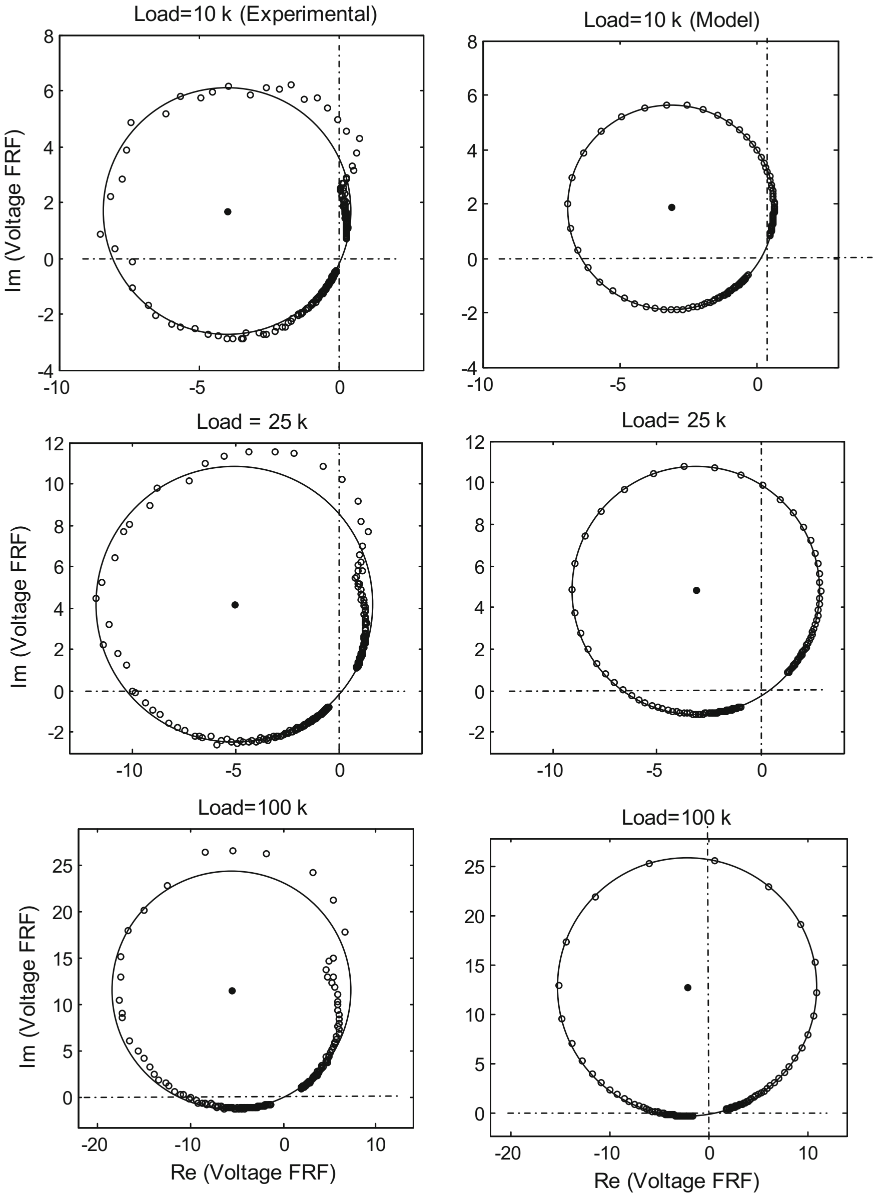


Fig. 3.19 Nyquist plot representation of Voltage FRF (using circle fit through data points; ω increases clockwise from origin) [4]

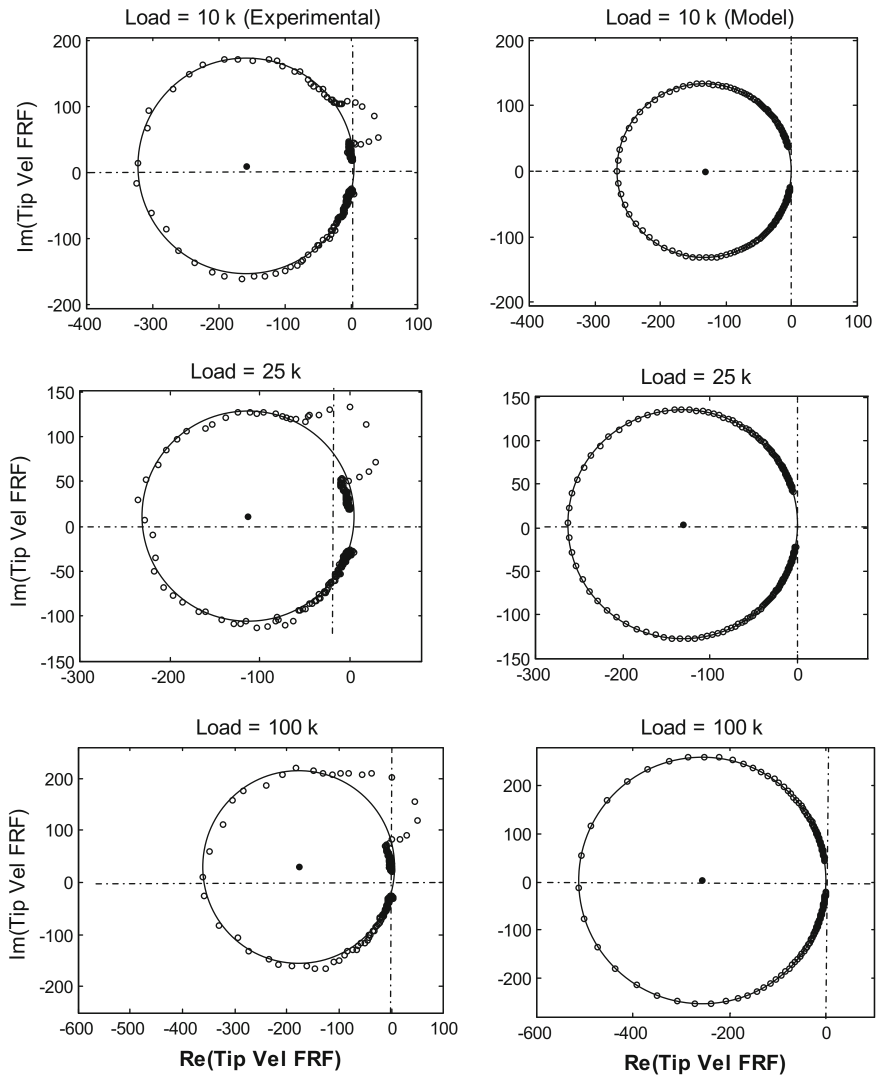


Fig. 3.20 Nyquist plot representation of tip velocity FRF (including circle fit through data points; ω increases clockwise from origin) [4]

- The bonding between the piezolayers and shim was assumed to be perfect, and the impedance of the adhesive used was taken to be negligible [4]

It is important to note that the bimorphs were manufactured by Piezo Systems Inc, who provided the parameters of Table 3.1. The deviations from above-mentioned assumptions may be equivalent to unaccounted small internal electrical impedance. However, its mechanism would most likely be far more complex than that of simple additional impedance, and its modelling is beyond the scope of the present research as the study only restricted to a simple resistive load [4].

3.4 Conclusions

The chapter presents theoretical and experimental study of a distributed parameter model of a clamp-free bimorph PVEH excited by base vibration. The experimental work successfully validates the accuracy of the distributed parameter model at relatively higher resonance frequency energy harvester. Frequency response functions of current, voltage, power and tip response were investigated for different electrical loads ranging from 10^2 Ω to 10^6 Ω. The results of experimental and theoretical graphs showed the variation in resonance frequency, resonant voltage, resonant power and resonant tip response of the energy harvester at different loads. At very low electrical loads, it was found that the electrical effect can only be regarded as a pure viscous damper. The mathematical model predicted a 6% change in the resonance frequency of the harvester as the load was shifted from the short-circuit to the open-circuit conditions, and this was validated by the experimental results. In between short-circuit and open-circuit load conditions, the energy harvesting effect produces additional damping that decreases the amplitude of the tip response. It is clearly shown that the electrical load that generates maximum power is not the same as the load which introduces maximum mechanical damping. The load that produced maximum resonant power was much higher than the load that gave minimum tip response. The latter load was much closer to the load that gave a minimum turning point on the resonant power versus load graph, since power generation depends on the mechanical deformation [4]. Nyquist plots have been used to present a comprehensive validation of the FRFs and to provide a self-validating means of estimating the mechanical damping. The modal-based theoretical analysis used in this chapter will be compared and verified against a completely different theoretical method in the next chapter.

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Chapter 4
Modelling of Energy Harvesting Beams Using Dynamic Stiffness Method (DSM)

4.1 Background

The previous chapter used the analytical modal analysis method (AMAM), originally developed by Erturk and Inman in order to model and analyse base-excited piezoelectric energy harvesting beams with the aim of predicting the electrical output for given base motion input [1]. So far in PVEH literature, the application of AMAM has been restricted to simple cantilevered uniform-section beams (Fig. 4.1) [1, 2]. This chapter compares two alternative modelling techniques for energy harvesting beams and uses these techniques in a theoretical study of a bimorph. One of the methods is an application of the dynamic stiffness method (DSM) to the modelling of energy harvesting beams [3]. The DSM, or its analogue the mechanical impedance method [4, 5], is a powerful technique for the derivation of the FRFs of a distributed parameter uniform-section structural element and assemblies of such elements. The dynamic stiffness matrix of a structural element is based on the exact solution of the wave equation of motion, i.e. obviates the need for modal or basis-function transformation. Hence, its application to a piezoelectric beam allows an independent means of verification of the AMAM model of [1] and Chap. 3. It is noted that the DSM works with force/displacement FRFs while the mechanical impedance method works with force/velocity FRFs. Hence, these methods differ merely by a factor of $j\omega$ (where ω is the circular frequency and $j = \sqrt{-1}$) [4, 5]. The dynamic stiffness matrix of a uniform-section beam could be used in the modelling of beams with arbitrary boundary conditions or assemblies of beams of different cross sections [3].

It is important to note that some of the equations presented in Chap. 3 for a bimorph case are re-expressed here to cover the general case (i.e. unimorph and bimorph with series or parallel connectivity).

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For both methods presented, the Euler-Bernoulli model with piezoelectric coupling is used, and the external electrical load is represented by generic linear impedance. Moreover, damping is considered in greater depth here than in Chap. 3.

4.2 Modelling

The modelling in this section refers to the systems in Fig. 4.1. It is assumed that the electrodes are infinitely thin, flexible, have negligible resistance and run along the whole length 'l' of the beam (the electrodes are indicated as thick black lines bracketing the piezolayers in Fig. 4.1 and elsewhere). The subscripts 'p' and 'sh' will be used to refer to the piezo and shim layers, respectively. The Euler-Bernoulli beam model is used, along with the standard piezoelectric constitutive relations:

$$D_3 = d_{31}\sigma_p + \varepsilon_{33}^T E_3 \tag{4.1}$$

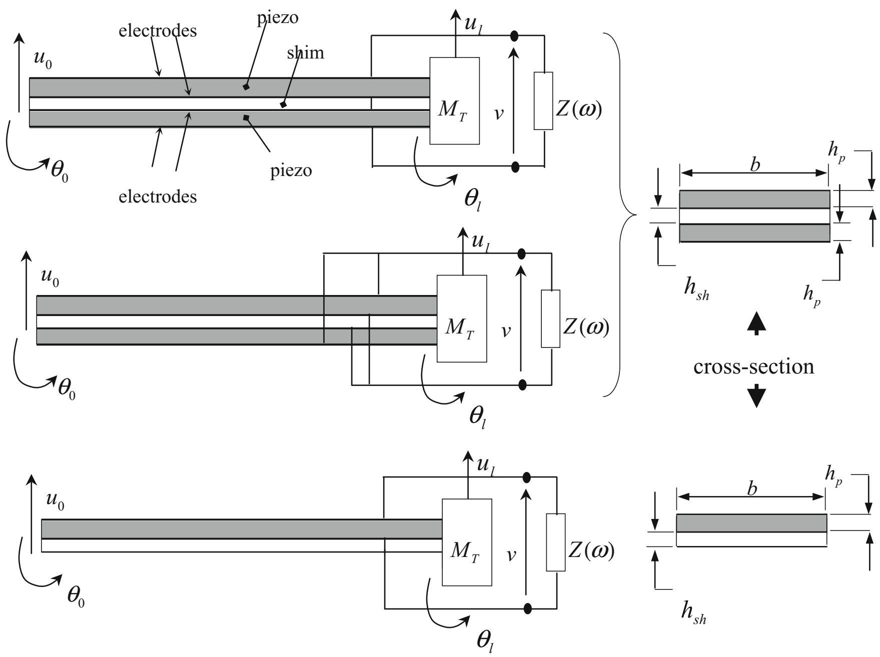


Fig. 4.1 Base-excited piezoelectric energy harvesting beams (series-connected bimorph, upper figure; parallel-connected bimorph, middle figure; and unimorph, lower figure) [3]

$$\delta_p = \frac{\sigma_p}{Y_p} + d_{31}E_{31} \tag{4.2}$$

However, Eq. (4.1) and (4.2) will be modified to include material damping in the following analysis.

4.2.1 General Equations

The mathematical model presented here follows [3] and is reproduced here for completeness. The equation of motion of the beam can be written as:

$$m\frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 M}{\partial x^2} + c_a\frac{\partial u}{\partial t} = 0 \tag{4.3}$$

where $u(x,t)$ is the displacement at location x at time t, c_a is the viscous damping coefficient of the surrounding medium per unit length, m is the mass per unit length and $M(x,t)$ is the bending moment. Modifying Eq. (4.1) to include damping, the stresses can be expressed as:

$$\sigma_p = Y_p\delta + c_p\dot{\delta} \pm d_{31}Y_pE_3 \tag{4.4}$$

$$\sigma_{sh} = Y_{sh}\delta + c_{sh}\dot{\delta} \tag{4.5}$$

where c_p and c_{sh} represent Kelvin-Voigt damping coefficients and E_3 is given by:

$$E_3 = -v(t)/\left(ah_p\right) \tag{4.6}$$

v being the voltage appearing across the electrical load and a is either 1 or 2:

$$a = \begin{cases} 2 & \text{bimorph(piezos in series)} \\ 1 & \text{bimorph(piezos in parallel), unimorph} \end{cases} \tag{4.7}$$

In Eq. (4.4), the minus sign applies for a unimorph. In the case of a bimorph, the minus and plus signs apply for the upper and lower layers, respectively (layers connected in series have the same field E_3 and so are poled in opposite directions, i.e. have coefficients $\pm d_{31}$; layers connected in parallel have equal and opposite fields and so are poled in the same direction). The strain δ at a distance z from the neutral axis is given by

$$\delta = -z\partial^2 u/\partial x^2 \tag{4.8}$$

Substituting Eqs. (4.6) and (4.8) into Eqs. (4.4) and (4.5) and integrating the product $-\sigma z$ with respect to z, over the cross section yields the bending moment as:

$$M(x,t) = B\frac{\partial^2 u}{\partial x^2} + A\frac{\partial^3 u}{\partial x^2 \partial t} + \vartheta\, v(t) \tag{4.9}$$

where:

$$B = Y_p I_p + Y_{sh} I_{sh} \tag{4.10}$$

$$A = c_p I_p + c_{sh} I_{sh} \tag{4.11}$$

I denotes the second moment of area about the neutral axis. For a bimorph, the electrical coupling term ϑ is given by:

$$\vartheta = -d_{31} Y_p b(h_p + h_{sh})\big/ a \tag{4.12}$$

For a unimorph, the expression for ϑ is given in reference [3]. Substituting (4.9) into (4.3) gives the equation of motion (wave equation) of the beam as:

$$B\frac{\partial^4 u}{\partial x^4} + A\frac{\partial^5 u}{\partial x^4 \partial t} + c_a\frac{\partial u}{\partial t} + m\frac{\partial^2 u}{\partial t^2} = 0 \tag{4.13}$$

Substituting for the stress from Eq. (4.4) into Eq. (4.2) gives the following expression for D_3:

$$D_3 = d_{31} Y_p \delta + d_{31} c_p \dot{\delta} + \varepsilon_{33}^S E_3 \tag{4.14}$$

$$\varepsilon_{33}^S = \varepsilon_{33}^T - d_{31}^2 Y_p \tag{4.15}$$

The previous research in [1] (on which Chap. 3 was based) omits the second term in Eq. (4.14). This could have been inadvertent since in [1], material damping was simply introduced into the wave Eq. (4.13) (in contrast to this chapter, where the material damping term in the wave equation was justified through prior derivation via modification of Eq. (4.1)). Alternatively, the omission of the second term in Eq. (4.14) by [1] could have been deliberate since it was deemed to be small and c_p difficult to quantify. This term will also be omitted from the following analysis. However, the validity of this omission is discussed in Sect. 4.3.3. Considering the strain at the mid-section of the upper piezolayer, for which $z = h_{pc}$, integrating over its area and differentiating with respect to time yields the current i:

$$i(t) = f\beta \int_0^l \frac{\partial^3 u}{\partial x^2 \partial t} dx - \frac{f}{a} C_p \dot{v} \tag{4.16}$$

where

$$\beta = -d_{31} Y_p h_{pc} b \tag{4.17}$$

$$C_p = \varepsilon_{33}^S bl/h_p \tag{4.18}$$

$$f = \begin{cases} 1 & \text{bimorph (piezos in series), unimorph} \\ 2 & \text{bimorph(piezos in parallel)} \end{cases} \tag{4.19}$$

4.2.2 *Dynamic Stiffness Method (DSM)*

The DSM assumes harmonic excitation. Let $\tilde{u}$ denote the complex amplitude of u, i.e.

$u(x,t) = \mathrm{Re}\{\tilde{u}(x)\mathrm{e}^{\mathrm{j}\omega t}\}$. Similar notation is used for the other time-varying quantities. The equation of motion (4.13) hence reduces to:

$$\frac{d^4\tilde{u}}{dx^4} - k^4\tilde{u} = 0 \tag{4.20}$$

where wave number k is given by:

$$k = \omega^{\frac{1}{2}} \left\{ \frac{m}{\widehat{B} \big/ [1 - \mathrm{j}c_a/(m\omega)]} \right\}^{\frac{1}{4}} \tag{4.21}$$

$$\widehat{B} = B(1 + \mathrm{j}\omega A/B) \tag{4.22}$$

What is required is the dynamic stiffness matrix **D** of the beam with tip mass (Fig. 4.2a):

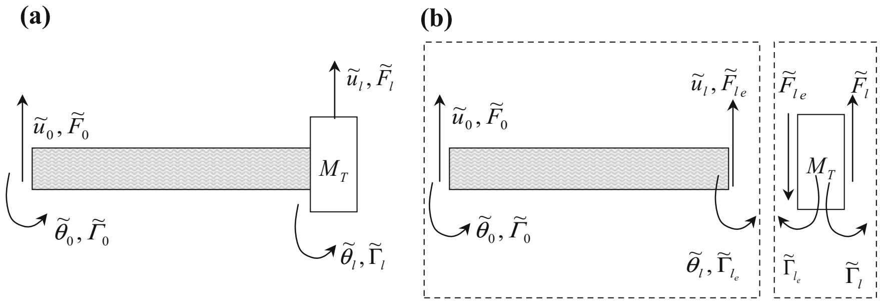


Fig. 4.2 Free body diagram of EH beam (damping forces not shown): **a** overall System; **b** exploded view [3]

$$\mathbf{f} = \mathbf{D}\mathbf{u}, \mathbf{f} = \begin{bmatrix} \tilde{F}_0 & \tilde{\Gamma}_0 & \tilde{F}_l & \tilde{\Gamma}_l \end{bmatrix}^{\mathrm{T}}, \mathbf{u} = \begin{bmatrix} \tilde{u}_0 & \tilde{\theta}_0 & \tilde{u}_l & \tilde{\theta}_l \end{bmatrix}^{\mathrm{T}} \tag{4.23}$$

where **f** and **u** are vectors of the complex amplitudes of the external excitations and displacements, respectively. The dynamic stiffness $\mathbf{D_e}$ of the beam itself is first derived (Fig. 4.2b), where:

$$\mathbf{f_e} = \mathbf{D_e}\mathbf{u}, \mathbf{f_e} = \begin{bmatrix} \tilde{F}_0 & \tilde{\Gamma}_0 & \tilde{F}_{l_e} & \tilde{\Gamma}_{l_e} \end{bmatrix}^{\mathrm{T}} \tag{4.24}$$

The solution to Eq. (4.20) is:

$$\tilde{u}(x) = C_1 \cosh kx + C_2 \sinh kx + C_3 \cos kx + C_4 \sin kx \tag{4.25}$$

The complex amplitudes of the bending moment and shear force are given by:

$$\tilde{M}(x) = \widehat{B}\tilde{u}'' + \vartheta\,\tilde{v} \tag{4.26}$$

$$\tilde{Q}(x) = \tilde{M}' = \widehat{B}\tilde{u}''' \tag{4.27}$$

If Z denotes the impedance of the electrical load, $\tilde{v} = \tilde{i}Z$ and hence, from Eq. (4.16):

$$\tilde{v} = G\int_0^l \tilde{u}''dx = G\left(\tilde{\theta}_l - \tilde{\theta}_0\right), \tag{4.28}$$

$$G = \frac{\mathrm{j}\omega f\beta}{\mathrm{j}\omega(f/a)C_p + 1/Z} \tag{4.29}$$

$\mathbf{D_e}$ is then derived as follows. Using Eq. (4.25) and applying displacement boundary conditions $\left(\tilde{u} = \tilde{u}_0, \tilde{u}' = \tilde{\theta}_0 \text{ at } x = 0; \tilde{u} = \tilde{u}_l, \tilde{u}' = \tilde{\theta}_l \text{ at } x = l\right)$, one obtains:

$$\mathbf{u} = \mathbf{A}\mathbf{c} \tag{4.30}$$

where $\mathbf{c} = \begin{bmatrix} C_1 & C_2 & C_3 & C_3 \end{bmatrix}^{\mathrm{T}}$ and **A** is a 4×4 matrix.

Using Eqs. (4.26) and (4.27) and applying force/moment boundary conditions $\left(\tilde{Q} = \tilde{F}_0, \tilde{M} = -\tilde{\Gamma}_0 \text{ at } x = 0; \tilde{Q} = -\tilde{F}_{le}, \tilde{M} = \tilde{\Gamma}_{l_e} \text{ at } x = l\right)$, one obtains:

$$\mathbf{f} = \mathbf{B}\mathbf{c} \tag{4.31}$$

where $\mathbf{B}$ is a 4×4 matrix. Eliminating $\mathbf{c}$ from (4.30) and (4.31) gives:

$$\mathbf{D_e} = \mathbf{BA^{-1}} \equiv \begin{bmatrix} s_1 & s_2 & s_3 & s_4 \\ s_2 & s_5 & -s_4 & s_6 \\ s_3 & -s_4 & s_1 & -s_2 \\ s_4 & s_6 & -s_2 & s_5 \end{bmatrix} + \vartheta G \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \tag{4.32}$$

where:

$$s_1 = \widehat{B}k^3(\cosh kl \, \sin kl + \sinh kl \, \cos kl)/\Delta \tag{4.33}$$

$$s_2 = \widehat{B}k^2(\sinh kl \, \sin kl)/\Delta \tag{4.34}$$

$$s_3 = -\widehat{B}k^3(\sin kl + \sinh kl)/\Delta \tag{4.35}$$

$$s_4 = -\widehat{B}k^2(\cos kl - \cosh kl)/\Delta \tag{4.36}$$

$$s_5 = \widehat{B}k(\cosh kl \, \sin kl - \sinh kl \, \cos kl)/\Delta \tag{4.37}$$

$$s_6 = \widehat{B}k(\sinh kl - \sin kl)/\Delta \tag{4.38}$$

$$\Delta = 1 - \cosh kl \, \cos kl \tag{4.39}$$

The first term of Eq. (4.32) is recognisable as the dynamic stiffness matrix of the beam for no electrical coupling, achievable under short-circuits conditions ($Z \to 0$, see Eq. (4.29)).

The dynamic stiffness matrix of the beam with tip mass is given by:

$$\mathbf{D} = \mathbf{D_e} + \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{D_m} \end{bmatrix}, \mathbf{D_m} = \begin{bmatrix} -\omega^2 M_T + \mathrm{j}\omega(c_a/m)M_T & 0 \\ 0 & -\omega^2 I_T \end{bmatrix} \tag{4.40}$$

where M_T, I_T denote the mass and moment of inertia, respectively. It is noted that the coefficient of the ambient damping on the tip mass is taken to be $(c_a/m)M_T$ (i.e. damping proportional to mass). This assumption is *chosen* to enable direct comparison with the modal method of Sect. 4.2.3, which *requires* it to uncouple the modal equations.

The receptance matrix $\mathbf{R}$ of the system in Fig. 4.2a is given by:

$$\mathbf{u} = \mathbf{Rf}, \mathbf{R} = \{R_{ij}\} = \mathbf{D}^{-1} \tag{4.41}$$

For the systems in Fig. 4.1, there are no excitations at the tip. By setting $\tilde{F}_l, \tilde{\Gamma}_l = 0$ in Eq. (4.41), one can express the tip displacement and rotation in terms of $\tilde{u}_0, \tilde{\theta}_0$.

Then, using Eq. (4.28), the voltage can also be expressed in terms of $\tilde{u}_0, \tilde{\theta}_0$. The following expressions are then applicable for the systems in Fig. 4.1:

$$\tilde{v} = G[(R_{41}R_{22} - R_{42}R_{21})/\alpha]\tilde{u}_0 + G[(-R_{41}R_{12} + R_{42}R_{11} - \alpha)/\alpha]\tilde{\theta}_0 \tag{4.42}$$

$$\tilde{u}_l = [(R_{31}R_{22} - R_{32}R_{21})/\alpha]\tilde{u}_0 + [(-R_{31}R_{12} + R_{32}R_{11})/\alpha]\tilde{\theta}_0 \tag{4.43}$$

where $\alpha = R_{11}R_{22} - R_{12}R_{21}$.

4.2.3 Analytical Modal Analysis Method (AMAM)

In this approach, u is expressed as:

$$u = u_0 + x\theta_0 + u_{flex} \tag{4.44}$$

In Eq. (4.44), $u_0 + x\theta_0$ is the rigid body component, and u_{flex} is the displacement relative to this straight line. $u_{flex}(x, t)$, therefore, defines cantilever-type flexure and can be expressed as a modal series using mass-normalised modes $\phi_r(x)$ describing undamped free vibration of the electrically uncoupled beam with tip mass M_T and the left end fixed in both translation and rotation:

$$u_{flex}(x, t) = \sum_{r=1}^{N} \eta_r(t)\phi_r(x) \tag{4.45}$$

In Eq. (4.45), the η_r's define the modal coordinates. The mode-shapes satisfy the orthogonality conditions, which, for negligible moment of inertia of the tip mass, are:

$$B\int_0^l \phi_r\phi_s^{iv}dx - B\{\phi_s\phi_r''\}\big|_{x=l} = \begin{cases} 0 & (r \neq s) \\ \omega_r^2 & (r = s) \end{cases} \tag{4.46}$$

$$m\int_0^l \phi_r\phi_s dx + M_T\phi_r(l)\phi_s(l) = \begin{cases} 0 & (r \neq s) \\ 1 & (r = s) \end{cases} \tag{4.47}$$

The expression for $\phi_r(x)$ is:

$$\phi_r(x) = \psi_r/\sqrt{M_r} \tag{4.48}$$

$$M_r = m_b \left\{ (1/l) \int_0^l \psi_r^2 dx + (M_T/m_b)\psi_r^2(l) \right\} \tag{4.49}$$

$$\psi_r(x) = \cosh(\lambda_r/l)x - \cos(\lambda_r/l)x - \upsilon_r\{\sinh(\lambda_r/l)x - \sin(\lambda_r/l)x\} \tag{4.50}$$

$$\upsilon_r = \frac{\sinh\lambda_r - \sin\lambda_r + \lambda_r(M_T/m_b)(\cosh\lambda_r - \cos\lambda_r)}{\cosh\lambda_r + \cos\lambda_r + \lambda_r(M_T/m_b)(\sinh\lambda_r - \sin\lambda_r)} \tag{4.51}$$

where $m_b = ml$ is the beam mass and the λ_r's are the roots of the equation:

$$1 + \cos\lambda_r \cosh\lambda_r + (M_T/m_b)\lambda_r(\cos\lambda_r \sinh\lambda_r - \sin\lambda_r \cosh\lambda_r) = 0 \tag{4.52}$$

The corresponding natural frequencies ω_r are given by:

$$\omega_r = (\lambda_r/l)^2\sqrt{(B/m)} \tag{4.53}$$

The information in Eqs. (4.48)–(4.51) is given here since Erturk and Inman [1] do not supply the expression for the mass-normalisation constant M_r (Eq. (4.49)) or any reference for its calculation.

It is noted that the substitution of the bending moment (Eq. (4.9)) into Eq. (4.3) led to the loss of the electrical coupling term from the resulting Eq. (4.13). Substitution of Eqs. (4.44, 4.45) into Eq. (4.13) would merely yield the modal equations for the specific case of no electrical coupling. In order to obtain the modal equations with electrical coupling, Eq. (4.9) needs to be recast by considering the case where the electrode extends from $x = x_1$ to $x = x_2$ [3]. The electrical term of Eq. (4.9) is then $\vartheta v(t)[H(x - x_1) - H(x - x_2)]$, where $H(x)$ is the Heaviside Function. Substituting the revised expression for M into Eq. (4.3) and setting $x_1 = 0$ and $x_2 = l$ yields an equation of motion that is transformable into electrically coupled modal equations:

$$B\frac{\partial^4 u}{\partial x^4} + A\frac{\partial^5 u}{\partial x^4 \partial t} + \vartheta v(t)[\delta'(x) - \delta'(x - l)] + c_a\frac{\partial u}{\partial t} + m\frac{\partial^2 u}{\partial t^2} = 0 \tag{4.54}$$

where $\delta(x)$ is the Dirac-Delta function (not to be confused with the strain symbol). It is noted that the version of this equation quoted in [1] has an additional tip mass inertia term. This term is not required, and its inclusion does not actually lead to the uncoupling of the modes. It is also worth noting that although DSM works with Eq. (4.13), the electrical term enters the analysis through the application of the moment boundary conditions (Eq. (4.31)).

Substituting Eqs. (4.44, 4.45) in (4.54), applying the orthogonality conditions of mode-shapes (Eqs. (4.46, 4.47)), considering the forces on the tip mass and assuming that the ambient damping force on the tip mass is $(c_a/m)M_T$ yields the following equations:

$$\ddot{\eta}_r + 2\zeta_r\omega_r\dot{\eta}_r + \omega_r^2\eta_r + \chi_r v = -m_b\left(\bar{\gamma}_r^u\ddot{u}_0 + \bar{\gamma}_r^\theta\ddot{\theta}_0\right) - \breve{c}_a\left(\bar{\gamma}_r^u\dot{u}_0 + \bar{\gamma}_r^\theta\dot{\theta}_0\right) \tag{4.55}$$

where

$$\zeta_r = A\omega_r/(2B) + c_a/(2m\omega_r) \tag{4.56}$$

$$\breve{c}_a = c_a l, \quad \chi_r = \vartheta\phi_r'(l) \tag{4.57a, b}$$

$$\bar{\gamma}_r^u = (1/l)\int_0^l \phi_r\,dx + (M_t/m_b)\phi_r(l) \quad \bar{\gamma}_r^\theta = (1/l)\int_0^l x\phi_r dx + l(M_t/m_b)\phi_r(l) \tag{4.58a, b}$$

For harmonic excitation, substituting Eqs. (4.44, 4.45) into Eq. (4.28) yields:

$$\tilde{v} = G\int_0^l \tilde{u}''_{flex}dx = \frac{GC_p}{\beta}\sum_{r=1}^N \varphi_r\tilde{\eta}_r \tag{4.59}$$

$$\varphi_r = \left(\beta/C_p\right)'_r\phi(l) \tag{4.60}$$

Also, from Eq. (4.55):

$$\tilde{\eta}_r = \left[\left(m_b\omega^2 - \mathrm{j}\omega C_a\right)\left(\bar{\gamma}_r^u\tilde{u}_0 + \bar{\gamma}_r^\theta\tilde{\theta}_0\right) - \chi_r\tilde{v}\right]\Big/\left(\omega_r^2 - \omega^2 + \mathrm{j}2\zeta_r\omega_r\omega\right) \tag{4.61}$$

An expression for $\tilde{v}$ is obtained by solving Eqs. (4.59, 4.61). The resulting expression is then back substituted in Eq. (4.61) and, using Eqs. (4.45, 4.46), an expression for $\tilde{u}_l$ is obtained. The resulting expressions for $\tilde{v}$ and $\tilde{u}_l$ are directly comparable to the ones in Eqs. (4.42, 4.43):

$$\tilde{v} = \left[\frac{\frac{GC_p}{\beta}\sum_{r=1}^N\left\{\frac{\varphi_r\bar{\gamma}_r^u\left(m_b\omega^2 - \mathrm{j}\omega\breve{c}_a\right)}{\omega_r^2 - \omega^2 + \mathrm{j}2\zeta_r\omega_r\omega}\right\}}{1 + \frac{GC_p}{\beta}\sum_{r=1}^N\left\{\frac{\varphi_r\chi_r}{\omega_r^2 - \omega^2 + \mathrm{j}2\zeta_r\omega_r\omega}\right\}}\right]\tilde{u}_0 + \left[\frac{\frac{GC_p}{\beta}\sum_{r=1}^N\left\{\frac{\varphi_r\bar{\gamma}_r^\theta\left(m_b\omega^2 - \mathrm{j}\omega\breve{c}_a\right)}{\omega_r^2 - \omega^2 + \mathrm{j}2\zeta_r\omega_r\omega}\right\}}{1 + \frac{GC_p}{\beta}\sum_{r=1}^N\left\{\frac{\varphi_r\chi_r}{\omega_r^2 - \omega^2 + \mathrm{j}2\zeta_r\omega_r\omega}\right\}}\right]\tilde{\theta}_0 \tag{4.62}$$

$$\tilde{u}_l = \left[1 + \sum_{r=1}^{N}\left\{\bar{\gamma}_r^u - \chi_r\left[\frac{\frac{GC_p}{\beta}\sum_{r=1}^{N}\left\{\frac{\varphi_r\bar{\gamma}_r^u}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega}\right\}}{1+\frac{GC_p}{\beta}\sum_{r=1}^{N}\left\{\frac{\varphi_r\chi_r}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega}\right\}}\right]\right\}\frac{\left(m_b\omega^2-\mathrm{j}\omega\breve{c}_a\right)\phi_r(l)}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega}\right]\tilde{u}_0$$
$$+\left[l + \sum_{r=1}^{N}\left\{\bar{\gamma}_r^\theta - \chi_r\left[\frac{\frac{GC_p}{\beta}\sum_{r=1}^{N}\left\{\frac{\varphi_r\bar{\gamma}_r^\theta}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega}\right\}}{1+\frac{GC_p}{\beta}\sum_{r=1}^{N}\left\{\frac{\varphi_r\chi_r}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega}\right\}}\right]\right\}\frac{\left(m_b\omega^2-\mathrm{j}\omega\breve{c}_a\right)\phi_r(l)}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega}\right]\tilde{\theta}_0 \tag{4.63}$$

These formulae, like the DSM formulae of the previous section, cover all the cases in [1, 6] (shown in Fig. 4.1), and one can switch easily from one system to the other by using the appropriate values for a and f, the appropriate expression for ϑ and the appropriate structural parameters.

It is noted that ambient damping contributes to both the system damping (Eq. (4.56)) and the system excitation (right-hand side of Eq. (4.55)). This latter contribution is neglected in [1], which results in another slight difference between Eq. (4.54) and its version in [1].

4.3 Theoretical Analysis of Cantilevered Bimorph

The theoretical study in this section concerns the following issues (i) the cross verification of DSM and AMAM; (ii) the effect of the type and magnitude of the electrical impedance, as well as series and parallel connection of the piezoelectric layers; (iii) the interaction between the harvester and the structure to which it is attached; and (iv) the effect of damping-related assumptions [3]. It is important to note that this analysis focuses the electro-mechanical response of the device. Optimisation of the parameters for maximal energy output is outside the scope of the analysis of this section. It is also noted that a thorough experimental validation of the AMAM was presented in Chap. 3 of this book (also in reference [6]).

The system analysed is a bimorph with parameters given in Table 4.1. Unless otherwise stated, the results refer to a series-connected bimorph (Fig. 4.1).

Table 4.1 Parameters of bimorph

h_p (mm)	0.267	l (mm)	60
h_{sh} (mm)	0.300	density of piezo $(\mathrm{kg/m^3})$	7800
Y_p (GPa)	62	density of shim $(\mathrm{kg/m^3})$	2700
Y_{sh} (GPa)	72	d_{31} (m/V)	-320×10^{-12}
b (mm)	25	ε_{33}^T (F/m)	3.3646×10^{-8}

A moderately sized tip mass $M_T = 0.5m_b$ is included to capture all features of the modelling procedure. The damping ratios for the first two modes are taken as $\zeta_1 = 0.0166, \zeta_2 = 0.0107$, unless otherwise stated. Specification of two damping ratios allows calculation of A and c_a from Eq. (4.56), enabling the application of the DSM. Knowledge of A and c_a also allows the calculation of all the remaining ζ_r 's, enabling multimodal application of the AMAM.

4.3.1 Verification: DSM Vs AMAM

Figure 4.3 shows the frequency response of the amplitude of the voltage generated per unit amplitude base acceleration and no base rotation (i.e. modulus of coefficient of $\tilde{u}_0$ in Eqs. (4.42) or (4.62), divided by ω^2) and a purely resistive impedance $Z = R = 100\text{k}\Omega$. The AMAM solution with one mode (i.e. $N = 1$ in Eq. (4.62)),

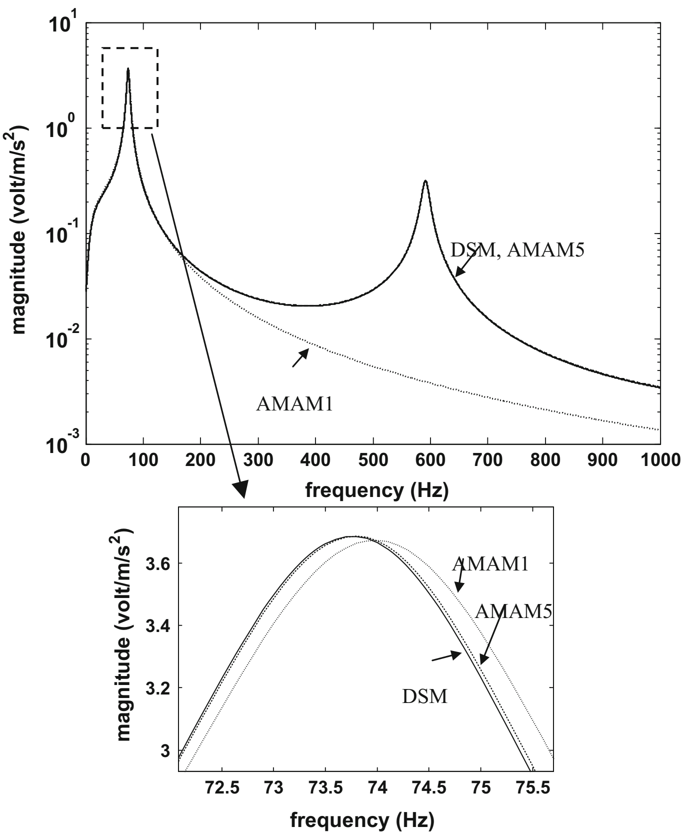


Fig. 4.3 Comparison of voltage FRF by DSM and AMAM [3]

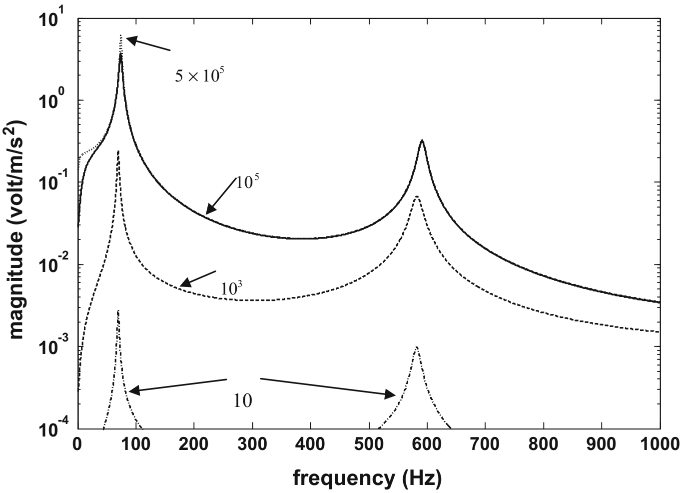


Fig. 4.4 FRFs for generated voltage with increasing values of resistive loads R, in Ohms [3]

referred to as 'AMAM1', diverges from the DSM solution beyond 150 Hz. The close-up of Fig. 4.3 shows that the one-mode AMAM slightly overestimates the first resonance frequency.

The AMAM solution converges towards the DSM solution as more modes are taken. The result for the 5-mode AMAM ('AMAM5') is virtually indistinguishable from the DSM result. The same holds for all other results presented in this chapter.

4.3.2 *Effect of Electrical Impedance*

First considered is the case of a purely resistive load $Z = R$. Figure 4.4 shows the voltage frequency response function (FRF), defined in the previous section, for increasing values of R. A resistance R of 10 Ω is very close to short-circuit conditions [3]. The voltage increases as R increases until open-circuit conditions are reached.

Figure 4.5 shows the transmissibility connecting u_l and u_0 for no base rotation (i.e. coefficient of $\tilde{u}_0$ in Eqs. (4.43) or (4.63)). It is noted that the resonance locations of the transmissibility coincide with those of the voltage FRF. Close-ups of the two marked resonance regions show that increasing the impedance stiffens the system, raising its first two resonance frequencies by 6.6% and 1.6%, respectively. There is negligible increase in the resonance frequencies beyond a resistance of 500 k Ω. The resistive load also significantly affects the transmissibility peaks. The mean power dissipated is $|\tilde{v}|^2 \big/ (2R)$ and is zero for both pure short circuit $(R \to 0 \Rightarrow |\tilde{v}| = 0)$ and pure open circuit $(R \to \infty)$. Hence, the peaks at these two

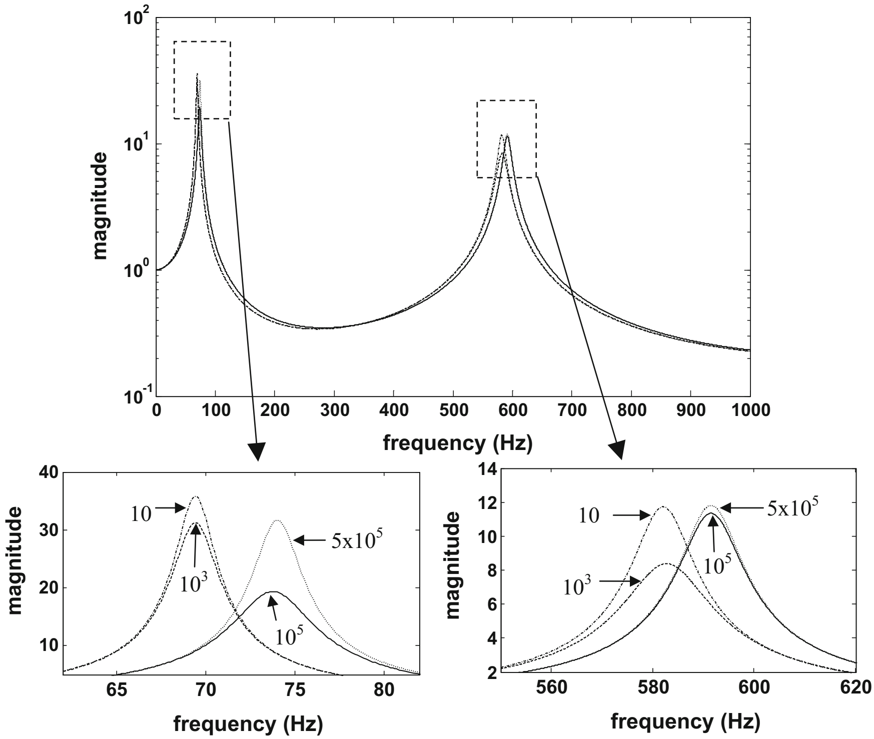
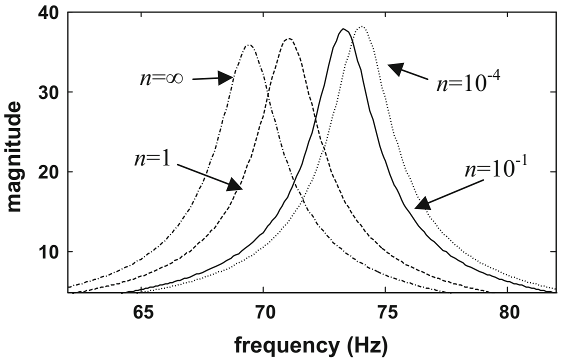


Fig. 4.5 Tip-to-base transmissibility for different resistive loads R in Ohms [3]

conditions are at similar levels. In between these conditions, there is a reduction in the peak due to the power dissipated, i.e. the electrical effect introduces additional damping.

Next considered is the case of a purely capacitive load for which $Z = 1/(j\omega C)$. Expressing the capacitance $C = nC_p$, where C_p is the effective capacitance of a piezolayer (Eq. (4.18)), then $n = \infty$ and $n = 0$, respectively, correspond to the short-circuit and open-circuit conditions. It is important to note that n is a multiplication factor of the capacitances and should not be confused with the mode number symbol. $n = 10^{-4}$ gives practically open-circuit conditions, and the voltage FRF is practically the same as that for the highest resistance in Fig. 4.4. The transmissibility FRFs for four values of n are basically similar to those in Fig. 4.5 except for the marked resonance regions. Comparing the close-ups in Fig. 4.6 with those in Fig. 4.5, one observes that: (a) the stiffening effect is still present for the capacitive load, with the two resonance frequencies increasing by the same amounts as for the resistive load as the impedance is raised from short- to open-circuit condition; and (b) there are no reductions in the peaks between short- and open-circuit conditions for the capacitive load. The latter observation is easily



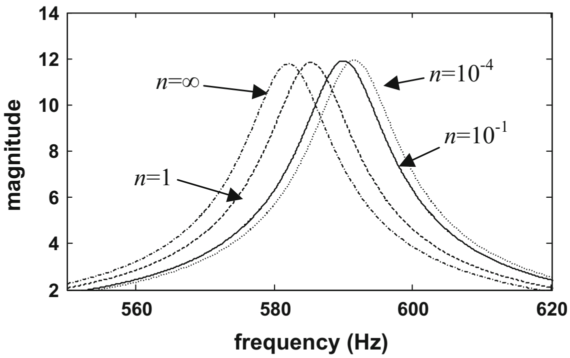


Fig. 4.6 Tip-to-base transmissibility peaks for different capacitive loads, C = *nCp* [3]

explained by the fact that the mean power dissipated is $\left|\tilde{i}\right|^2 \mathrm{Re}\{Z\}/2$ and so is zero for a capacitive load.

The slight monotonic increase in the peaks in Fig. 4.6 is due to the increase in resonance frequency: this can be understood by considering that the value at resonance of the transmissibility of a simple one-degree-of-freedom system is $1 + \mathrm{j}m_0\omega_0/c_0$, where ω_0, m_0 and c_0 are the natural circular frequency, mass and damping coefficient, respectively.

The shift in resonance between the short- and open-circuit conditions can be explained by considering single-mode approximations of the voltage and transmissibility FRFs. Assuming the modes are well-spaced, for excitation frequencies ω in the region of any particular natural frequency ω_r, the main contributions to the summations in Eqs. (4.62, 4.63) come from the r^{th} mode. Dispensing with the summation signs in Eqs. (4.62, 4.63) and substituting for G/β from Eq. (4.29) yields the following approximations for the voltage and transmissibility FRFs:

$$\left.\frac{\tilde{v}}{-\omega^2\tilde{u}_0}\right|_{\tilde{\theta}_0=0} \approx \frac{-\mathrm{j}\omega f C_p Z \varphi_r \bar{\gamma}_r^u m_b}{\left[1+\mathrm{j}\omega C_p Z(f/a)\right]\left(\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega\right)+\mathrm{j}\omega f C_p Z \varphi_r \chi_r} \tag{4.64}$$

$$\left.\frac{\tilde{u}_l}{\tilde{u}_0}\right|_{\tilde{\theta}_0=0} \approx 1+\frac{\left[1+\mathrm{j}\omega C_p Z(f/a)\right]\bar{\gamma}_r^u m_b \omega^2 \phi_r(l)}{\left[1+\mathrm{j}\omega C_p Z(f/a)\right]\left(\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega\right)+\mathrm{j}\omega f C_p Z_r \varphi \chi_r} \tag{4.65}$$

It is noted that the $\mathrm{j}\omega\breve{c}_a$ term has been omitted from the above equations. This is a valid assumption that is discussed in Sect. 4.3.3.

For a small resistive impedance *R*, the above expressions approximate to:

$$\left.\frac{\tilde{v}}{-\omega^2\tilde{u}_0}\right|_{\tilde{\theta}_0=0} \approx \frac{-\mathrm{j}\omega f C_p R \varphi_r \bar{\gamma}_r^u m_b}{\omega_r^2-\omega^2+\mathrm{j}2\left\{\zeta_r+\frac{f C_p R \varphi_r \chi_r}{2\omega_r}\right\}\omega_r\omega} \tag{4.66}$$

$$\left.\frac{\tilde{u}_l}{\tilde{u}_0}\right|_{\tilde{\theta}_0=0} \approx 1+\frac{\bar{\gamma}_r^u m_b \omega^2 \phi_r(l)}{\omega_r^2-\omega^2+\mathrm{j}2\left\{\zeta_r+\frac{f C_p R_r \varphi \chi_r}{2\omega_r}\right\}\omega_r\omega} \tag{4.67}$$

Hence, it is evident from the denominators of Eqs. (4.66, 4.67) that a *small resistive* impedance is merely equivalent to a viscous damper. This explains the reduction in the resonance peaks as the load is increased from 10 Ωto 1 k Ω with little or no shift in resonance frequency (see Fig. 4.5).

For a large impedance, the limiting forms of Eqs. (4.64, 4.65) are:

$$\left.\frac{\tilde{v}}{-\omega^2\tilde{u}_0}\right|_{\tilde{\theta}_0=0} \approx \frac{-a\varphi_r \bar{\gamma}_r^u m_b}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega+a\varphi_r\chi_r} \tag{4.68}$$

$$\left.\frac{\tilde{u}_l}{\tilde{u}_0}\right|_{\tilde{\theta}_0=0} \approx 1+\frac{\bar{\gamma}_r^u m_b \omega^2 \phi_r(l)}{\omega_r^2-\omega^2+\mathrm{j}2\zeta_r\omega_r\omega+a\varphi_r\chi_r} \tag{4.69}$$

Hence, an approximate expression for the open-circuit resonance of mode no. *r* is

$$(\omega_{oc})_r \approx \sqrt{\omega_r^2+a\varphi_r\chi_r} = \sqrt{\omega_r^2+a\theta\frac{\beta}{C_p}\left\{\phi_r'(l)\right\}^2} \tag{4.70}$$

Substitution of the parameter values into the above expression yields the first two open-circuit resonance values of the bimorph as 74.3 Hz and 591.8 Hz, which roughly agree with the frequency locations of the open-circuit peaks in Figs. 4.5 and 4.6 (74.0 Hz and 591.5 Hz by DSM or AMAM5).

Equation (4.70) shows that the open-circuit resonances of a bimorph are independent of the wiring of the piezolayers (series or parallel connection) since the value of $a\vartheta$ is the same in either case (see Eq. (4.12)). This contradicts the statement in a recent (2010) paper by Zhu et al. [7] that a wider tuning range can be achieved with a bimorph if its layers are wired in parallel. With regard to the voltage generated, Eq. (4.68) shows that the open-circuit resonance peak voltages of a parallel-connected bimorph $(a = 1)$ are half those of a series-connected one $(a = 2)$.

4.3.3 Effect of Mechanical Damping

It is pertinent to illustrate the influence of the mechanical damping on the voltage generated and electrical power dissipated. The results presented above pertained to structural and ambient damping values $(A = A_{nom}, c_a = c_{a_{nom}})$ that resulted in damping ratios of $1.66\%, 1.07\%$ for the first two modes. For a given beam (i.e. fixed A), the damping ratios of all modes can be altered by modifying the damping capacity of the surrounding medium (i.e. c_a). Figures 4.7a,b, respectively, show the variation (with a resistive load) of the first resonant value of the voltage FRF and the corresponding electrical power dissipated for four different ambient damping levels $(c_a = 0.5c_{a_{nom}}, 0.75c_{a_{nom}}, c_{a_{nom}}, 1.5c_{a_{nom}},$ respectively) which respectively result in $\zeta_1 = 0.88\%, 1.27\%, 1.66\%, 2.43\%$. The power is specific mean power, obtained by dividing the mean power $\left(= (\text{voltage FRF})^2 \big/ (2R)\right)$ by the total volume of the piezolayers. It is noted that the local shallow minimum in the power curves was observed theoretically and experimentally in [6] to correspond closely to the resistance value that minimises the transmissibility connecting u_l and u_0. As expected, the mechanical damping reduces the voltage generated for a given load.

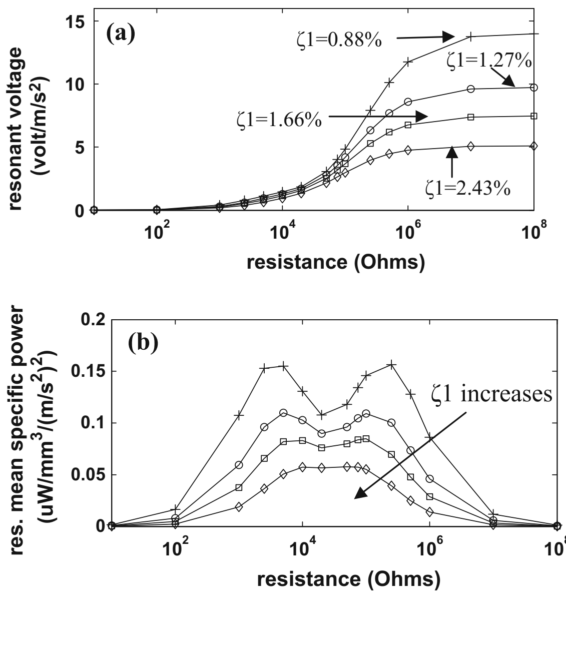


Fig. 4.7 Mechanical damping effects, at 1st resonance frequency, on **a** voltage and **b** mean specific power [3]

For a purely resistive load and no mechanical damping, the voltage FRF is infinite at resonance only for open-circuit condition (this can be seen from substituting Eq. (4.70) into Eqs. (4.68, 4.69)).

For a purely capacitive load and no mechanical damping, the voltage FRF is infinite at resonance for all load conditions other than short circuit. This can be seen by setting $Z = 1/(\mathrm{j}\omega C)$ in Eq. (4.64) and observing that the resonances occur at

$$\left(\omega_{capac\,load}\right)_r = \sqrt{\omega_r^2 + \frac{f\left(C_p/C\right)\varphi_r\chi_r}{\left\{1 + \left(C_p/C\right)(f/a)\right\}}} \tag{4.71}$$

The same statements (pertaining to no mechanical damping) also apply for the transmissibility, except that this would also be infinite at pure short circuit.

4.3.4 Duality of Energy Harvesting Beam and Vibration Neutraliser

In AMAM, the input excitation is a prescribed vibration at the base of the harvester. For this reason, very little mention, if any, has been made of the fact that the energy harvesting beams in Fig. 4.1 are, in many practical applications, mechanical absorbers of the vibration at their base (i.e. their point of attachment to the host structure). Mechanical vibration absorbers can be either tuned mass dampers (TMDs) or neutralisers [8].

The tuned frequency of an absorber is defined as its fundamental resonance with its base fixed. TMDs are tuned to suppress a specific mode of the host structure over a wide range of frequencies. Neutralisers are tuned to a specific excitation frequency and suppress the vibration at this frequency by planting an anti-resonance at their point of attachment to the host structure [8]. In contrast to the TMD, the attenuation of a neutraliser is dynamic rather than dissipative, and any damping mechanism within the neutraliser degrades the attenuation. The tuned frequency of an energy harvesting beam is defined in a similar manner to an absorber, and the harvester is typically tuned to a specific ambient frequency. This means that it doubles as a neutraliser. Hence, the base vibration is affected (indeed minimised) by the harvester. The DSM lends itself more readily than AMAM to investigate this practical situation. This is done by studying the FRFs $\tilde{u}_0/\tilde{F}_0$ (attachment point receptance) and $\tilde{v}/\tilde{F}_0$ (voltage per unit base force) for the direction-fixed base/free-end system considered in the previous section (illustrated in Fig. 4.8) [3]. By setting $\tilde{\theta}_0 = 0$ in Eq. (4.41), in addition to $\tilde{F}_l, \tilde{\Gamma}_l = 0$, one can express $\tilde{\Gamma}_o$ in terms of $\tilde{F}_0$. Hence, for the system in Fig. 4.8:

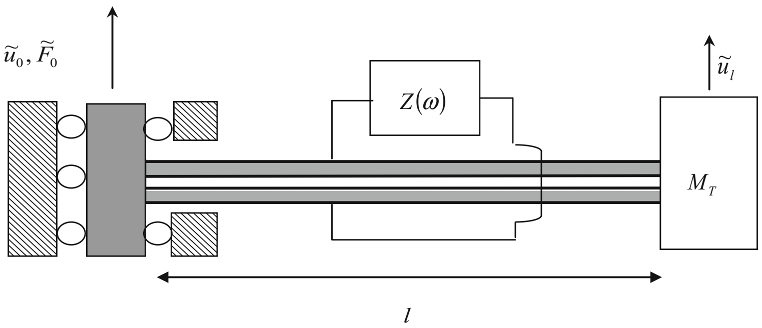


Fig. 4.8 Direction-fixed base/tip free cantilever beam having end mass [3]

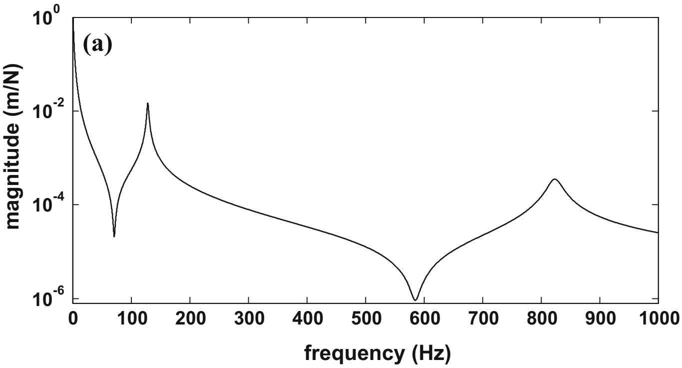
$$\tilde{u}_0/\tilde{F}_0 = R_{11} - R_{12}R_{21}/R_{22} \tag{4.72}$$

$$\tilde{v}/\tilde{F}_0 = G\tilde{\theta}_l = G(R_{41} - R_{42}R_{21}/R_{22}) \tag{4.73}$$

These two FRFs are plotted in Figs. 4.9a,b. It is clear that the anti-resonances in $\tilde{u}_0/\tilde{F}_0$ (Fig. 4.9a) correspond to the resonances in the voltage FRF and transmissibility (Figs. 4.4, 4.5). The resonances in $\tilde{v}/\tilde{F}_0$ (and $\tilde{u}_0/\tilde{F}_0$) are the 'free-body' resonances of the system in Fig. 4.8 (these resonances correspond to the symmetric modes of vibration of the electrically coupled centrally excited free-free beam obtained by reflecting the system in Fig. 4.8 about a vertical line through its left-hand end).

Figures 4.10 shows that the dissipation introduced by a resistive load reduces the depth of the anti-resonance of $\tilde{u}_0/\tilde{F}_0$. This is not the case for a capacitive load: in fact, there would be a slight monotonic increase in depth of the anti-resonance as the load is increased from short circuit to open circuit due to the stiffening effect (as already observed for the transmissibility in Fig. 4.6). For a capacitive load and no mechanical damping, the base vibration is completely cancelled at the tuned frequency, and the voltage FRF is infinite. The voltage generated is still finite, however, and is calculated by multiplying the finite values of $\tilde{v}/\tilde{F}_0$ and $\tilde{F}_0$.

Zhu et al. [7] explore ways of tuning the resonant frequency of the energy harvesters in Fig. 4.1 to the frequency of the ambient vibrations, thereby maximising energy output. One of the methods considered is the adjustment of the electrical load, which they explore through the elementary single degree of freedom theoretical analysis of Roundy et al. [9]. They state that the most feasible tuning approach is to adjust capacitive loads since resistive loads reduce the efficiency of power transfer, and load inductances are difficult to adjust. Based on the above analysis, the tunability is also useful for adaptive vibration neutralisation in response to a variable excitation frequency. In cantilever-type neutralisers, this adaptation is commonly achieved by mechanical means (e.g. varying the cross section or repositioning an attached mass) [8]. If the beam is piezoelectric, the



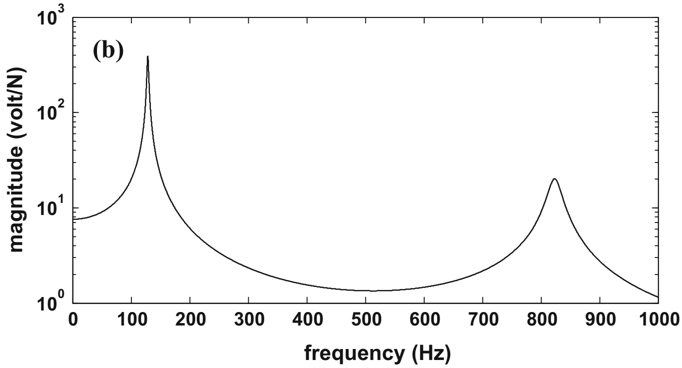


Fig. 4.9 Force-referenced FRFs for a capacitive load $C = C_p$: **a** base displacement per unit base force $\tilde{u}_0/\tilde{F}_0$ **b** voltage per unit base force $\tilde{v}/\tilde{F}_0$

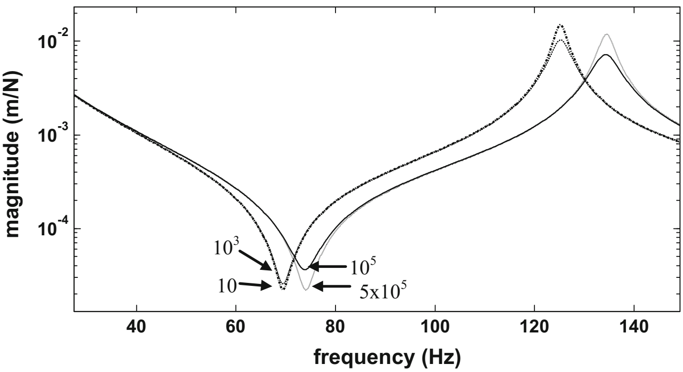


Fig. 4.10 Base displacement per unit base force $\tilde{u}_0/\tilde{F}_0$ for different values of resistive loads R in Ohms

retuning is readily achievable through variable impedance although its use would be restricted to applications that require only a narrow tuning frequency range. Hence, one can envisage a variable capacitance adaptive vibration neutraliser that doubles as an adaptive energy harvester. The formulae of Eq. (4.70) provide a guideline for optimising the parameters to widen the tuning range as much as possible.

4.3.5 Effect of Damping-Related Assumptions

It is pertinent to investigate the potential influence of damping-related assumptions on the above-presented results. Two issues are considered: (a) the omission of c_p from Eq. (4.14) and (b) the consideration or otherwise of the ambient damping c_a.

4.3.5.1 Effect of Omission of C_p Term from Eq. (4.14)

It was mentioned in Sect. 4.2.1 that the analysis here follows the same practice in [1, 6] of omitting the $d_{31}c_p\dot{\delta}$ term in Eq. (4.14). However, the above formulation allows the re-introduction of its effect by simply replacing β (Eq. (4.17)) in all equations by a complex version $\widehat{\beta} = -d_{31}\widehat{Y}_p h_{pc} b$ where the complex modulus $\widehat{Y}_p = Y_p + \mathrm{j}\omega c_p$. The value of c_p is difficult to determine since $\zeta_{1,2}$ determine A and not its individual constituents in Eq. (4.56). However, from Eq. (4.11), it is evident that the value of c_p is bounded by A/I_p. By using this value for c_p in $\widehat{Y}_p$, it was verified that there was negligible change to the results presented above.

4.3.5.2 Effect of Omission of the Ambient Damping C_a

It was stated at the end of Sect. 4.2.3 that ambient damping contributes to both the system damping (Eq. (4.56)) and the system excitation (right-hand side of Eq. (4.55)). The latter contribution can be easily verified as negligible for most practical applications by rerunning the AMAM simulations with the term $\mathrm{j}\omega\breve{c}_a$ omitted from Eqs. (4.62) and (4.63). On the other hand, for a given ζ_1, the consideration or otherwise of c_a will have a profound effect on the damping of the second and higher modes. This is due to its effect on the estimate obtained for the structural damping constant A.

The values for the first two modal damping ratios ζ_1 and ζ_2 can be obtained individually through the experimental methods described in [1, 6]. Their knowledge enables calculation of A and c_a from Eq. (4.56) and hence the remaining ζ_r's, as mentioned previously. If, for the sake of argument, ζ_1 and ζ_2 are found to be of approximately similar magnitude (as in the previous analysis), then

$$A = \frac{2B}{\omega_1 + \omega_2}\zeta_1 \tag{4.74}$$

$$\zeta_r = \zeta_1 \frac{\omega_r}{\omega_1}\left\{\left(1 + \frac{\omega_1\omega_2}{\omega_r^2}\right) \Big/ \left(1 + \frac{\omega_2}{\omega_1}\right)\right\}(c_a \neq 0 \,\text{and}\, \zeta_1 = \zeta_2) \tag{4.75}$$

If, on the other hand, only ζ_1 is determined experimentally and c_a is taken to be zero:

$$A = \frac{2B}{\omega_1}\zeta_1 \tag{4.76}$$

$$\zeta_r = \zeta_1 \frac{\omega_r}{\omega_1}(c_a = 0) \tag{4.77}$$

Notice that although the estimates for A are different for the two cases, the FRF levels at the first resonance are unaffected since both cases have the same value of ζ_1. However, comparing Eqs. (4.75) and (4.77), it is clear that, for $r \geq 2, \zeta_r$ is considerably greater for the latter case than the former case. Hence, the FRF levels at the resonances of these modes are significantly damped for the latter case. Hence, the existence of c_a needs to be recognised if the FRF resonance peak levels of the second and higher modes are to be reliably predicted.

4.4 Extension of DSM

4.4.1 One-Dimensional Assembly of Beam Segments

The AMAM analysis can be readily extended to the case where the electrode extends from $x = x_1$ to $x = x_2$ rather than the entire beam length l by slight modification to C_p, χ_r and φ_r [3]. However, it would still be restricted to a system with a uniform cross section throughout and the boundary conditions of Fig. 4.1. In the DSM approach, each electrode portion would be regarded as a separate beam element that is part of a one-dimensional assembly of beam elements. Moreover, these elements can have different cross sections, and arbitrary boundary conditions can be readily accommodated. In contrast to the Transfer Matrix Method [10], the method of assembly of the global DSM is the same as that used to assemble the global finite element mass and stiffness matrices of a structure. Hence, the DSM can easily be combined with the FE techniques for those applications where the structure to be analysed comprises of a beam assembly incorporated into a more complicated structure. In the example below, the application of the proposed DSM method to an assembly of beams is presented. Figure 4.11a and b show the assembly and the exploded view of two beams (beam A and beam B), respectively.

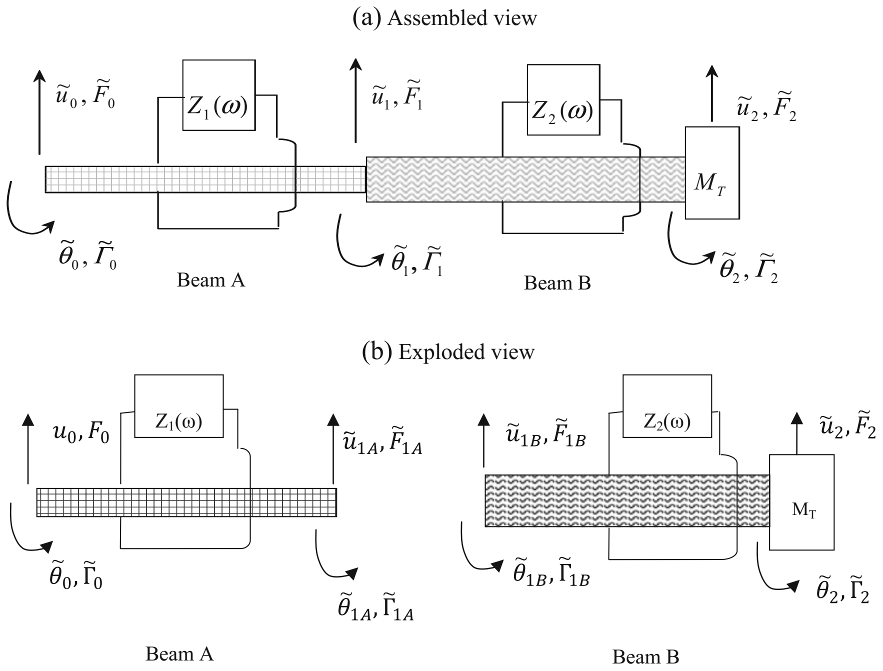


Fig. 4.11 Piezoelectric beam segments with external loads: assembled and exploded view [3]

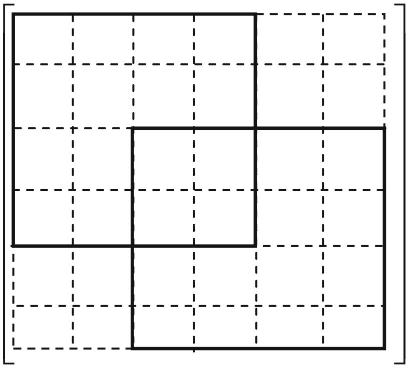
The DSM matrix of each beam can be determined separately using the method explained in Sect. 4.2.2. Following the steps of Sect. 4.2.2, the DSM matrices of beam A and beam B of Fig. 4.11b can be written as:

$$\begin{bmatrix} \tilde{F}_0 \\ \tilde{\Gamma}_0 \\ \tilde{F}_{1A} \\ \tilde{\Gamma}_{1A} \end{bmatrix} = [D_A] \begin{bmatrix} \tilde{u}_0 \\ \tilde{\theta}_0 \\ \tilde{u}_1 \\ \tilde{\theta}_1 \end{bmatrix} \tag{4.78a}$$

$$\begin{bmatrix} \tilde{F}_{1B} \\ \tilde{\Gamma}_{1B} \\ \tilde{F}_2 \\ \tilde{\Gamma}_2 \end{bmatrix} = [D_B] \begin{bmatrix} \tilde{u}_1 \\ \tilde{\theta}_1 \\ \tilde{u}_2 \\ \tilde{\theta}_2 \end{bmatrix} \tag{4.78b}$$

Next, with reference to Fig. 4.11a, the 6×6 global dynamic stiffness matrix $\mathbf{D_g}$ can be assembled from the elemental 4×4 matrices of both beams (Eqs. (4.78a and 4.78b)) using the following familiar technique [4]:

$$\begin{bmatrix} \tilde{F}_0 \\ \tilde{\Gamma}_0 \\ \tilde{F}_1 \\ \tilde{\Gamma}_1 \\ \tilde{F}_2 \\ \tilde{\Gamma}_2 \end{bmatrix} =$$



$$\begin{bmatrix} \tilde{u}_0 \\ \tilde{\theta}_0 \\ \tilde{u}_1 \\ \tilde{\theta}_1 \\ \tilde{u}_2 \\ \tilde{\theta}_2 \end{bmatrix} \tag{4.78c}$$

where $\tilde{F}_1 = \tilde{F}_{1A} + \tilde{F}_{1B}$ and $\tilde{\Gamma}_1 = \tilde{\Gamma}_{1A} + \tilde{\Gamma}_{1B}$.

Following this method, inertia attachments at the nodes can be accounted for by the insertion of the appropriate terms. The matrix is then assembled/processed by taking into account the boundary conditions and any absence of force/moment excitations at intermediate nodes.

4.4.1.1 Example: Direction-Fixed Base and Direction-Fixed Tip

As an example, one can consider modifying the system in Fig. 4.8 by fixing the direction of its right end, segmenting its electrodes into two halves and connecting equal external impedances Z to each segment as shown in Fig. 4.12. The tip constraint would create a point of maximum slope at the midpoint P, thereby increasing the voltage generated (from Eq. (4.28)). The arrangement in Fig. 4.12 can be realised in practice through a symmetric set-up comprising a beam of length $2l$ direction-fixed at both ends with a central mass of $2\hat{M}_T$, electrodes segmented into four equal parts, and an external impedance of Z connected to each segment as shown in Fig. 4.12. With reference to Fig. 4.12, the 6×6 global dynamic stiffness matrix $\mathbf{D_g}$ is assembled as in Eq. (4.78). By inverting this matrix equation and setting $\tilde{F}_1, \tilde{\Gamma}_1, \tilde{F}_2, \tilde{\theta}_0$ and $\tilde{\theta}_2$ to zero, one can express $\tilde{F}_0, \tilde{\Gamma}_0$ and $\tilde{\Gamma}_2$ in terms of $\tilde{u}_0$. The

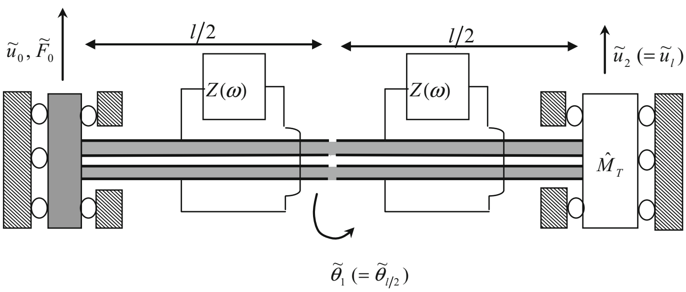


Fig. 4.12 Direction-fixed base/direction-fixed tip energy harvester having segmented electrodes [3]

maximum slope $\tilde{\theta}_1$ can then be expressed in terms of $\tilde{u}_0$, enabling the calculation of the voltage FRF from Eq. (4.28).

The performance of this direction-fixed tip system (Fig. 4.12) was compared with that of the previous free-tip system (Fig. 4.8) for the same bimorph parameters shown in Table 4.1 and a series piezoconnection for each electrode segment. The structural and ambient damping parameters (A and c_a) were the same for both systems. The tip mass $\hat{M}_T$ of the direction-fixed tip system was raised to $2.5785m_b$ (from $0.5m_b$) so that both systems were tuned to the same frequency under short-circuit conditions. Figures 4.13a, b show the variation with resistive load of the value of the voltage FRF (per electrode segment) at the first resonance and the corresponding specific mean power. It is clear that the voltage generated by one electrode of length *l/2* of the direction-fixed tip system is significantly greater than that produced by one electrode of length l of the free-tip system. Consequently, the

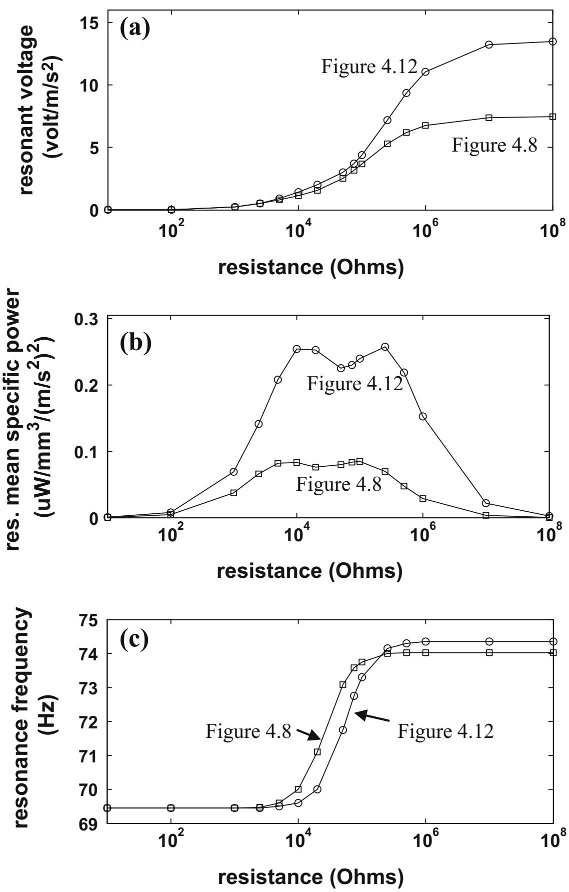


Fig. 4.13 Comparison of systems in Figs. 4.12 and 4.8 over a range of resistive loads (Z): **a** voltage amplitude at first resonance; **b** mean specific power at first resonance; **c** first resonant frequency [3]

mean power per unit piezovolume is much greater. Figure 4.13c shows that the tuned frequency range of the direction-fixed tip system is marginally greater than that of the free-tip system.

4.4.2 Other Considerations

A piezoelectric beam element can be incorporated as a member of a two-/three-dimensional assembly if the matrix in Eq. (4.32) is expanded to include longitudinal vibration. Since the format of the elemental dynamic stiffness matrix is similar to an FE matrix, it can be incorporated into the FE model of a complex structure and used in frequency domain analysis. It is noted that the above-described matrix assembly procedure assumes that the individual piezoelectric segments are not electrically connected. The above-described assembly process would need to be modified to accommodate electrically interconnected segments.

It is noted that the FRFs derived by DSM (or any other method) can be used in problems where the base-excitation u_0 is non-harmonic. In such cases, the FRFs define the ratio of Fourier transforms of the respective output to input (for deterministic excitation) or the ratio of the cross/auto power spectra (for random excitation). Hence, solution is achievable by appropriate transformation between the time and frequency domains. It is noted by Adhikari et al. [11] that in many applications, the ambient vibration is random and broadband. The accurate solution provided by DSM over a wide frequency range makes it more useful in such applications than single degree of freedom or single-mode approximations. One limitation of the DSM is that it assumes a linear impedance. Hence, nonlinear elements used in AC–DC rectification cannot be accommodated by the DSM in its present form.

4.5 Conclusions

This chapter presented two alternative modelling techniques for energy harvesting beams and used these techniques in a theoretical study of a bimorph. One of the methods was an application of the dynamic stiffness method (DSM) to the modelling of energy harvesting beams. This method was based on the exact solution of the wave equation and so obviated the need for modal transformation. The other method was a much-needed reformulation of AMAM that condensed the analysis to encompass all previously analysed systems. The AMAM is restricted to uniform-section cantilevered systems, but the DSM could be used in the modelling of beams with arbitrary boundary conditions or assemblies of beams. The Euler-Bernoulli model with piezoelectric coupling was used, and the external electrical load was represented by generic linear impedance. Simulations verified that, with a sufficient number of modes included, the AMAM result converged to the DSM result. A theoretical study of a bimorph quantified the tuning range of its resonance frequencies under variable impedance. It was demonstrated that tuning range was the same for both series- and parallel-connected layers. It was also shown that the electrical effect can only be regarded as equivalent to viscous damping if the electrical load is small and purely resistive. The neutralising effect of a tuned harvester on the vibration at its base was investigated using DSM, and the findings suggest the use of variable capacitance for the dual function of adaptive vibration neutralisation/energy harvesting. The application of DSM to more complex systems was illustrated. For the case studied, a significant increase in the power generated was achieved for a given working frequency through the application of a tip rotational restraint, the use of segmented electrodes and a resized tip mass

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Chapter 5
A Theoretical Analysis of an 'Electromechanical' Beam Tuned Mass Damper

5.1 Background

In the previous chapter, it was observed that energy harvesting piezoelectric beams are, in many practical applications, absorbers of the vibration at their base. This chapter exploits this principle in order to develop a novel type of vibration absorber [1]. The usefulness of this absorber is best appreciated following an outline on the basic principles of the tuned vibration absorber (TVA).

As mentioned in Sect. 2.3.3, a TVA is typically an auxiliary system whose parameters can be tuned to attenuate the vibration of a host structure. As illustrated by Von Flotow et al. [2], the auxiliary system can be represented by an equivalent spring-mass-damper system also known as 'mechanical' TVA. It is shown in [3] that any arbitrary auxiliary structure can be precisely represented by an equivalent two-degree-of-freedom (2-DOF) model as shown in Fig. 5.1b; the effective part of that 2-DOF system is a spring-mass-damper system. The mass m_a of the auxiliary structure (2-DOF) is divided into two components, an effective mass $m_{a,eff}$ and a redundant mass $m_{a,red}$ as shown in Fig. 5.1b. The component of the redundant mass is simply added to the host structure and does not contribute any inertia towards the tuning of the auxiliary structure as TVA. Typically, the tuned frequency ω_a of the TVA is defined as its first undamped natural frequency with its base fixed. The mechanical TVA suppresses the vibration of the host structure at its point of attachment through the application of an opposite interface force ($F(t)$ as shown in Fig. 5.1 [3]).

As discussed in Sect. 2.3.3, a TVA can be used in two discrete ways, both resulting in diverse optimal tuning criteria and design requirements:

(a) It can be tuned to fully suppress the vibration amplitude of a system at a specific troublesome excitation frequency, in which case it acts as a notch filter.
(b) It can be tuned to attenuate the modal contribution from a specific troublesome natural frequency Ω_s of the host structure over a wide range of excitation frequencies.

S. Rafique, *Piezoelectric Vibration Energy Harvesting*,
https://doi.org/10.1007/978-3-319-69442-9_5

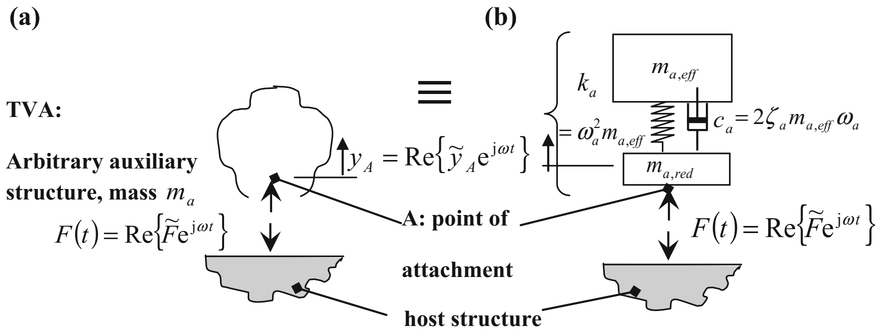

Fig. 5.1 **a** Actual arbitrary TVA, **b** Equivalent two-degree-of-freedom model of TVA [1]

When used in application (a), a mechanical TVA is referred to as a tuned vibration neutraliser (or undamped TVA) and the TVA is tuned to the excitation frequency; i.e., the condition $\omega_a = \omega$ defines optimal tuning [1]. In this condition, the neutraliser plants an anti-resonance at its point of attachment to the host structure, and the vibration of the host structure is suppressed over a very small bandwidth centred around ω_a. In the absence of damping in the neutraliser, a total attenuation of the vibration of the host structure is achieved. The performance of the neutralizer degrades as the damping is increased. In practice, a neutraliser can be easily developed in the form of a simple beam-like structure [4], as illustrated in Fig. 2.14. Such beam-like configurations permit adaptability and easy retuning of the device under variable conditions through the modification in the effective beam cross section or beam span [1]. The detailed procedure illustrating the derivation of the equivalent two-degree-of-freedom model of beam structures as shown in Fig. 2.14 is presented in [4].

As mentioned above, when used in application (b), a mechanical TVA is referred to as a tuned mass damper (TMD) and for this case ω_a is tuned to a frequency that is slightly lower than that of the targeted vibration mode Ω_s [1]. Neglecting the redundant mass of the TVA and any damping in the original host structure, the optimal tuning condition is given by [1, 5]:

$$\frac{\omega_a}{\Omega_s} = \frac{1}{1+\mu} \tag{5.1}$$

$$\mu = m_{a,eff}/M_A^{(s)} \tag{5.2}$$

The ratio of the effective mass of the TVA to the mass related to the host structure's targeted mode is denoted by μ. As illustrated in Fig. 5.1(b), a TMD needs optimal amount of damping in order to suppress the contribution of the targeted mode of the host structure, at the point of attachment, over a wide band of excitation frequencies. The exact amount of viscous damping ratio needed to optimally tune the TVA can be determined by [5]:

$$\xi_{opt} = \sqrt{\frac{3\mu}{8(1+\mu)^3}} \tag{5.3}$$

At the tuned conditions, the TVA suppresses the response of the targeted resonance peak of the original host structure, and the single peak response is replaced by two damped resonance peaks whose vibration levels are approximately inversely proportional to $\sqrt{\mu}$ separated by a shallow trough at $\omega_a = \omega$ [1]. The use of TVAs is an effective way of suppressing the response of the troublesome vibration mode; however, it is extremely difficult to provide correct amount of damping in conventional TVAs. Moreover, once the damping level is implemented in conventional TVA, it is difficult to adjust the response to a variation in the system parameters [1]. Additionally, the requirement of exact damping means that simple beam-like designs as shown in Fig. 2.14 are not easy to realise for TMDs.

As demonstrated by Von Flotow [2] that a TVA can also be designed and developed by other physics. The most pertinent in this context is the 'electrical' TVA [6], which is generally used for application (b) as discussed above. In such a device, the function of the auxiliary structure is provided by a piezoelectric shunt circuit. As shown in Fig. 2.15 of Chap. 2, a piezoelectric patch is directly bonded to the vibrating host structure and is connected across an external inductor-resistor circuit to generate the effect similar to the conventional TVA [6].

Typically, a piezoelectric patch is employed to convert the vibration energy of the host structure into electrical energy and produces a capacitor effect into the circuit, turning it into an R-L-C circuit [1]. The generated electrical energy is then dissipated most efficiently in the attached resistor R as Joule heating when the electrical resonance produced by the inductor-capacitor components is in the vicinity of the frequency of the targeted mode. The optimal resistance value in the R-L-C circuit ensures that the contribution of the targeted mode at a chosen location is attenuated over a wide band of excitation frequencies [1].

The electrical TVA has the following distinct advantages relative to conventional mechanical TVA: (a) performance is relatively less temperature dependent; (b) compact and durable; (c) level of damping is easily controllable to adapt to optimum vibration suppression [6–9]. However, the theoretical analysis needed to determine the optimal design parameters of the electrical TVA is not tractable to complex generic host structures. Such analysis involves development of the coupled electromechanical equations of the host structure with the attached shunted piezoelectric patch and their transformation to modal space, in the similar approach as the AMAM analysis of Chap. 3. The transfer function of the modal vibration of the TVA-host structure at a specified position could then be extracted and optimised. Owing to these modelling complexities, the application of electrical TVA has only been demonstrated for simple cantilever like host structure beams as shown in Fig. 2.15. On the other hand, the classical theory of the conventional

TMD is easily applicable to host structure of any complexity since the only host structure parameters it requires are the frequency Ω_s and mass $M_A^{(s)}$ of the targeted mode [1].

5.1.1 Distinct Aspects of This Chapter's Work

In Chap. 4, it was demonstrated that piezoelectric energy harvesting beams can also be used as *mechanical* absorbers to suppress the vibration of the host structure at the point of attachment. In Chap. 4 (Sect. 4.3.4), the concept of an adaptive dual vibration neutraliser/energy harvester beam with variable capacitance load was investigated. This chapter presents the theoretical analysis of the concept of a dual EH/TMD beam device, wherein a suitably shunted energy harvesting beam is used as a TMD on an arbitrary structure. The required level of optimum damping is provided by the electrical energy harvesting effect of the vibrating TMD as in Fig. 5.2. The proposed 'electromechanical' TVA or 'electromechanical TMD' combines the advantages of classical mechanical TMD and its electrical analogue (the piezoelectric shunt circuit), reducing their relative disadvantages [1]:

- It mimics conventional TMD theory, thereby making it tractable to generic host structures
- It keeps the advantages of the electrical TVA, predominantly with regard to the accurate application and adjustment of the required amount of damping
- It has the beam-like design that is widely used by adaptive vibration absorbers as of Fig. 2.14; apart from being simple and compact, such a design helps in retuning of the device as needed [1].

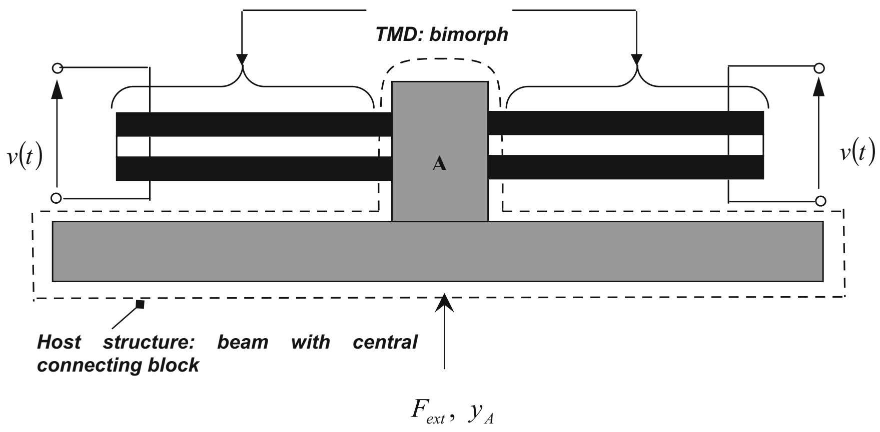

Fig. 5.2 A proposed electromechanical tuned mass damper [1]

The AMAM piezoelectric beam model developed in Chap. 3 and the DSM model of Sect. 4.2.3 are used in the analysis. An equivalent lumped-parameter model of a beam-like piezoelectric TMD under short-circuited conditions is shown in Fig. 5.2; the terminals short circuited such that $v(t) \equiv 0$, removing the electric coupling effect, is first derived and validated. This condition allows the calculation of the optimal amount of viscous damping, needed according to the classical theory, for a given application. This hypothetical optimally damped electrically uncoupled system is then used as a benchmark to calculate the optimal R-L-C parameters in the actual (electrically coupled) TMD using optimisation program written in MATLAB [1]. The final result is also verified by the dynamic stiffness method (DSM) technique developed in Chap. 4. As shown in Fig. 5.2, the host structure in this study is selected to be a free-free beam with an attachment block in the middle and the TMD is targeted at dampening its first flexural mode [1]. It is further stated that the theory developed in this chapter can be applicable to any arbitrary host structure.

In addition to the advantages mentioned above, the proposed electromechanical TVA retains the potential of energy storage through the use of an AC-DC rectification device. However, such nonlinear analysis is beyond the scope of this book. The mathematical derivation of a dual function energy harvesting TVA presented in this chapter follows the model [1].

This chapter is structured as follows. The theory is presented in Sect. 5.2. Section 5.2.1 presents the derivation of the benchmark hypothetical optimally damped mechanical system. Section 5.2.2 and 5.2.3 presents the theory for the electrically coupled TMD with different circuit configurations. Section 5.3 presents and discusses the results of simulations, which are verified by the DSM.

5.2 Theory

The aim of the study is to suppress the targeted resonance peak of the frequency response function (FRF or receptance) connecting the response at the attachment point y_A to the external excitation F_{ext} on the host structure, i.e. $\left| r_{y_A F_{ext}}^{host+TMD}(\omega) \right| << \left| r_{y_A F_{ext}}^{host}(\omega) \right|$ for the range of excitation frequencies ω over which the targeted host structure mode is dominant, where $r_{y_A F_{ext}}^{host+TMD}(\omega)$, $r_{y_A F_{ext}}^{host}(\omega)$ are defined as the complex ratio $\tilde{y}_A / \tilde{F}_{ext}$, with and without the TMD attached, respectively, for harmonic functions $y_A = \mathrm{Re}\{\tilde{y}_A e^{j\omega t}\}$, $F_{ext} = \mathrm{Re}\{\tilde{F}_{ext} e^{j\omega t}\}$ [1]. Figure 5.2 demonstrates the boundaries of the TMD and the host structure. The receptance of the system can be written as

$$r_{y_A F_{ext}}^{host+TMD}(\omega) = \frac{r_{y_A F_{ext}}^{host}(\omega)}{1 + r_{y_A F_{ext}}^{host}(\omega) \Big/ r_{y_A F}^{TMD}(\omega)} \tag{5.4}$$

By ignoring the damping in the original host structure, its receptance can be written as [10]:

$$r_{y_A F_{ext}}^{host}(\omega) = \sum_{s=1}^{\infty} \frac{\left\{ \overset{\frown}{\phi}{}_A^{(s)} \right\}^2}{\Omega_s^2 - \omega^2} \tag{5.5}$$

where Ω_s is the circular frequency of the sth mode and $\overset{\frown}{\phi}{}_A^{(s)}$ the corresponding mass-normalised mode-shape at the degree of freedom being targeted (the vertical displacement at A in Fig. 5.1). The targeted mode of the original host structure is its first flexural mode, of frequency Ω_2, since the first mode is the rigid body translation mode (i.e. $\Omega_1 = 0$) [1]. In Eq. (5.4), $r_{y_A F}^{TMD}(\omega) = \tilde{y}_A / \tilde{F}$ is the attachment point receptance of the TMD, and F is the force at the interface between the TMD and the host structure (Fig. 5.1). With reference to Eq. (5.4), the appropriate adjustment of $r_{y_A F}^{TMD}(\omega)$ (through optimisation of the TMD) allows the suppression of targeted resonance peak of $r_{y_A F_{ext}}^{host+TMD}(\omega)$ in Eq. (5.4).

5.2.1 Benchmark Model and Its Validation

The classical theory of Den Hartog [5] is used to derive the relation to calculate the optimal level of hypothetical viscous damping that will be used as the benchmark for the performance of the electromechanical TMD [1].

5.2.1.1 Optimal Damping Calculation

The contribution of the modal parameters (modal mass and modal stiffness) of the targeted mode of the host structure can be determined using standard modal theory [10] and can be represented as mass $M_A^{(2)} = 1 \Big/ \left\{ \overset{\frown}{\phi}{}_A^{(2)} \right\}^2$ and stiffness $K_A^{(2)} = \Omega_s^2 M_A^{(2)}$ as shown in Fig. 5.5(a). The dynamics of the original host structure at the targeted degree of freedom can be precisely modelled in this form for excitation frequencies ω in the vicinity of Ω_2, where the targeted mode is dominant (see Eq. (5.5)). The optimal tuning Eq. (5.1), by Den Hartog [5], neglects the redundant mass of the absorber as shown in Fig. 5.1(b). In the present analysis, the tuning condition in Eq. (5.1) is corrected to account for $m_{a,red}$, and addition of the equivalent lumped-parameter TMD model in Fig. 5.1(b) to the system in Fig. 5.3(a) results in the system in Fig. 5.3(b) [1].

The harmonic analysis by Den Hartog [5] can then be applied to the system in Fig. 5.3(b). The corrected optimal tuning condition is written as:

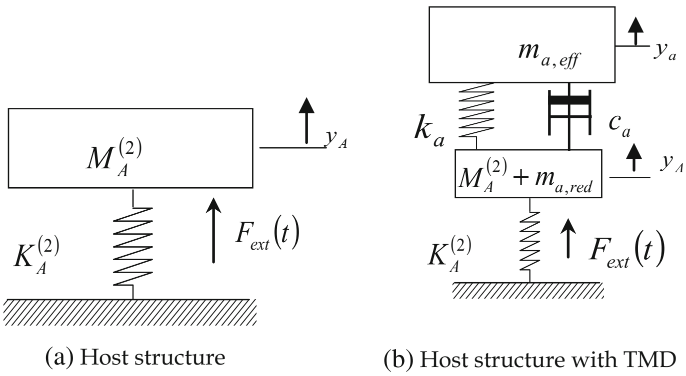

Fig. 5.3 Dynamic 'modal' model of the host structure **a** without, **b** with TMD for frequencies ω in the surrounding of Ω_2[1]

$$\frac{\omega_{a_{opt}}}{\Omega_2'} = \frac{1}{1+\mu} \tag{5.6}$$

$$\Omega_2' = \Omega_2 \sqrt{M_A^{(2)} \Big/ \left\{ M_A^{(2)} + m_{a,red} \right\}} \tag{5.7}$$

and the corrected μ is:

$$\mu = m_{a,eff} \Big/ \left\{ M_A^{(2)} + m_{a,red} \right\} \tag{5.8}$$

The optimal value of the viscous damping ratio is then given by Eq. (5.3). The optimal value of the viscous damping coefficient is then calculated by using optimal values for ζ_a and ω_a [1]:

$$c_a = 2\zeta_a m_{a,eff} \omega_a \tag{5.9}$$

Figure 5.4 shows the host structure attached to the two-degree-of-freedom model of the electrically uncoupled TVA beams (that were shown in Fig. 5.2). The expression for $r_{y_A F}^{TMD}(\omega)$ to use within Eq. (5.4) for this case is denoted by $\left\{ r_{y_A F}^{TMD}(\omega) \right\}_{uncoup}^{2-DOF}$. From elementary harmonic analysis:

$$\left\{ r_{y_A F}^{TMD}(\omega) \right\}_{uncoup}^{2-DOF} = \frac{-m_{a,eff}\omega^2 + k_a + \mathrm{j}\omega c_a}{-m_{a,eff}\omega^2 (k_a + \mathrm{j}\omega c_a) - m_{a,red}(-m_{a,eff}\omega^2 + k_a + \mathrm{j}\omega c_a)} \tag{5.10}$$

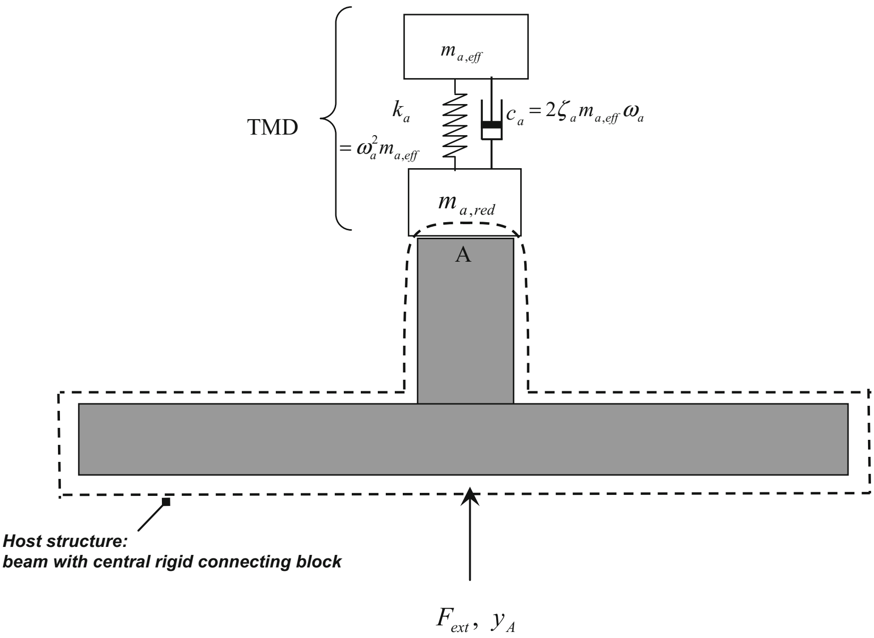

Fig. 5.4 Equivalent 2-DOF TMD attached to host structure (electrically uncoupled) [1]

where

$$k_a = m_{a,eff}\omega_a^2, m_{a,eff} = \bar{R}m_a, m_{a,red} = (1 - \bar{R})m_a \tag{5.11a–c}$$

m_a is the total mass of the two overhanging portions of the bimorphs of the TMD. Following the work in [3], for an overhanging beam without tip mass, the fraction of effective mass can be expressed as:

$$\bar{R} = 60.49\% \tag{5.12}$$

The benchmark performance is acquired by solving Eq. (5.4) for $r_{y_A F}^{TMD}(\omega)$ calculated from Eq. (5.10) with optimal values used for ω_a and c_a, as per Eqs. (5.3), (5.9) and (5.11a–c).

5.2.1.2 Verification of Benchmark Model

The theory presented above is compared and validated using the equivalent system of Fig. 5.4 against the proposed electromechanical system of Fig. 5.2, for electrically uncoupled conditions and having the same damping values.

As presented in Figs. 5.4 and 5.1(b), the interface force $F(t)$ between the TMD and the host structure is the sum of the shear forces at the clamped end of each overhanging portion of the TMD beams [1]:

$$F(t) = 2\frac{\partial M}{\partial x}\bigg|_{x=0} \tag{5.13}$$

where M is the bending moment at position x along the length of beam at time t. This is given by Eq. (4.9), which is reproduced here:

$$M(x,t) = B\frac{\partial^2 u}{\partial x^2} + A\frac{\partial^3 u}{\partial x^2 \partial t} + \vartheta\, v(t) \tag{5.14}$$

where $u(x,t)$ is the absolute displacement at location x at time t, B and A are the bending stiffness and damping constants of a composite piezoelectric beam, given in Eqs. (3.3) and (3.4), and ϑ is the electromechanical coupling parameter given by Eq. (4.12) [1].

Hence, substituting Eq. (5.14) in (5.13)

$$F(t) = 2\left\{B\frac{\partial^3 u}{\partial x^3} + A\frac{\partial^4 u}{\partial x^3 \partial t}\right\}\bigg|_{x=0} \tag{5.15}$$

Ignoring any rotational motion effects at the attachment point A:

$$u = y_A(t) + \sum_{r=1}^{N} \eta_r(t)\phi_r(x) \tag{5.16}$$

where the series summation defines clamp-free cantilever-type flexure of either overhanging portion and is expressed as a modal series, and $\phi_r(x)$ defines mass-normalised modes illustrating undamped free vibration of either of the electrically uncoupled beams with the base fixed in both translation and rotation ('clamped-free'). The η_rs define the modal coordinates [1]. The mode shapes $\phi_r(x)$ are given by:

$$\phi_r(x) = \frac{1}{\sqrt{ml}}[\cosh(\lambda_r/l)x - \cos(\lambda_r/l)x - \sigma_r\{\sinh(\lambda_r/l)x - \sin(\lambda_r/l)x\}] \tag{5.17}$$

$$\sigma_r = \frac{\sinh\lambda_r - \sin\lambda_r}{\cosh\lambda_r + \cos\lambda_r} \tag{5.18}$$

where m is the mass per unit length and λ_r's are the roots of the equation:

$$1 + \cos\lambda_r \cosh\lambda_r = 0 \tag{5.19}$$

The corresponding natural frequencies ω_r are given by:

$$\omega_r = (\lambda_r/l)^2\sqrt{(B/m)} \tag{5.20}$$

Substituting Eq. (5.16) into Eq. (5.15) and assuming harmonic vibration $\eta_r = \mathrm{Re}\{\tilde{\eta}_r e^{j\omega t}\}$, the complex amplitude of the interface force F is given by:

$$\tilde{F} = 2B\left(1 + \frac{j\omega A}{B}\right)\sum_{r=1}^{N}\tilde{\eta}_r\phi_r'''(0) \tag{5.21}$$

By rearranging Eq. (5.21), the expression for $r_{y_AF}^{TMD}(\omega)$ can be written as:

$$r_{y_AF}^{TMD}(\omega) = \frac{\tilde{y}_A}{\tilde{F}} = \frac{1}{2B\left(1+\frac{j\omega A}{B}\right)\sum_{r=1}^{N}\frac{\tilde{\eta}_r}{\tilde{y}_A}\phi_r'''(0)} \tag{5.22}$$

It is noted that Eqs. (5.15), (5.21) and (5.22) apply for any voltage $v(t)$ across the piezo. Specialising the study of this section for the case of no electrical coupling (i.e. the electrode terminals short circuited, $v(t) = 0$), then from Eq. (3.26) or (4.61), in the short-circuited condition:

$$\frac{\tilde{\eta}_r}{\tilde{y}_A} = \frac{m\omega^2\gamma_r^u}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{5.23}$$

where

$$\gamma_r^u = \int_{x=0}^{l}\phi_r(x)dx = \frac{2\sigma_r}{\lambda_r}\sqrt{\frac{l}{m}} \tag{5.24}$$

and, from Eqs. (4.76), (4.77)

$$A = \frac{2B}{\omega_1}\zeta_1, \quad \zeta_r = \zeta_1\frac{\omega_r}{\omega_1} \tag{5.25a, b}$$

The expression for $r_{y_AF}^{TMD}(\omega)$ to use within Eq. (5.4) for this case is denoted by $\left\{r_{y_AF}^{TMD}(\omega)\right\}_{uncoup}$ and is obtained by substituting Eq. (5.23) into Eq. (5.22):

$$\left\{r_{y_AF}^{TMD}(\omega)\right\}_{uncoup} = \frac{\tilde{y}_A}{\tilde{F}} = \frac{1}{2Bm\omega^2\left(1+\frac{j\omega A}{B}\right)\sum_{r=1}^{N}\frac{\phi_r'''(0)\gamma_r^u}{\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega}} \tag{5.26}$$

Table 5.1 Modal parameters of host structure [11]

$\Omega_1/(2\pi)$ (Hz)	0	$M_A^{(1)} = 1\Big/\left\{\widehat{\phi}_A^{(1)}\right\}^2$ (kg)	0.278
$\Omega_2/(2\pi)$ (Hz)	127.7	$M_A^{(2)} = 1\Big/\left\{\widehat{\phi}_A^{(2)}\right\}^2$ (kg)	0.469

It is noted from Eq. (5.23) and Eqs. (5.25a, b) that the effect of ambient (air) damping has been neglected. As discussed in Sect. 4.3.5.2, the air damping contributes to both system damping and system excitation. The latter effect is negligible, as discussed in Sect. 4.3.5.2. Moreover, in Sect. 4.3.5.2, it was shown that, for a given experimentally determined value of ζ_1, the omission of the ambient damping only affects the FRF levels at the resonances of the higher clamped-free modes [1]. By definition, for the short-circuited TMD bimorph absorber, the tuned frequency ω_a and absorber damping ζ_a (Fig. 5.1(b)):

$$\omega_a \equiv \omega_1, \zeta_a \equiv \zeta_1 \tag{5.27a, b}$$

i.e. the bimorph TMD is designed to be effective around its first clamped-free mode. Hence, the neglect of air damping throughout the analysis is justified.

The theory in this section is verified by considering the test case whose parameters are given in Tables 5.1 and 5.2. The host structure parameters in Table 5.1 were obtained experimentally. It is noted that the accuracy of modal parameters of Table 5.1 is illustrated later in Chap. 6 (Fig. 6.3). The electrical parameters in Table 5.2 are used later in Sect. 5.3.

From Eqs. (5.8), (5.11a–c), the mass ratio was calculated as $\mu = 1.86\%$. Also, from Eqs. (5.27a, b), (5.20), (5.6) and (5.7), it is evident that ω_1 (i.e. ω_a) $\approx \omega_{a_{opt}}$; i.e., the electrically uncoupled system is approximately optimally tuned. The optimal damping ratio $\zeta_{a_{opt}}(= 8.24\%)$ is calculated using Eq. (5.3) and the optimal

Table 5.2 Parameters of either beam of the electromechanical TMD [12]

Property	Units	Value
Overhanging length of the beam, l	mm	58.75
Width of the beam, b	mm	25
Thickness of each piezoelectric layer, h_p (upper & lower layers)	mm	0.267
Thickness of the shim (substrate), h_{sh}	mm	0.285
Young's Modulus of the Piezoelectric, Y_p	GPa	66
Young's Modulus of the shim, Y_{sh}	GPa	72
Density of the piezoelectric material	Kg/m^3	7800
Density of the shim material	Kg/m^3	2700
Piezoelectric constant, d_{31}	pm/volt	−190
Relative dielectric constant (at constant stress)		1800

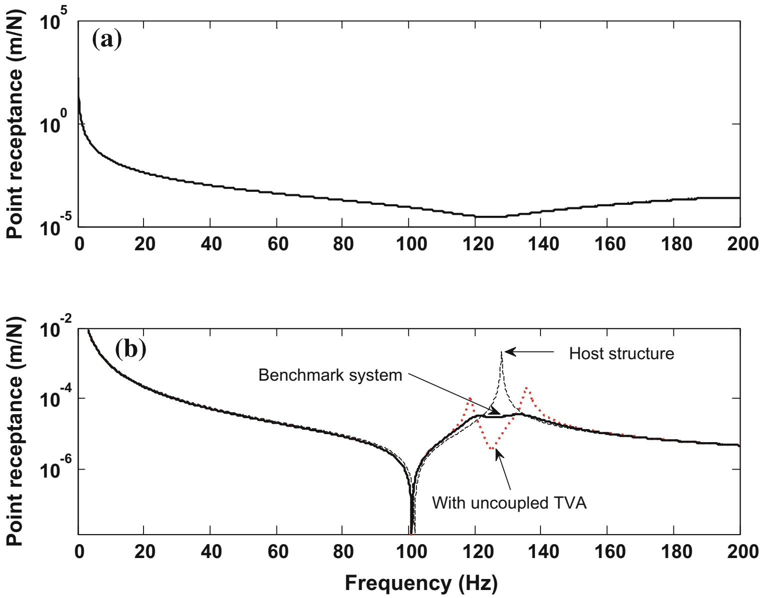

Fig. 5.5 **a** Receptances of the TVAs alone **b** benchmark model (thick solid line); optimum by AMAM model (thick black dashed line) host without TMD (thin black dashed line); host structure with uncoupled TVA (red dotted line)

value of the damping coefficient then computed according to Eq. (5.9) [1]. The thick solid black curve in Fig. 5.5(a) shows the equivalent lumped-parameter TMD receptance $\left\{r^{TMD}_{y_A F}(\omega)\right\}^{2-DOF}_{uncoup}$ with optimal parameters, denoted by $\left\{r^{TMD}_{y_A F}(\omega)\right\}^{2-DOF}_{uncoup,opt}$. This receptance was used in Eq. (5.4), along with the unmodified host structure receptance shown by thin dash black line in Fig. 5.5(b) (calculated using Eq. (5.5)), to yield the resulting modified host structure receptance, shown as the thick solid black line in Fig. 5.5(b).

The thick black dashed line in Fig. 5.5(a) shows the (exact) distributed parameter TMD receptance for the same level of damping, denoted by $\left\{r^{TMD}_{y_A F}(\omega)\right\}_{uncoup,opt}$. In this case, Eq. (5.26) was used, with ζ_1 (i.e. ζ_a) set to $\zeta_{a_{opt}}$ and the values of A and ζ_r computed accordingly, as per Eqs. (5.25a, b). It is noted that, due to series summation in the denominator of Eq. (5.26), a suitably large number of modes N has to be taken to ensure convergence ($N = 300$ in this study) [1]. The thick black dashed line in Fig. 5.5(b) shows the corresponding modified host structure receptance. The excellent agreement between the thick black dashed and solid lines in Figs. 5.5(a) and 5.5 (b) verifies the benchmark model (the thick black dashed line is not visible in Fig. 5.5(a,b) due to the excellence of the agreement).

It is important to note that the above derived optimal level of viscous damping ($\zeta_1 = \zeta_{a_{opt}}$) in the TMD is of course fictitious and not physical, i.e. in reality ζ_1 (i.e. ζ_a) $\neq \zeta_{a_{opt}}$. In fact, the actual value of ζ_1 was found by experiment to be only 1%, whereas $\zeta_{a_{opt}} = 8.24\%$. With this value of ζ_1 and the actual values of A and ζ_r computed accordingly (as per Eqs. (5.25a, b)), the actual modified host structure receptance for the short-circuited TMD is given by the red dash line in Fig. 5.5(b) [1]. However, as will be shown in the remainder of the chapter, through the use of appropriately tuned circuitry across the terminals in Fig. 5.2, the electromechanical TMD is capable of producing a frequency response that closely mimics the benchmark response given by the solid line in Fig. 5.5(b) [1].

5.2.2 Derivation of Coupled Electromechanical Receptance of TMD by AMAM

In this section, the analytical modal analysis method (AMAM) is used to derive the equations for the TMD receptance $r_{y_A F}^{TMD}(\omega)$ of the proposed electromechanical TMD for each of the four different circuit configurations as shown in Figs. 5.8, 5.9, 5.10 and 5.11. The TMD receptances calculated for each of the four circuits can then be used in Eq. (5.4) to calculate the receptance of the host structure $r_{y_A F_{ext}}^{host+TMD}(\omega)$. Later on, in Sect. 5.3, it will be shown that the appropriate adjustment of $r_{y_A F}^{TMD}(\omega)$ through the use of optimised R-L-C (resistor, inductor and a capacitor) parameters allows the suppression of targeted resonance peak of $r_{y_A F_{ext}}^{host+TMD}(\omega)$ in Eq. (5.4).

The TMD receptance $r_{y_A F}^{TMD}(\omega)$ is given by Eq. (5.22), but the expression for $\tilde{\eta}_r/\tilde{y}_A$ in the denominator of this expression is different for each type of circuit. The following analysis derives $\tilde{\eta}_r/\tilde{y}_A$ for each case.

The value of complex amplitude of the modal coordinate for the electrically coupled case, as derived in Chap. 3, can be expressed as:

$$\tilde{\eta}_r = \frac{\left(m\omega^2\gamma_r^u\tilde{y}_A - \chi_r\tilde{v}\right)}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{5.28}$$

Hence, in order to derive the expression for $\tilde{\eta}_r/\tilde{y}_A$, the expression for $\tilde{v}$, the complex amplitude of the output voltage, needs to be derived by considering the relevant circuit equations.

Prior to analysing the circuits, some general electrical relations pertaining to the generated charge q and the current i are presented. The relation for charge q, generated by a vibrating piezo element, can be obtained by integrating the piezoelectric constitutive relation of Eq. (3.12), over the whole electrode area:

$$q = \frac{f}{a} C_p \left\{ \sum \Psi_r \eta_r - v(t) \right\} \tag{5.29}$$

where C_p is the capacitance of one layer of piezoelectric material, defined by Eq. (3.24) or (4.18); also, from Eqs. (4.7) and (4.19)

$$a = \begin{cases} 2 & \text{bimorph (piezos in series)} \\ 1 & \text{bimorph (piezos in parallel),unimorph} \end{cases}$$

$$f = \begin{cases} 1 & \text{bimorph (piezos in series),unimorph} \\ 2 & \text{bimorph (piezos in parallel)} \end{cases}$$

and Ψ_r is a constant that can be written as:

$$\Psi_r = -\frac{a d_{31} Y_p h_{pc} h_p}{\varepsilon_{33}^S l} \int_0^l \frac{d^2 \phi_r}{dx^2} dx = -\frac{a d_{31} Y_p h_{pc} h_p}{\varepsilon_{33}^S l} \frac{d\phi}{dx}\bigg|_0^l \tag{5.30}$$

The electric current from the piezo can then be written as:

$$i(t) = \frac{dq}{dt} = \frac{f\, C_p}{a} \left\{ \sum_{r=1}^{\infty} \Psi_r \dot{\eta}_r - \frac{dv}{dt} \right\} \tag{5.31}$$

From Eqs. (5.29), (5.31), it is seen that the equivalent internal capacitance of the bimorph is $(f/a)C_p$.

In the circuits of Figs. 5.6, 5.7, 5.8 and 5.9, for ease of modelling, the external capacitor C is taken as a multiple of the capacitance of one layer of piezoelectric material, i.e. $C = nC_p$. From Figs. 5.6, 5.7, 5.8 and 5.9, it is noted that the external

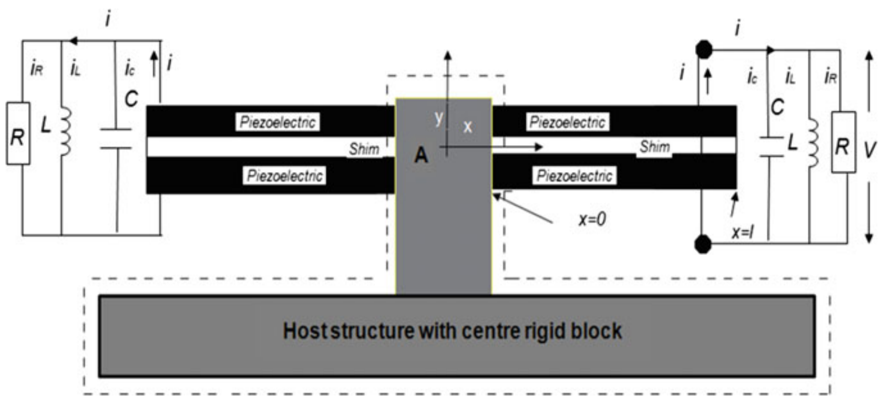

Fig. 5.6 A double circuit with parallel R-L-C circuit configuration

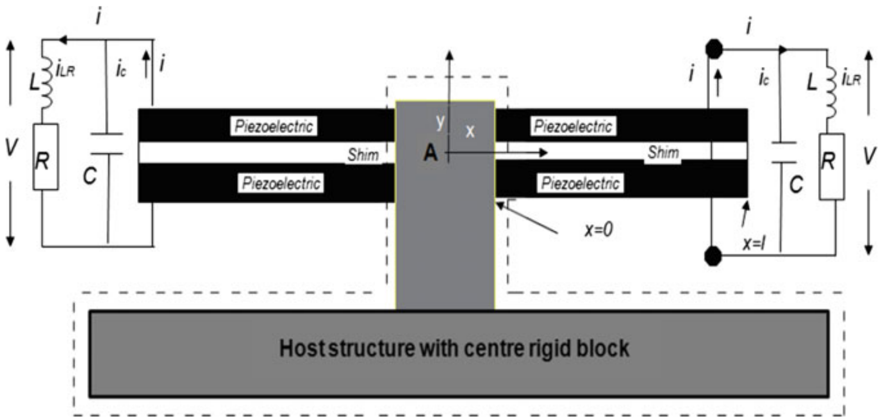

Fig. 5.7 A double circuit: parallel C and series R-L circuit configuration

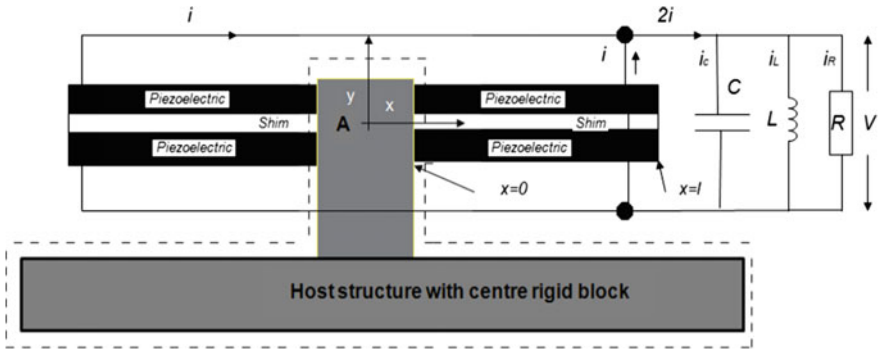

Fig. 5.8 A single circuit: parallel R-L-C circuit configuration

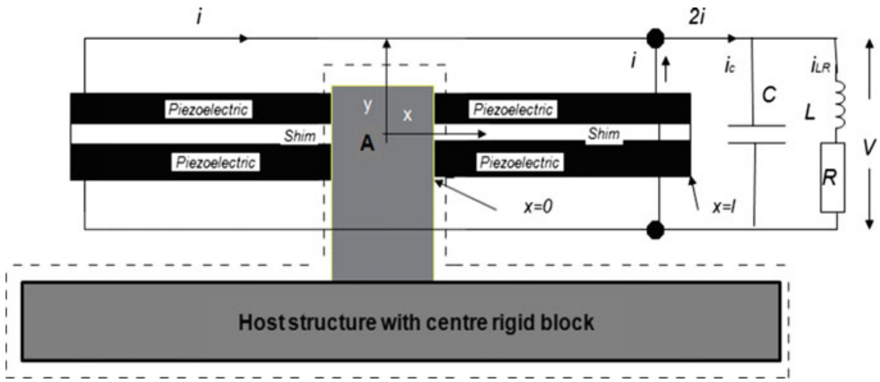

Fig. 5.9 A single circuit: parallel C and series R-L circuit configuration

capacitor C is always in parallel with one bimorph (Figs. 5.6, 5.7) or both bimorphs (Figs. 5.8, 5.9). The parallel connection of C and one or both bimorphs adds up their capacitances, resulting in a higher total capacitance of the system. It is found from the analysis of this chapter that the higher capacitance of the system reduces the size of the inductor L required to tune the circuit.

5.2.2.1 Double Circuit: Derivation for R-L-C in Parallel

In this case, each side of the EH beam is connected to a separate but symmetric R-L-C circuit as shown in Fig. 5.6. In this arrangement, all the (R-L-C) circuit parameters are parallel to each other, and the generated current i is now divided into three different paths, i.e. through capacitor, through inductor and through resistor. It can be observed in Fig. 5.6 that the voltage across the resistor, capacitor and inductor is same. However, the total current is the sum of all three current components and can be written as:

$$i = i_C + i_L + i_R \tag{5.32}$$

According to Ohm's law, the current in the resistor is $i_R = \frac{v(t)}{R}$, and the current through the capacitor is $i_C = C\frac{dv}{dt}$, and then the current through the inductor can be written as:

$$i_L = dq/dt - i_C - i_R \tag{5.33}$$

Substituting relations for i_C and i_R into Eq. (5.33) will yield:

$$i_L = \frac{dq(t)}{dt} - C\frac{dv(t)}{dt} - \frac{v(t)}{R} \tag{5.34}$$

Equation (5.34) represents the segment of the generated current i passing through the inductor. The voltage across the inductor is:

$$v(t) = L\frac{di_L}{dt} \tag{5.35}$$

Substituting Eq. (5.34) into Eq. (5.35) and using Eq. (5.29) will yield the following expression for the voltage generated across the terminals:

$$v(t) = L\frac{f}{a}C_p\left\{\sum \Psi_r \ddot{\eta}_r - \frac{d^2v(t)}{d^2t}\right\} - LC\frac{d^2v(t)}{dt^2} - \frac{L}{R}\frac{dv(t)}{dt} \tag{5.36}$$

Rewriting Eq. (5.36) in terms of complex amplitudes, putting $C = nC_p$ and rearranging yields:

$$\tilde{v} = \frac{-\frac{f}{a}\omega^2 \sum_{r=1}^{\infty} \Psi_r \tilde{\eta}_r}{\left[\frac{1}{C_pL} + \frac{j\omega}{C_pR} - \frac{f\omega^2}{a} - n\omega^2\right]} \tag{5.37}$$

Substituting $\tilde{\eta}_r$ from Eq. (5.28) into (5.37), rearranging and solving for $\tilde{v}$ will yield:

$$\tilde{v} = \frac{-\frac{f}{a} m\omega^4 \tilde{y}_A \sum_{r=1}^{\infty} \frac{\Psi_r \gamma_r^u}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}}{\left[\frac{1}{C_pL} + \frac{j\omega}{C_pR} - \frac{f\omega^2}{a} - n\omega^2\right] - \frac{\frac{f}{a}\omega^2 \sum_{r=1}^{\infty} \Psi_r \chi_r}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}} \tag{5.38}$$

By substituting Eq. (5.38) back into Eq. (5.28) and rearranging, one obtains:

$$\frac{\tilde{\eta}_r}{\tilde{y}_A} = \left(\gamma_r - \chi_r \frac{-\frac{f}{a}\omega^2 \sum_{r=1}^{\infty} \frac{\Psi_r \gamma_r^u}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}}{\left[\frac{1}{C_pL} + \frac{j\omega}{C_pR} - \frac{f\omega^2}{a} - n\omega^2\right] - \frac{\frac{f}{a}\omega^2 \sum_{r=1}^{\infty} \Psi_r \chi_r}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}} \right) \times \frac{m\omega^2}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{5.39}$$

The TMD receptance $r_{y_A F}^{TMD}(\omega)$ is then obtained by substituting this expression into the denominator of Eq. (5.22).

5.2.2.2 Double Circuit: Derivation for Parallel C and Series R-L

In this case, as in the previous one, each side of the EH beam is connected to a separate but symmetric R-L-C circuit. However, the resistor and inductor are now connected together in series (i.e. in a single branch) but in parallel with the capacitor, as shown in Fig. 5.7. The electric current dq/dt (i.e. i) generated by the vibrating piezo beam is then divided into two paths, through the external capacitor and through resistor-inductor branch.

The current through the resistor-inductor branch can now be written as:

$$i_{LR} = \frac{dq}{dt} - i_C \tag{5.40}$$

The voltage generated across the R-L branch can then be written as:

$$v(t) = i_{LR}R + L\frac{di_{LR}}{dt} \tag{5.41}$$

By substituting Eqs. (5.40) and (5.31) into Eq. (5.41) and putting $i_C = Cdv/dt$ will yield the following expression for the generated voltage:

$$v(t) = R\frac{f\,C_p}{a}\left\{\sum_{r=1}^{\infty}\Psi_r\dot{\eta}_r - \frac{dv}{dt}\right\} - RC\frac{dv(t)}{dt} + L\frac{f\,C_p}{a}\left\{\sum_{r=1}^{\infty}\Psi_r\ddot{\eta}_r - \frac{d^2v}{dt^2}\right\} - LC\frac{d^2v(t)}{dt^2} \tag{5.42}$$

Rewriting Eq. (5.42) in terms of complex amplitudes, putting $C = nC_p$ and rearranging yields:

$$\tilde{v} = \frac{\frac{f}{a}(j\omega R - \omega^2 L)\sum_{r=1}^{\infty}\Psi_r\tilde{\eta}_r}{\left\{\frac{1}{C_p} + j\omega R\left(\frac{f}{a}+n\right) - L\omega^2\left(\frac{f}{a}+n\right)\right\}} \tag{5.43}$$

Substituting $\tilde{\eta}_r$ from Eq. (5.28) into (5.43), rearranging and solving for $\tilde{v}$ will yield:

$$\tilde{v} = \frac{\frac{f}{a}m\omega^2\tilde{y}_A(j\omega R - \omega^2 L)\sum_{r=1}^{\infty}\frac{\left(\gamma_r^u\Psi_r\right)}{(\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega)}}{\left[\left\{\frac{1}{C_p} + j\omega R\left(\frac{f}{a}+n\right) - L\omega^2\left(\frac{f}{a}+n\right)\right\} + \frac{f}{a}(j\omega R - \omega^2 L)\sum_{r=1}^{\infty}\frac{\Psi_r\chi_r}{(\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega)}\right]} \tag{5.44}$$

By substituting Eq. (5.44) back into Eq. (5.28) and rearranging, one obtains:

$$\frac{\tilde{\eta}_r}{\tilde{y}_A} = \left\{\gamma_r^w - \chi_r\frac{\frac{f}{a}(j\omega R - \omega^2 L)\sum_{r=1}^{\infty}\frac{\left(\gamma_r^u\Psi_r\right)}{(\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega)}}{\left[\left\{\frac{1}{C_p} + j\omega R\left(\frac{f}{a}+n\right) - L\omega^2\left(\frac{f}{a}+n\right)\right\} + \frac{f}{a}(j\omega R - \omega^2 L)\sum_{r=1}^{\infty}\frac{\Psi_r\chi_r}{(\omega_r^2-\omega^2+j2\zeta_r\omega_r\omega)}\right]}\right\} \times \frac{m\omega^2}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{5.45}$$

The TMD receptance $r_{y_AF}^{TMD}(\omega)$ is then obtained by substituting this expression into the denominator of Eq. (5.22).

5.2.2.3 Single Circuit: Parallel R-L-C

In the previous two cases, both piezoelectric cantilever beams are connected to two separate but symmetric R-L-C circuits on either side as can be seen in Figs. 5.6 and 5.7. However, in this section and the following one, both bimorphs are first connected to each other in parallel, which adds their generated charge and internal capacitances, before being connected to an external R-L-C circuit as shown in Fig. 5.8 and Fig. 5.9. The practical advantages of these latter configurations are discussed in Sect. 5.3.3 and, in more detail, in Chap. 6, Sects. 6.3.1 and 6.3.5. The required voltage and modal mechanical response $\tilde{\eta}_r$ equations are derived in the same way as in the previous cases.

By assuming a perfect mechanical and electrical symmetry on both sides, the voltage and the modal response $\tilde{\eta}_r$ equations for this configuration can be obtained simply by replacing f by $2f$ in double-circuit Eqs. (5.38), (5.39), (5.44) and (5.45). This is evident from Fig. 5.8 or 5.9, where the currents generated by each piezo are added together before delivery to the external circuit.

For the parallel R-L-C single-circuit case, shown in Fig. 5.8, the equations for voltage $\tilde{v}$ and $\tilde{\eta}_r/\tilde{y}_A$ can then be obtained by replacing f by $2f$ in the double-circuit Eqs. (5.38), (5.39):

$$\tilde{v} = \frac{-\frac{2f}{a} m\omega^4 \tilde{y}_A \sum_{r=1}^{\infty} \frac{\Psi_r \gamma_r^u}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}}{\left[\frac{1}{C_p L} + \frac{j\omega}{C_p R} - \frac{2f\omega^2}{a} - n\omega^2\right] - \frac{\frac{2f}{a}\omega^2 \sum_{r=1}^{\infty} \Psi_r \chi_r}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}} \tag{5.46}$$

$$\frac{\tilde{\eta}_r}{\tilde{y}_A} = \left(\gamma_r - \chi_r \frac{-\frac{2f}{a}\omega^2 \sum_{r=1}^{\infty} \frac{\Psi_r \gamma_r^u}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}}{\left[\frac{1}{C_p L} + \frac{j\omega}{C_p R} - \frac{2f\omega^2}{a} - n\omega^2\right] - \frac{\frac{2f}{a}\omega^2 \sum_{r=1}^{\infty} \Psi_r \chi_r}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega}} \right) \times \frac{m\omega^2}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{5.47}$$

The TMD receptance $r_{y_A F}^{TMD}(\omega)$ is then obtained by substituting this expression into the denominator of Eq. (5.22).

5.2.2.4 Single Circuit: Parallel C and Series R-L

Figure 5.9 shows the configuration of a single circuit, with parallel C and series R-L connection.

The equations for voltage $\tilde{v}$ and $\tilde{\eta}_r/\tilde{y}_A$ can be obtained by replacing f by $2f$ in Eqs. (5.44) and (5.45), yielding:

$$\tilde{v} = \frac{\frac{2f}{a} m\omega^2 Y_0 (j\omega R - \omega^2 L) \sum_{r=1}^{\infty} \frac{(\gamma_r^u \Psi_r)}{(\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega)}}{\left[\left\{\frac{1}{C_p} + j\omega R\left(\frac{2f}{a} + n\right) - L\omega^2\left(\frac{2f}{a} + n\right)\right\} + \frac{2f}{a}(j\omega R - \omega^2 L) \sum_{r=1}^{\infty} \frac{\Psi_r \chi_r}{(\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega)}\right]} \tag{5.48}$$

$$\frac{\tilde{\eta}_r}{\tilde{y}_A} = \left\{ \gamma_r^w - \chi_r \frac{\frac{2f}{a}(j\omega R - \omega^2 L) \sum_{r=1}^{\infty} \frac{\left(\gamma_r^u \Psi_r\right)}{(\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega)}}{\left[\left\{ \frac{1}{C_p} + j\omega R\left(\frac{2f}{a} + n\right) - L\omega^2\left(\frac{2f}{a} + n\right) \right\} + \frac{2f}{a}(j\omega R - \omega^2 L) \sum_{r=1}^{\infty} \frac{\Psi_r \chi_r}{(\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega)} \right]} \right\} \times \frac{m\omega^2}{\omega_r^2 - \omega^2 + j2\zeta_r\omega_r\omega} \tag{5.49}$$

The TMD receptance $r_{y_A F}^{TMD}(\omega)$ is then obtained by substituting this expression into the denominator of Eq. (5.22).

5.2.3 *Derivation of Electromechanical TMD Receptance Using DSM*

In this section, the DSM is used as an alternative method to obtain the TMD receptance $r_{y_A F}^{TMD}(\omega)$ for the four circuit configurations shown in Fig. 5.6 to 5.9. This provides a means of independent verification of the AMAM analysis of Sect. 5.2.2. The procedure extends the novel concept analysed in Chap. 4 (Sect. 4.3.4) for the case of an external capacitor alone (i.e. a dual energy harvester/vibration neutraliser) to relatively complicated R-L-C circuits (resulting in a dual EH/TMD beam device).

By the DSM method, the TMD receptance $r_{y_A F}^{TMD}(\omega)$ can be obtained from Eq. (4.72) by setting $\tilde{u}_0 = \tilde{y}_A$ and $\tilde{F}_0 = \tilde{F}/2$ (in order to account for the two symmetric bimorphs constituting the TMD) and rearranging:

$$r_{y_A F}^{TMD}(\omega) = \frac{\tilde{y}_A}{\tilde{F}} = \frac{1}{2}\left(R_{11} - \frac{R_{12}R_{21}}{R_{22}}\right) \tag{5.50}$$

The receptance terms within the brackets are obtained by inverting the dynamic stiffness matrix of one of the TMD bimorph beams at each excitation frequency ω (see Eqs. (4.41), (4.40)). The dynamic stiffness matrix (Eq. (4.32), (4.40)) has a frequency-dependent electrical coupling term G, defined in Eqs. (4.28) and (4.29). This coupling term will be different for the different circuit configurations. The expression for G for each type of circuit is derived in the following subsections.

The TMD receptances calculated for each of the four circuits by Eq. (5.50) can then be used in Eq. (5.4) in lieu of the AMAM expressions of Sect. 5.2.2 to determine the modified receptance of the host structure $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for a given R-L-C configuration and setting.

5.2.3.1 DSM : Double Circuit, Parallel R-L-C Circuit

For the circuit shown in Fig. 5.6, expressing Eqs. (5.34) and (5.35) in terms of complex amplitudes and combining the resulting equations, the impedance Z of the external circuit connected to one of the two bimorphs is given by:

$$Z = \frac{\tilde{v}}{\tilde{i}} = \frac{1}{1/(\mathrm{j}\omega L) + \mathrm{j}\omega C + 1/R} \tag{5.51}$$

Substituting this expression for Z into the expression for G in Eq. (4.29), putting $C = nC_p$ and simplifying, yields the relevant expression for G:

$$G = \frac{-\omega^2 f\beta}{\left\{\frac{1}{L} + \frac{j\omega}{R} - \omega^2 C_p\left(\frac{f}{a} + n\right)\right\}} \tag{5.52}$$

5.2.3.2 DSM: Double Circuit, Parallel C and Series R-L Circuit

For the circuit shown in Fig. 5.7, expressing Eqs. (5.40) and (5.41) in terms of complex amplitudes and combining the resulting equations, the impedance Z of the external circuit connected to one of the two bimorphs is given by:

$$Z = \frac{\tilde{v}}{\tilde{i}} = \frac{R + \mathrm{j}\omega L}{1 - \omega^2 LC + \mathrm{j}\omega CR} \tag{5.53}$$

Substituting this expression for Z into the expression for G in Eq. (4.29), putting $C = nC_p$ and simplifying, yields the relevant expression for G:

$$G = \frac{f\beta(j\omega R - \omega^2 L)}{C_p\left\{\frac{1}{C_p} + \frac{j\omega}{R}\left(\frac{f}{a} + n\right) - L\omega^2\left(\frac{f}{a} + n\right)\right\}} \tag{5.54}$$

5.2.3.3 DSM: Single Circuit, Parallel R-L-C Circuit

The single-circuit configuration of the parallel R-L-C circuit is shown in Fig. 5.8. As discussed in Sect. 5.2.2.3, the expression for G for this case can simply be obtained by replacing f by $2f$ in the corresponding double-circuit case (i.e. Equation (5.52)):

$$G = \frac{-\omega^2 2f\beta}{\left\{\frac{1}{L} + \frac{j\omega}{R} - \omega^2 C_p\left(\frac{2f}{a} + n\right)\right\}} \tag{5.55}$$

5.2.3.4 DSM: Single Circuit, Parallel C and Series R-L Circuit

The single-circuit configuration of the parallel R-L-C circuit is shown in Fig. 5.9. As discussed in Sect. 5.2.2.3, the expression for G this case can simply be obtained by replacing f by $2f$ in the corresponding double-circuit case (i.e. equation (5.54)):

$$G = \frac{2f\beta(j\omega R - \omega^2 L)}{C_p\left\{\frac{1}{C_p} + \frac{j\omega}{R}\left(\frac{2f}{a} + n\right) - L\omega^2\left(\frac{2f}{a} + n\right)\right\}} \tag{5.56}$$

5.3 Simulations: FRFs of Host with Energy Harvesting TVA

The aim of using the electromechanical TMD is to suppress the targeted resonance peak of the FRF connecting the response at the attachment point y_A to the external excitation F_{ext} on the host structure over a range of excitation frequencies ω over which the targeted host structure mode is dominant. In the following subsections, the effect of the electromechanical TMD is investigated. The parameters of the host structure (free-free beam) and the bimorphs of the electromechanical TMD are given in Tables 5.1 and 5.2, respectively. The piezoelectric layers of each bimorph are taken to be series connected.

In summary, the theoretical study investigates the following issues (i) the interaction between the harvester and the structure to which it is attached; (ii) the effect on the vibration attenuation of the type and magnitude of the electrical impedance, as well as the effect of either series or parallel connection of the R-L combination; (iii) the generation of an effect equivalent to that of the optimally damped benchmark model using electrical R-L-C circuitry. This latter is the primary goal, since, as observed at the end of Sect. 5.2.1.2, the damping ratio of the fictitious benchmark model $\zeta_{a_{opt}} = 8.24\%$, whereas the actual damping ratio of the TMD $\zeta_a (= \zeta_1)$ is only 1%.

The simulations of this study use the four different R-L-C circuit configurations shown in Figs. 5.6, 5.7, 5.8 and 5.9. The coupled FRFs of the host structure with the TMD attached $r_{y_A F_{ext}}^{host+TMD}(\omega)$ are investigated for a broad range of R-L-C circuit parameters. The resistor and capacitor values are changed such that the circuit impedance ranges from the short circuit (low impedance) to open circuit (high impedance) condition [1]. For this study, the resistor values are varied in the range 100 Ω to 1 MΩ. For a given resistance in this range, the ratio n of the external capacitor C to the internal capacitance of a piezoelectric layer C_p is varied from 0 (high external capacitor impedance) to 5 (low external capacitor impedance). For each given R-C combination, the effect of the inductor L is investigated by considering the two cases:

(a) inductor omitted from the circuit;
(b) optimised inductor included.

In case (a), the effect of the inductor is omitted by setting $L \to \infty$ in the equations of the parallel circuits in Fig. 5.6 (Sect. 5.2.2.1) and Fig. 5.8 (Sect. 5.2.2.3) and $L = 0$ in the equations of the series R-L circuits in Fig. 5.7 (Sect. 5.2.2.2) and Fig. 5.9 (Sect. 5.2.2.4). In case (b), the inductor was optimised using the MATLAB optimisation toolbox [13]. In this case, the MATLAB function *fgoalattain©* was used in order to find, for given R and C, the value of L, denoted by $L_{opt}\big|_{RC}$, which, as far as possible, makes the objective function $\left|r^{host+TMD}_{y_A F_{ext}}(\omega)\right|$ less than or equal to the target (benchmark) function $\left|\left\{r^{host+TMD}_{y_A F_{ext}}(\omega)\right\}_{uncoup,opt}\right|$ (depicted by the thick solid line in Fig. 5.5(b)) over a wide frequency range [1]. The initial approximation values for $L_{opt}\big|_{RC}$ supplied to *fgoalattain©* denoted by $\hat{L}_{opt}\big|_{RC}$ was given by:

$$\hat{L}_{opt}\big|_{RC} = \frac{1}{C_p(n + f/a)\omega_a^2} \text{ (double circuits, Sections 5.2.2.1, 5.2.2.2)} \tag{5.57}$$

$$\hat{L}_{opt}\big|_{RC} = \frac{1}{C_p(n + 2f/a)\omega_a^2} \text{ (single circuits, Sections 5.2.2.3, 5.2.2.4)} \tag{5.58}$$

The reasoning behind the above approximations is that the peaks of $\left|r^{host+TMD}_{y_A F_{ext}}(\omega)\right|$ that need to be damped are roughly centred around $\omega = \omega_a$, the tuned frequency of the electrically uncoupled TMD (defined in Eq. (5.27a, b)) and the above expressions for $\hat{L}_{opt}\big|_{RC}$ therefore approximately correspond to an electrical resonance condition which amplifies the electrical effect (the capacitive and inductive terms cancel out in a part of the denominator in the Eqs. (5.38), (5.39), (5.44), (5.45), (5.46), (5.47), (5.48) and (5.49) at $\omega = \omega_a$ for $L = \hat{L}_{opt}\big|_{RC}$) [1].

After determining the optimal inductor values $L_{opt}\big|_{RC}$ for a range of R-C values, the combination of R, C, $L_{opt}\big|_{RC}$ values that best replicates or improves (vibration attenuation) the benchmark response $\left|\left\{r^{host+TMD}_{y_A F_{ext}}(\omega)\right\}_{uncoup,opt}\right|$ is selected. The selected combination is denoted by $\hat{R}_{opt}$, $\hat{C}_{opt}$, $\hat{L}_{opt}$ and is used as the input to *fgoalattain©* for one final optimisation process in order to determine overall optimal resistor-capacitor-inductor values R_{opt}, C_{opt}, L_{opt} for the selected circuit configuration [1].

It is important to note that the optimisation procedure was carried out using the expressions for $r^{host+TMD}_{y_A F_{ext}}(\omega)$ derived by AMAM (Sect. 5.2.2). However, the

AMAM-computed function for the parameters R_{opt}, C_{opt}, L_{opt} was then verified by recalculating, $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for these same R-L-C parameters using the DSM formulae of Sect. 5.2.3.

In each of the graphs presented in the following sections, in addition to the electrically coupled FRFs, $r_{y_A F_{ext}}^{host+TMD}(\omega)$, the following graphs are plotted for comparison purposes:

(i) $r_{y_A F_{ext}}^{host}(\omega)$(shown as a thin dashed black line), i.e. the host structure FRF without TMD attached;

(ii) $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$ (shown as a thick dotted black line): the benchmark (optimal) response, i.e. the host structure FRF with an attached electrically uncoupled TMD that has fictitious optimal mechanical damping (8.24%);

(iii) $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup}$ (shown as a thick solid black line): the host structure FRF with the attached electrically uncoupled TMD and the actual level of (mechanical) damping (1%).

5.3.1 Double Circuit: Coupled FRFs for Parallel R-L-C Connection

This section deals with the circuit configuration of Fig. 5.6. Figures 5.10(a,b), 5.11(a,b) and 5.12(a,b), respectively, show the results for three resistors, 100 Ω, 50 kΩ and 1 MΩ. In each figure, $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is plotted for three values of $C = 0C_p,\ 1C_p,\ 5C_p$ (shown, respectively, in thin blue, black and red). In the upper Figs. 5.10(a), 5.11(a) and 5.12(a), the inductor is omitted from the simulations (i.e. R-C only), whereas in the lower Figs. 5.10(b), 5.11(b) and 5.12(b), the optimised inductor $L_{opt}\big|_{RC}$ is included.

In Fig. 5.10, the resistance is very low (100 Ω). Since the R-C-L components are in parallel, this means that the net impedance is very low. Hence, the piezos are virtually short-circuited, and the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ plots virtually coincide with the electrically uncoupled response $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup}$ for all three external capacitance values [1]. This means that, at very low resistance, any value of inductor or capacitor has no effect in vibration attenuation and the system acts like a purely mechanical system. The anti-resonance point occurred between the two resonance peaks, in the receptance plots of the host structure, represents the tuned frequency of the piezoelectric TMD/energy harvester. In Fig. 5.10, where the electrical effect is negligible, this anti-resonance coincides with the tuned frequency of the electrically uncoupled TMD (defined by Eq. 5.27a, b and denoted by ω_a).

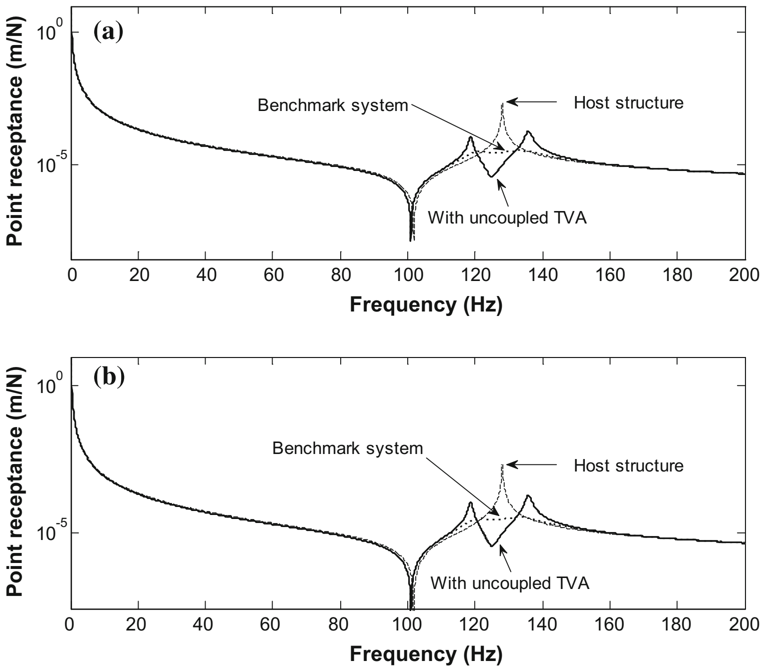

Fig. 5.10 Double-circuit parallel R-L-C at 100 Ω; **a** inductor omitted from the circuit; **b** optimised inductor included; (*C = 0Cp, 1Cp and 5Cp* = thin blue, black and red lines, respectively)

As the resistance in the circuit is increased, the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ response plots for both R-C and R-L-C circuits gradually emerge out from the shadow of the thick solid line representing $\left\{ r_{y_A F_{ext}}^{host+TMD}(\omega) \right\}_{uncoup}$ as clearly shown in Fig. 5.11. This phenomenon demonstrates that the electrical coupling is now influencing the system dynamics.

Furthermore, a closer look at Figs. 5.11(a) and 5.12(a) shows the change in the location of the anti-resonance between the two peaks as the impedance is increased (it is noted that the magnitude of the net circuit impedance increases with increase in R but decreases with increase in C) [1]. This anti-resonance is the effective tuned frequency of the electromechanical TMD, denoted by $\omega_a|_{coupled}$ which is similar to the electrically coupled resonance frequency of the base-excited bimorphs that make up the TMD [1]. This shift is aligned with what was observed in Chaps. 3 and 4.

As stated previously, the external capacitor C (when connected in parallel with the internal piezoelectric capacitance C_p) increases the net capacitance of the whole system which lowers the required optimum inductor. Figure 5.11(b) illustrates near optimum conditions since $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for the first two capacitors (i.e. $C = 0C_p$, $1C_p$, blue and black lines) with a resistor of 50 kΩ and optimised

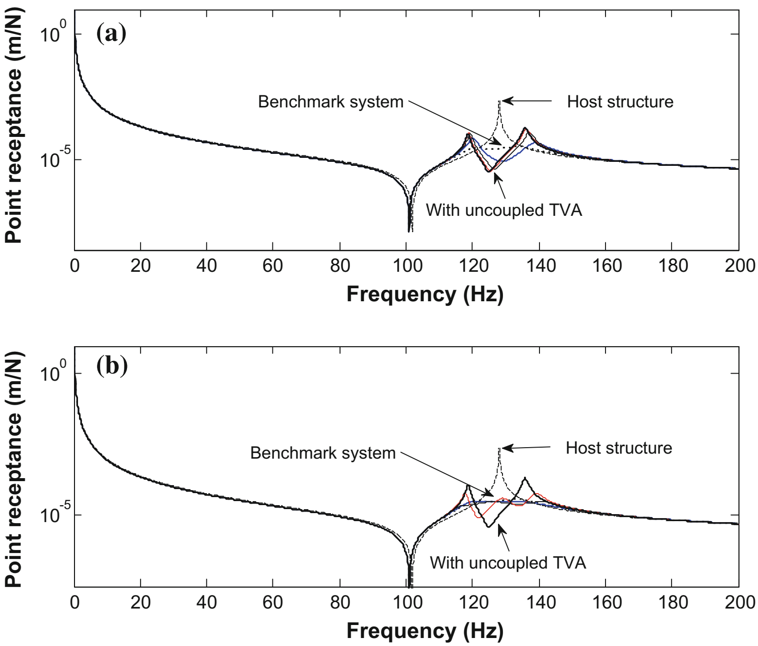

Fig. 5.11 Double-circuit parallel R-L-C at 50 kΩ; **a** inductor omitted from the circuit; **b** optimised inductor included; (*C = 0Cp, 1Cp and 5Cp* = thin blue, black and red lines, respectively)

inductors yields a flat plateau which closely matches to the benchmark curve $\left\{r^{host+TMD}_{y_A F_{ext}}(\omega)\right\}_{uncoup,opt}$. From the approximate expressions in Eqs. (5.57) and (5.58), it is clear that the optimised inductor required decreases with increasing capacitance C [1]. However, as can be noted in Fig. 5.11(b), increasing the capacitance to $C = 5C_p$ and optimising the inductor accordingly for the same resistance of 50 kΩ will result in a deterioration in the response $r^{host+TMD}_{y_A F_{ext}}(\omega)$, i.e. divergence from the benchmark curve [1]. Figure 5.12 shows that, for very high resistance (1 MΩ), the electrical effect does not contribute to the vibration suppression since there is negligible net power dissipation in the circuit due to small current in the resistor. For the R-C case (Fig. 5.12(a)), the $r^{host+TMD}_{y_A F_{ext}}(\omega)$ curve merely exhibits a shift to the right relative to the $\left\{r^{host+TMD}_{y_A F_{ext}}(\omega)\right\}_{uncoup}$ curve due to the stiffening produced by the increased impedance (high R, low C), as mentioned earlier. Figure 5.12(b) shows that the introduction of the optimised inductor $L_{opt}\big|_{RC}$

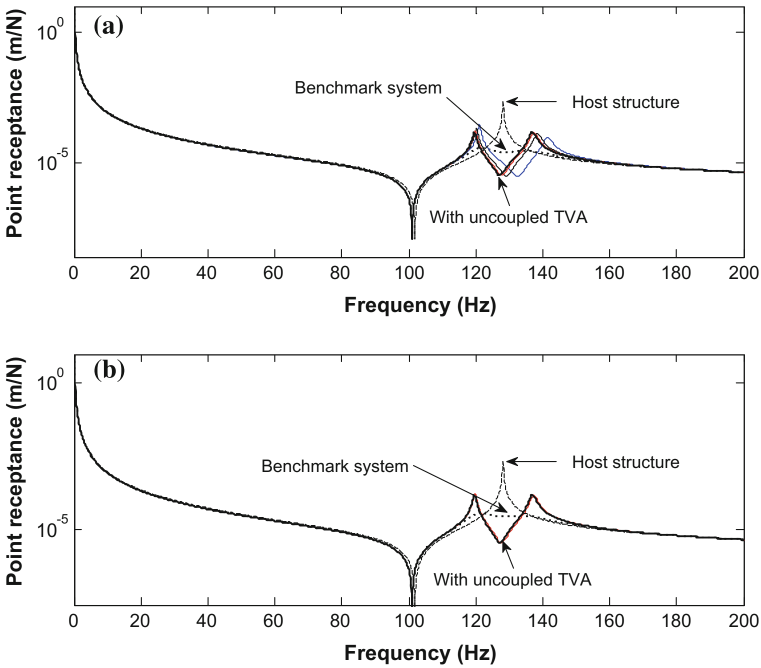

Fig. 5.12 Double-circuit parallel R-L-C at 1 MΩ; **a** inductor omitted from the circuit; **b** optimised inductor included; (*C = 0Cp, 1Cp and 5Cp* = thin blue, black and red lines, respectively)

eliminates this shift and all three curves for $r_{y_A F_{ext}}^{host+TMD}(\omega)$ merge back into the $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup}$ curve, since the inductor counteracts the increase in impedance from the other two components.

From the previous analysis, the R, C, $L_{opt}\big|_{RC}$ combination that most closely reproduces the benchmark response is that produced by $R = 50$ kΩ, $C = 1C_p$ (and $L_{opt}\big|_{RC}$ found accordingly, producing the thin black line in Fig. 5.11(b)). These R-L-C values were input into the final optimisation process to yield the overall optimal values $R_{opt} = 45$ kΩ, $C_{opt} = 1C_p$ and $L_{opt} = 13.5$H. Figure 5.13 shows that the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ curve obtained with these parameters matches excellently with the benchmark response $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$. The function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for these R-L-C parameters was also recomputed using the relevant DSM formulae of Sect. 5.2.3. The resulting DSM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is also plotted in Fig. 5.13 and is seen to match excellently with the AMAM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$.

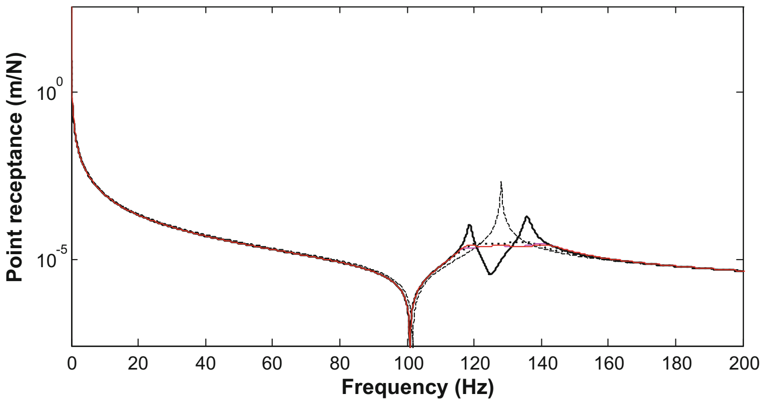

Fig. 5.13 Double-circuit parallel R-L-C receptance plots: optimum electrical by AMAM (magenta); optimum electrical by DSM (red); benchmark (thick dotted line); electrically uncoupled (thick black line); host without TMD (thin dashed line)

5.3.2 *Double Circuit: Coupled FRFs for Parallel C and Series R-L*

This section deals with the circuit configuration of Fig. 5.7. The advantage of using the resistor—inductor in one branch (R-L in series) over the previous parallel R-L-C configuration is that the series R-L combination increases the net impedance of the system. Hence, as shall be seen from the following results, the system attains tuned conditions at much lower values of R than in the previous case of the parallel R-L arrangement [1]. Therefore, the series R-L configuration is suitable for low resistor applications and the parallel R-L configuration is suitable for high resistor load applications.

Figures 5.14(a,b), 5.15(a,b), 5.16(a,b) respectively show the results for three resistors, 100 Ω, 2.5 kΩ, and 1 MΩ. In each figure, $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is plotted for three values of $C = 0C_p,\ 1C_p,\ 5C_p$ (shown respectively in thin blue, black, red line). In the upper Figs. 5.14(a), 5.15(a), 5.16(a) the inductor is omitted from the simulations (i.e. R-C only), whereas in the lower Figs. 5.14(b), 5.15(b), 5.16(b) the optimised inductor $L_{opt}\big|_{RC}$ is included. Figure 5.14 shows the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ plots at very low load (100 Ω). It can be seen from Fig. 5.14 (a) that at 100 Ω, and without an inductor, the system totally behaved like the electrically uncoupled system (since the bimorphs are practically short-circuited) and the responses for all three capacitor values combined into the curve of $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup}$.

However, Fig. 5.14(b) shows that the addition of the inductor $L_{opt}\big|_{RC}$ in series with the same resistor (100 Ω) resulted in a significant electrical effect due to a significant alteration in the circuit impedance (note that, in Fig. 5.14(b) the red and

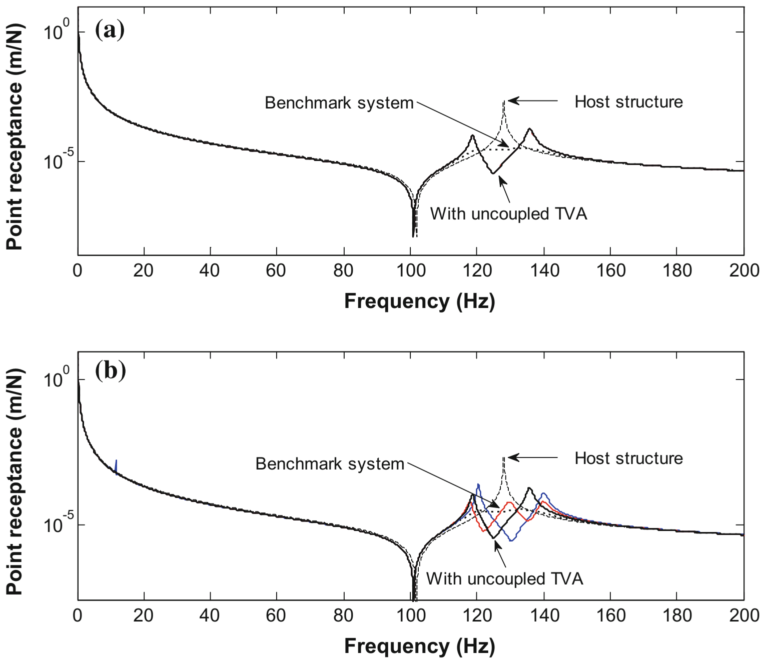

Fig. 5.14 Double circuit: parallel C and series R-L at 100 Ω; **a** inductor omitted from the circuit; **b** optimised inductor included; (*C = 0Cp, 1Cp and 5Cp* = thin blue, black and red lines, respectively)

blue curves diverge from the $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup}$ plot, in contrast to the corresponding plots in Fig. 5.10(b) for the parallel R-L-C circuit).

Figures 5.15(a,b) show the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ plots at 2.5 kΩ, for the three different capacitors. The plots in Fig. 5.15(a) are hardly different from those in Fig. 5.14(a) since the resistance is still relatively low. However, Fig. 5.15(b) shows that the inclusion of the inductor $L_{opt}\big|_{RC}$ at this low resistance results in the optimal condition being approximately achieved by one of the capacitor values ($C = 1C_p$) (the red curve in Fig. 5.15(b)) [1]. The near-optimal resistance in this case, 2.5 kΩ is seen to be much lower than the optimal one for the previous parallel R-L connection (45 kΩ).

Figures 5.16(a,b) shows that, at the high resistance (1 MΩ), there is no attenuation in the response $r_{y_A F_{ext}}^{host+TMD}(\omega)$ since there is no net power being dissipated in the circuit due to the negligible current in the resistor [1]. The $r_{y_A F_{ext}}^{host+TMD}(\omega)$ curves merely demonstrate any shift to the right relative to the $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup}$ curve due to the stiffening produced by the increased impedance (high R, low C). The

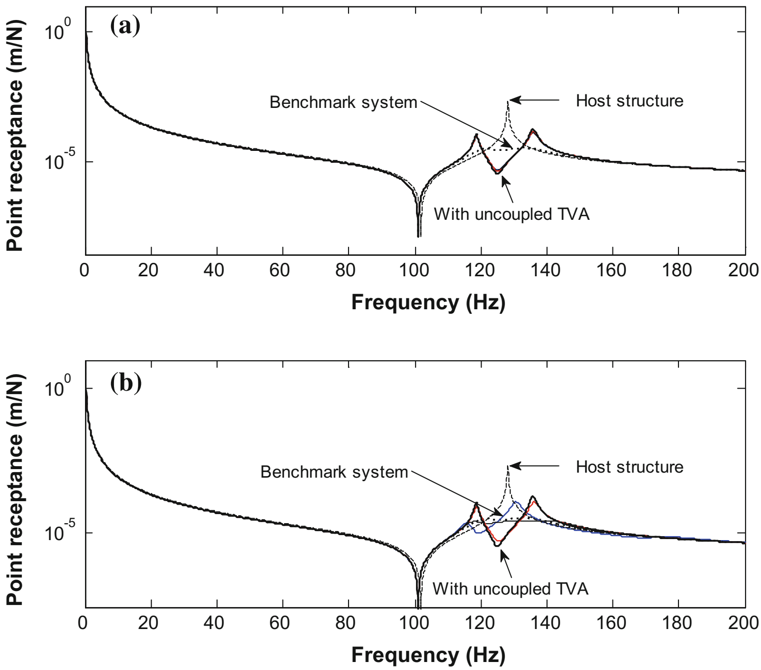

Fig. 5.15 Double circuit: parallel C and series R-L at 2.5 kΩ; **a** inductor omitted from the circuit; **b** optimised inductor included; (*C = 0Cp, 1Cp and 5Cp* = thin blue, black and red lines, respectively)

inductor has no effect in this case since it is in series with a very high resistance, and so has very small current through it [1].

From the previous analysis, the R, C, $L_{opt}\big|_{RC}$ combination that most closely reproduces the benchmark response is that produced by $R = 2.5$ kΩ, $C = 1C_p$ (and $L_{opt}\big|_{RC}$ found accordingly, producing the black line in Fig. 5.15(b)). These R-L-C values were input into the final optimisation process to yield the overall optimal values $R_{opt} = 2.25$ kΩ, $C_{opt} = 1.06C_p$ and $L_{opt} = 12.1$ H, and it can be noted from Fig. 5.17 that the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ curve obtained with these parameters matches excellently with the benchmark response $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$ [1]. The function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for these R-L-C parameters was also recomputed using the relevant DSM formulae of Sect. 5.2.3. The resulting DSM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is also plotted in Fig. 5.17 and is seen to match excellently with the AMAM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$.

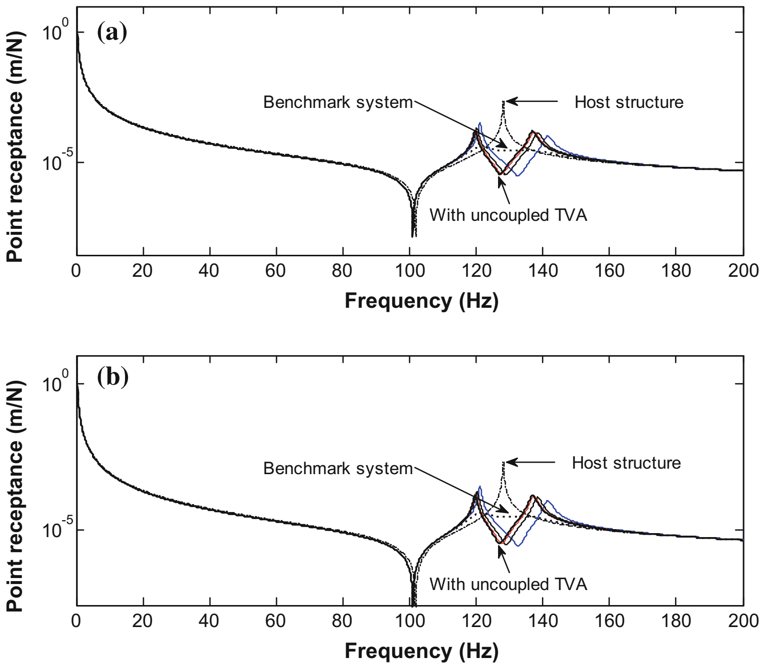

Fig. 5.16 Double circuit: parallel C and series R-L at 1 MΩ; **a** inductor omitted from the circuit; **b** optimised inductor included; (*C = 0Cp, 1Cp and 5Cp* = thin blue, black and red lines, respectively)

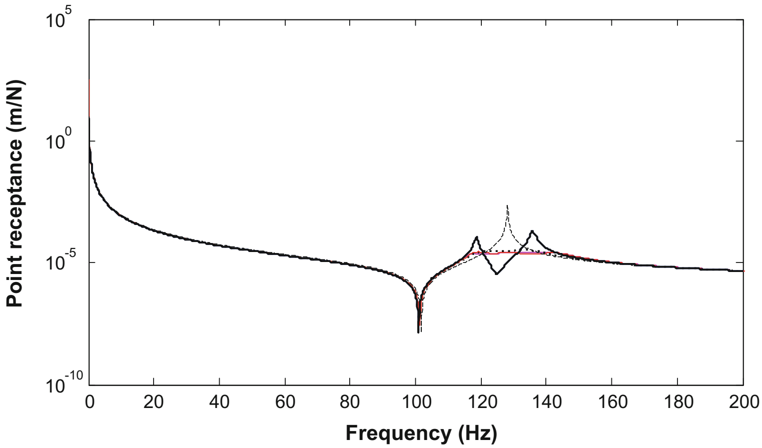

Fig. 5.17 Double-circuit parallel C and series R-L receptance plots: optimum electrical by AMAM (magenta); optimum electrical by DSM (red); benchmark (thick dotted line); electrically uncoupled (thick black line); host without TMD (thin dashed line)

5.3.3 Single Circuit: Coupled FRFs

In the previous two sections, receptances of the host structure for the double circuit with series and parallel RL configurations, as shown in Figs. 5.6 and 5.7, were discussed. In these configurations each side of the piezoelectric beam was connected to a separate circuit, thereby doubling the number of circuit components required. In this section, the concept of connecting both piezoelectric beams to a single circuit is analysed. The single-circuit configuration shown in Figs. 5.8 and 5.9 has three main advantages over the double circuits of Fig. 5.6 and 5.7: (i) The two bimorphs on either side of the TMD are in parallel with each other and so their the internal capacitances ($(f/a)C_p$, see Eqs. (5.29), (5.31)) are additive, resulting in a higher system capacitance, thereby requiring a smaller inductor for optimal tuning (as is evident from Eq. (5.58)); (ii) relatively much lower resistance is required to attain the tuned conditions; (iii) a single circuit reduces the number of components by half and is therefore much less expensive and requires less space [1]. Apart from these advantages, the receptance plots $r_{y_A F_{ext}}^{host+TMD}(\omega)$ of the single-circuit configurations demonstrate the same trends, over the impedance range considered, as observed previously in the corresponding double-circuit cases presented in Sects. 5.3.1 and 5.3.2 [1]. Therefore, only the results for the overall optimum circuit parameters will be presented in this section.

For the single circuit with parallel R-L-C connection (Fig. 5.8), the overall optimal parameters were found to be $R_{opt} = 22.25\ \text{k}\Omega$, $C_{opt} = 0.96C_p$ and $L_{opt} = 9.082$ H. Figure 5.18 shows that the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ curve obtained with these parameters matches very well with the benchmark response $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$. The

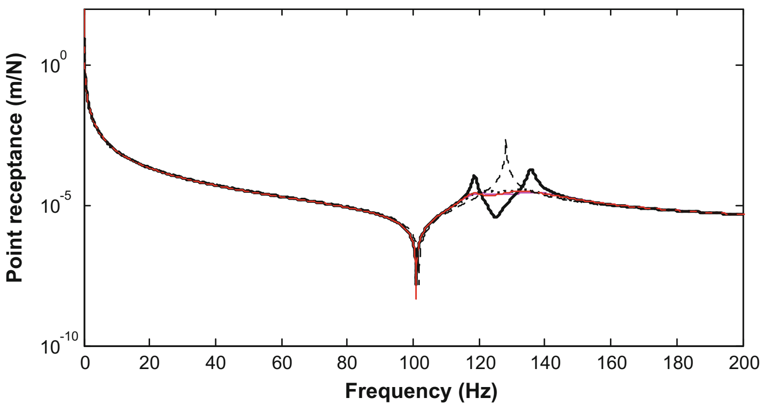

Fig. 5.18 Single-circuit parallel R-L-C receptance plots: optimum electrical by AMAM (magenta); optimum electrical by DSM (red); benchmark (thick dotted line); electrically uncoupled (thick black line); host without TMD (thin dashed line)

function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for these R-L-C parameters was also recomputed using the relevant DSM formulae of Sect. 5.2.3. The resulting DSM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is also plotted in Fig. 5.18 and is seen to match excellently with the AMAM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$. By comparing the result of Fig. 5.18 to the corresponding double-circuit parallel R-L-C configuration presented in Fig. 5.18, it can be noted that the circuit is now tuned at a resistor of 22.5 kΩ, which is significantly less than the optimum resistor of 45 kΩ for the double-circuit case shown in Fig. 5.13. Moreover, the size of the required optimum inductor is also reduced to 9.082H from 13.5H.

For the single circuit with series R-L connection (Fig. 5.9), the overall optimal parameters were found to be $R_{opt} = 1.80$ kΩ, $C_{opt} = 1.42C_p$ and $L_{opt} = 7.66$ H [1]. Figure 5.19 shows that the $r_{y_A F_{ext}}^{host+TMD}(\omega)$ curve obtained with these parameters matches very well with the benchmark response $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$. The function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for these R-L-C parameters was also recomputed using the relevant DSM formulae of Sect. 5.2.3. The resulting DSM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is also plotted in Fig. 5.19 and is seen to match excellently with the AMAM-computed function $r_{y_A F_{ext}}^{host+TMD}(\omega)$. By comparing the result of Fig. 5.19 to the corresponding double-circuit series R-L configuration presented in Fig. 5.17, the optimum resistance required in the present case is slightly lower (1.8 kΩ) than the corresponding double circuit (2.25 kΩ). In both these series R-L configurations, the optimal resistors are much less than those required by the double/single parallel R-L circuits.

Table 5.3 presents the summary of overall optimum parameters for both single and double-circuit configurations discussed in Sects. 5.3.1, 5.3.2 and 5.3.3. It is

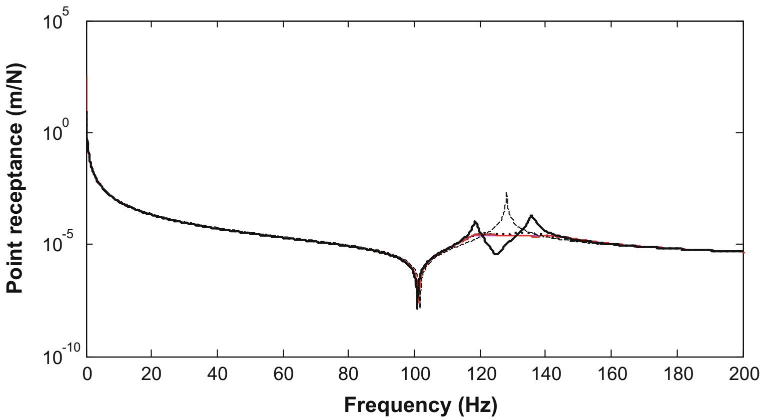

Fig. 5.19 Single-circuit parallel C and series R-L receptance plots: optimum electrical by AMAM (magenta); optimum electrical by DSM (red); benchmark (thick dotted line); electrically uncoupled (thick black line); host without TMD (thin dashed line)

Table 5.3 Optimum R-L-C parameters at tuned conditions

Circuit type	R-L connection	N	$R\ (k\Omega)$	$L\ (H)$
Double circuit	Parallel R-L	1	45	13.5
	Series R-L	1.06	2.25	12.1
Single circuit	Parallel R-L	0.96	22.5	9.082
	Series R-L	1.42	1.18	7.66

evident that the most convenient and economical circuit configuration is the single-circuit series R-L since it requires the smallest optimal resistor and inductor (and only one of each, being single circuit) [11].

It is important to note that the optimum R-L-C parameters given in Table 5.3 are not unique. In fact, numerical experiments with the optimisation procedure showed that, by changing one or more of the resistor, capacitor or inductor values in these tables and re-inputting into the optimisation algorithm, one could obtain a totally new set of optimum R-L-C parameters that result in an equally valid match with the benchmark response [1]. This gives a broad range of choices to the designer to design the EH/TMD device, to meet a specific requirement, in a variety of ways. However, it is found that, despite this non-uniqueness, the inter-dependence between the resistance, inductor and capacitor still follows the same trends observed in the previous sections.

5.4 Conclusions

This chapter has introduced, and investigated theoretically, the concept of an electromechanical tuned vibration absorber (TVA—or, more specifically, tuned mass damper, TMD). This device was created from two symmetric dual-function energy harvesting/TMD beams which were appropriately shunted in order to attenuate a vibration mode of a generic structure. The optimised damping of this TMD device was managed and supplied by the piezoelectric energy harvesting mechanism.

The proposed electromechanical TMD mimics classical TMD theory, thereby making it tractable to generic host structures. Moreover, it retains the advantages of the electrical TVA, particularly with regard to the precise application and adjustment of the required amount of damping for different circuit configurations. Furthermore, the beam-like design that is popular with adaptive vibration absorbers can be used, which, apart from being simple and compact, facilitates retuning of the device as needed.

It is noted that, in addition to the above-mentioned advantages, the electromechanical TMD holds the potential of energy storage through the use of an AC-DC rectification device. However, such nonlinear analysis is outside the scope of this book

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Chapter 6
Experimental Study of an Energy Harvesting Beam-Tuned Mass Damper

6.1 Experimental Set-up

The schematic diagram of the experimental set-up is as shown in Fig. 5.4 (previous chapter). This figure clearly demarcates the boundaries of the host structure and the TMD. The theory developed in the previous chapter has been developed for a generic host structure. However, for illustrative purposes (only), the host structure in this study is taken to be a free-free beam with an attachment block in the middle and the TMD is targeted at dampening its first flexural mode (see Fig. 5.4). This simple configuration reduced the manufacturing and experimental set-up intricacies. A photograph of the actual hardware is shown in Fig. 6.1. The electromechanical TMD, composed of two symmetrical shunted bimorph beams, is attached to the host structure. The host structure is mounted on an electrodynamic shaker which provides the external excitation (F_{ext} in Fig. 5.4). Appendix-B gives the brief details of the instrumentation used in the tests.

The main design and development activities performed in order to produce the experimental set-up of Fig. 6.1 can be summarised as follows:

- Experimental identification of the modal parameters (resonance frequency Ω_2, modal mass $M_A^{(2)}$ and damping) of the targeted mode of the host structure (free-free aluminium beam in this case);
- Design and procurement of the necessary bimorph beams that, when short-circuited, have a fundamental clamped-free frequency ω_a (Eq. 5.27(a)) that satisfies the TMD optimal tuning condition defined by Eq. (5.6) for a given prescribed mass ratio (defined by Eq. (5.8)).
- Design and development of R-L-C circuitry that can be easily adapted to any one of the four configurations in Figs. 5.8–5.11, based on calculations that optimise the circuit parameters such that the electromechanical TMD behaves

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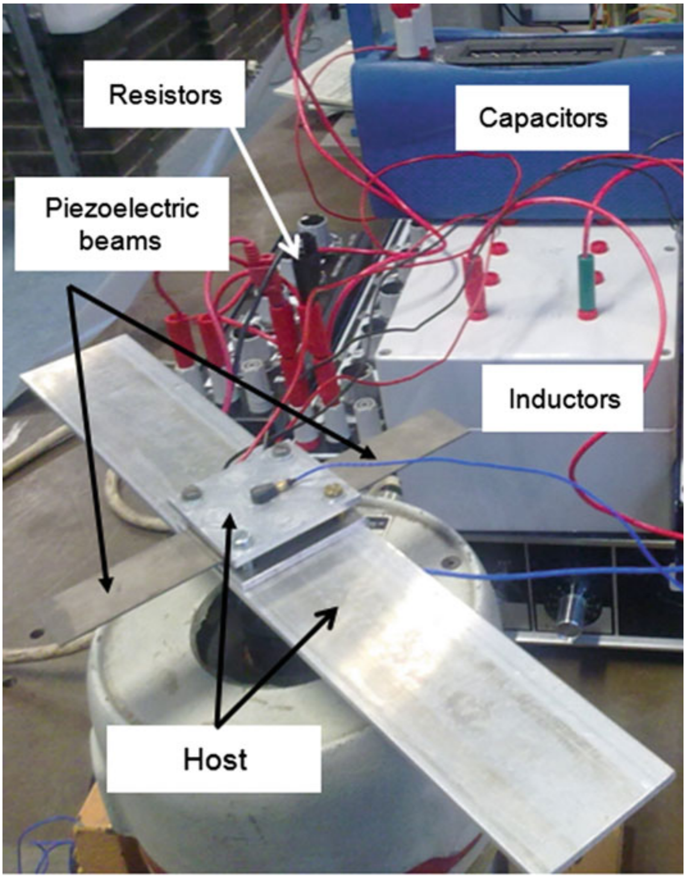

Fig. 6.1 Experimental set-up for EH/TMD beams attached to host structure

like an equivalent mechanical system with optimal damping defined by Eq. (5.3). For this study, variable resistor, capacitor and inductor boxes are used in order to tune and adjust the required value of the R-L-C circuit.

6.1.1 Experimental Determination of the Modal Parameters of the Host Structure

Figure 6.2 shows the experimental set-up for the determination of the modal parameters of the host structure (without the TMD). The host structure, free-free beam, was mounted on the shaker via a force gauge which measured the external force F_{ext}. A tiny accelerometer was attached at the centre of the connecting block. Random excitation was fed to the shaker and the receptance FRF $r^{host}_{y_A F_{ext}}(\omega)$ was

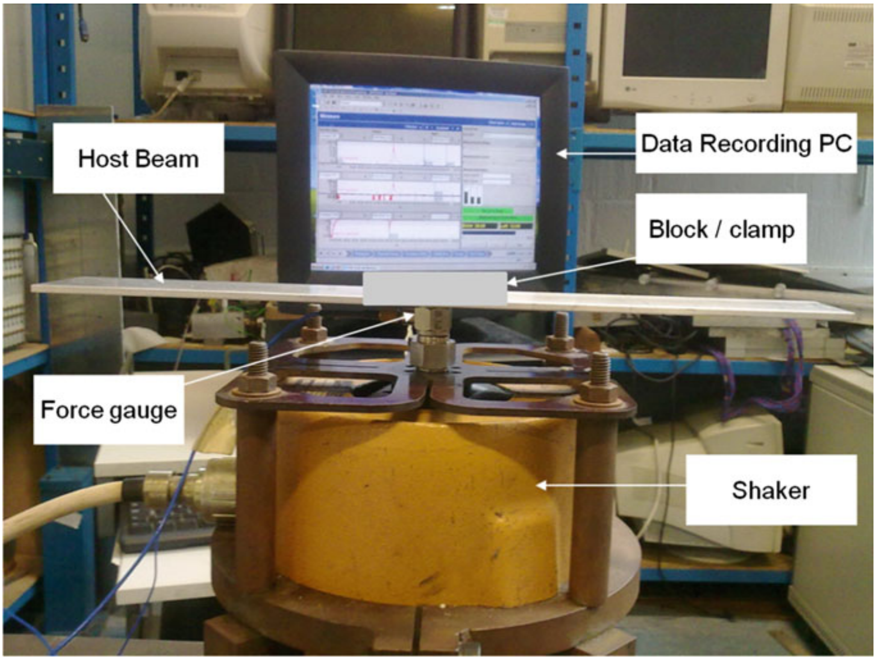

Fig. 6.2 Test set-up (see Appendix-C for detail)

measured. The random signal was generated by a pc-controlled data acquisition system which was also used to measure the FRF. A curve fitting method was then used to determine the modal parameters from the experimental FRF, based on the modal series expansion of Eq. (5.5), which neglects damping in the structure [1]. The series expansion was truncated beyond the second mode (which is the first flexural mode, the first mode simply defining rigid body motion). The model parameters were presented in Table 5.1. Figure 6.3 shows a comparison between the measured FRF and the one reconstructed from Eq. (5.5) using the identified model parameters. It is noted that, in this reconstruction, a damping term of $j2\widehat{\zeta}_2\Omega_2\omega$ was added to the denominator of the second term of the series expansion of Eq. (5.5) with $\widehat{\zeta}_2 = 0.14\%$. The agreement in resonance amplitude level in the flexural mode between the measured and reconstructed FRFs indicates that the damping present in the host structure is indeed negligible.

6.1.2 Design of the Bimorph TMD

According to Fig. 5.4, the TMD, of mass m_a, is comprised of the overhanging portions of the bimorphs. The mass of the overhanging part of each of the required bimorph beams is $m_a/2$ and their first clamped-free natural frequency under

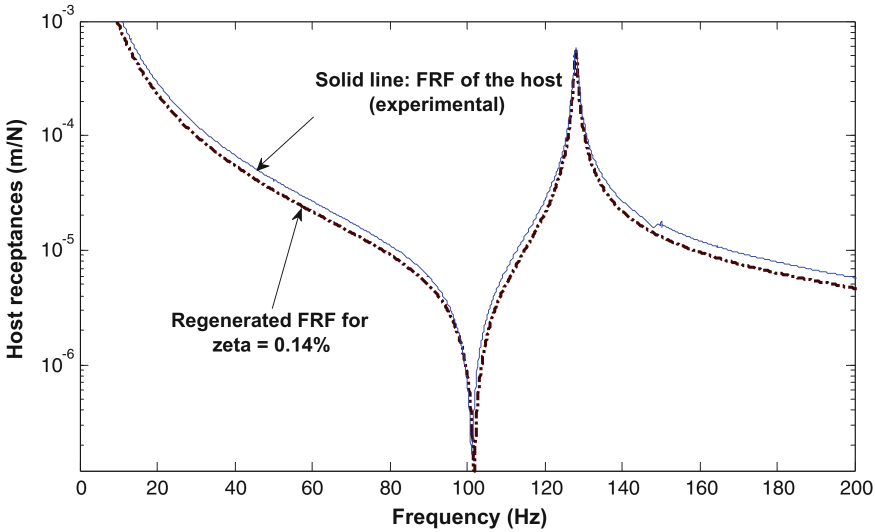

Fig. 6.3 Estimation of damping present in host structure, using regeneration of FRFs [1]

short-circuited conditions is the tuned frequency of the TMD ω_a (as per Eq. (5.27 (a)). Having determined $M_A^{(2)}$ and prescribing a mass ratio μ in the region of 2%, the required value of m_a was estimated from Eqs. (5.8), (5.11b, c) and (5.12). The required tuned frequency ω_a was then estimated from Eqs. (5.6) and (5.7) such that it was optimal. Knowing the required tuned frequency, mass and the density of the absorber, the geometrical parameters of length, width and height of the piezoelectric and the shim material is determined. The bimorphs were then specified giving due consideration to Eqs. (5.27a) and (5.20).

The above-described calculations determined the size and choice of the bimorphs, which were manufactured by Piezo Systems Inc. Each was made up of two PZT-5A4E layers bonded on top and bottom surfaces of an aluminium shim. Each bimorph had series-connected layers and its geometric, material and electromechanical properties, as provided by the manufacturer, are given in Table 5.2. It is noted that the total length of the bimorph as received from the manufacturer was 72.5 mm. Of this length, 58.75 mm was overhung to attain the required tuned frequency. The clamped length of the piezos could be regarded as a simple rigid mass addition to the host structure. Hence, with the TMD attached, the host structure parameters $M_A^{(1)}$, $M_A^{(2)}$, Ω_2 were very slightly corrected to account for this. The final value of the mass ratio recalculated from Eqs. (5.8), (5.11b, c), was $\mu = 1.86\%$. Also, from Eqs. (5.27a), (5.20), (5.6) and (5.7) it was evident that ω_1 (i.e. ω_a) $\approx \omega_{a_{opt}}$ i.e. the electrically uncoupled system was approximately optimally tuned. Using Eq. (5.3), the optimal damping ratio was calculated to be $\zeta_{a_{opt}} = 8.24\%$. This value of optimum damping was used in the benchmark model (Sect. 5.2.1), against which the experimental and theoretical performance of the electromechanical TMD was judged.

6.1.3 Design and Development of R-L-C Circuitry

The same four R-L-C circuits configurations used in the previous chapter, shown in Fig. 5.8–5.11, are used in the experimental study of the present chapter. It can be noted from Table 5.3 (Chap. 5) that, to validate the theory, the following components are required:

- Inductors in the range 7.66–13.45 H;
- Resistances in the range of 100 Ω–1000 kΩ;
- Capacitors in the range 76–400 nF.

The resistances and capacitances required were readily available in the form of variable resistance and capacitance boxes, respectively. However, inductors above 10 H are less readily available.

Inductors are difficult to fabricate for higher values (e.g. 10–1000 H), due to weight and space constraints. The weight and volume of the copper coil increases as the number of wire turns increases, and the inductance is a function of square of the number coils as shown in Eq. (6.1):

$$L = \frac{\mu_0 \mu_r N^2 A}{l} \tag{6.1}$$

where, L is the inductance in henries (H), μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ H/m), μ_r is a specific coefficient for the solenoid, N is number of turns on the coil, A is the area of the core in square metres (m^2) and l is the length of the core in metres.

In practice, artificial or virtual inductor circuits (e.g. (2), (3)) are often used instead of conventional inductors, where higher inductances are required and a traditional inductor is difficult to realise. Virtual inductors use operational amplifiers and other electronic circuit elements to simulate higher inductance effects. However, these virtual circuits require some external power source to produce higher inductances. Use of an external power source goes against the concept of the electromechanical TMD, which is a purely passive device. One could consider powering a virtual inductor circuit from the energy harvesting effect of the TMD. However, this would involve the design and development of highly complicated circuitry that is beyond the scope of this project.

Hence, given that the inductor requirement for this application was not that high (up to 20 H), a simple variable copper wire-wound inductor box was designed and fabricated by hand winding in the lab. Five inductors of different values were prepared and then soldered on a single-sided strip board (*Kelan—147,899*). The circuit board, containing these inductors, was fitted at the bottom of an *ABS* enclosure box, as shown in Fig. 6.4. The top cover of the box was drilled and 4-mm sockets were attached which were connected to the respective inductors fitted on the bottom side of the box. Different values of inductances could be achieved by changing their interconnections in different ways (series or parallel to each other).

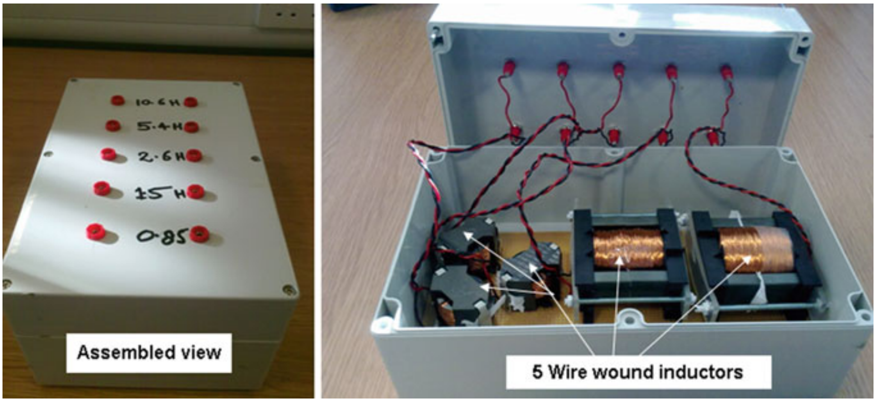

Fig. 6.4 Custom-made hand built inductor box

Table 6.1 Design parameters of the variable inductor box

Inductor (Henry)	No. of Wire Turns	Internal Resistance (Ohms)	Spacer or gap (mm)	Cores used (from *Farnell*)
10.66	2150	50	0.1	*EPCOS ETD59*
5.4	1520	35	0.1	*EPCOS ETD59*
2.66	1457	99	0.1	*EPCOS RM14*
1.5	1030	16	0.1	*EPCOS RM14*
0.8	728	11.5	0.1	*EPCOS RM14*

Table 6.1 provides the design parameters of the components used to build this variable inductor box. The custom-made variable inductor box served the experimental requirements reasonably well and produced excellent experimental results.

The inductor box provided variable inductances ranging from 0.5–20 H. Moreover, it was quite easy to switch between the inductances with a resolution of 0.5 H. The tolerance was theoretically $\pm$ 25% due to ferrite material variability.

6.2 Experimental Validation

In this section, experimental results are presented to validate the theory developed in the previous chapter. The complete set-up is shown in Fig. 6.1, 6.2 and 6.5. The TMD-host structure combination was mounted on the shaker through a force gauge. A tiny accelerometer was attached to the centre of the connecting block. The external excitation F_{ext} from the shaker was a random excitation signal of frequency bandwidth 0–320 Hz, generated by the pc-controlled data acquisition system, which also measured the receptance FRF $r_{y_A F_{ext}}^{host+TMD}(\omega)$ [4]. The magnitude of the

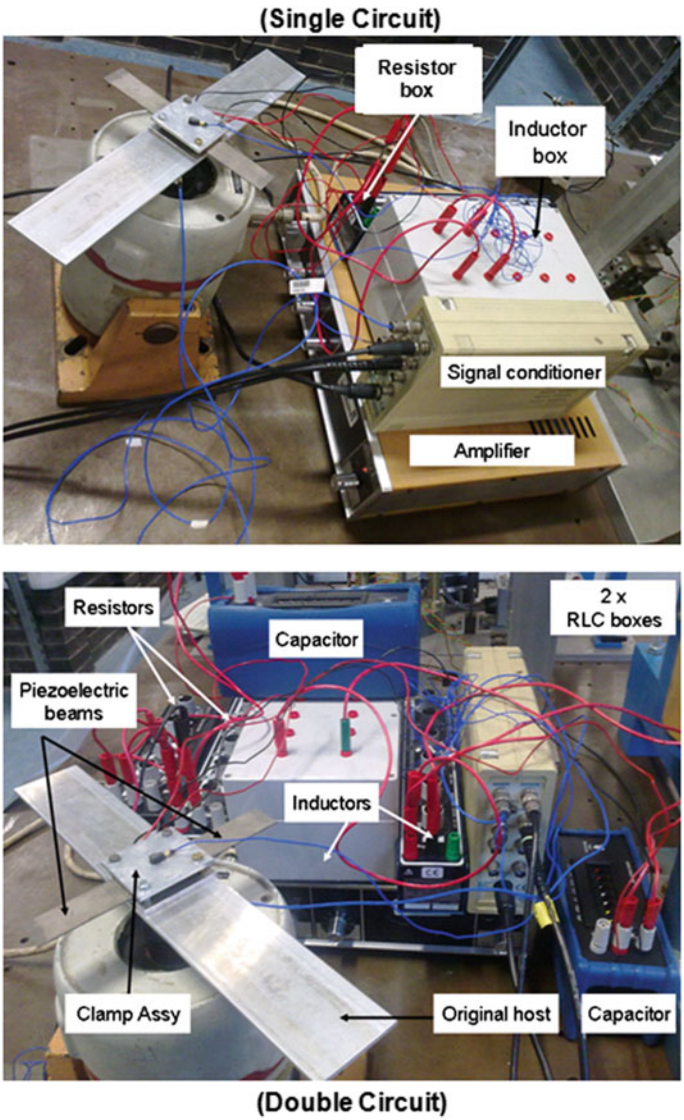

Fig. 6.5 Experimental set-up for single-circuit and double-circuit testing configuration

input excitation force was kept small to avoid the introduction of nonlinear effects. In the following sections, the FRFs $r^{host+TMD}_{y_A F_{ext}}(\omega)$ are presented for the four-circuit cases (Figs. 5.8–5.11) for a range of resistor, inductor and capacitor values. The bottom picture of Fig. 6.5 shows the hardware set-up for the double-circuit configurations, (Figs. 5.8, 5.9) and the top picture of Fig. 6.5 shows the hardware set-up for the single-circuit configurations (Figs. 5.10, 5.11). The latter configurations necessitate half the number of components of the former configurations, reducing the cost, weight and space requirements.

For each of the four-circuit configurations, the optimally tuned experimental result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is compared with [1]:

- the benchmark mechanical model $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$ (Fig. 5.7(b));
- the optimised simulation result (with R-L-C obtained using the optimisation procedure described in Sect. 5.3);
- the simulated result for the same R-L-C parameters (since the experimentally determined optimal R-L-C parameters will not precisely match the predicted optimal R-L-C parameters)

It is found that the experimental results exhibit the same trends predicted by the theory in all the cases in Chap. 5. It is also observed that any change in the circuit arrangement or the circuit parameters has a profound effect on the response of the host structure.

6.2.1 FRFs: Single-Circuit Configuration, Parallel R-L-C

The circuit diagram for this case is shown in Fig. 5.10. The terminals of both bimorphs are first connected in parallel to each other and then connected to a single external R-L-C circuit. In this arrangement, the internal capacitances $(f/a)C_p$ of the bimorphs add together, doubling the capacitance of the system, thereby reducing the size of the inductor required to achieve the tuned conditions, as discussed in the previous chapter. For the bimorphs used, the capacitance of each piezoelectric layer C_p was experimentally found to be 76 nF. The equivalent capacitance of the two bimorphs is $(2f/a)C_p$ (see Eq. (5.29) and (5.31). Hence, for series-connected piezo layers ($a = 2$, $f = 1$, Eqs. (4.7), (4.19)), the equivalent capacitance of the two bimorphs was 76 nF.

A preliminary estimate of the required inductor was obtained by the following relation, based on Eq. (5.58):

$$L \approx \frac{1}{C_p(n + 2f/a)(\Omega_2)^2} \tag{6.2}$$

where, Ω_2 is the required tuned frequency for the target structural vibration mode (127.7 Hz). The above equation was adapted from Eq. (5.58) by considering that $\Omega_2 \approx \omega_a$. Setting the external capacitor $C = nC_p$ to zero, the required approximate inductor was estimated to be about 20 H. Subsequent experimental investigations without the external capacitor showed that the optimal FRF was achievable with an inductor of 15.8 H. Hence, apart from reducing the number of components by about half, the elevated internal capacitance of the single-circuit capacitance eliminated

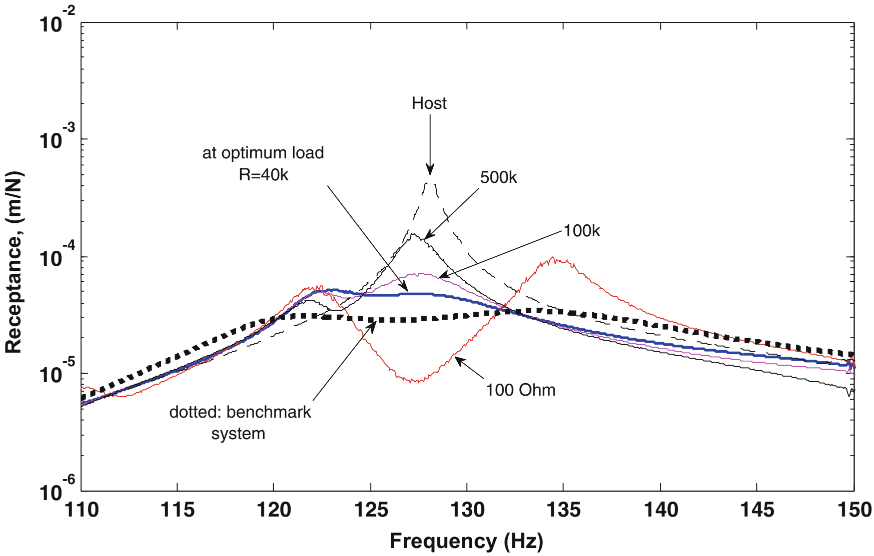

Fig. 6.6 Comparison of experimental results for single-circuit parallel R-L circuit

the need of an external capacitor. Figure 6.6 shows the receptances $r_{y_A F_{ext}}^{host+TMD}(\omega)$, for this value of inductance, no external capacitor and four values of resistance: short-circuit conditions (R = 100 Ω); R = 40 kΩ; R = 100 kΩ; and R = 500 kΩ.

It can be observed in Fig. 6.6 that at low resistances (100 Ω, i.e. short-circuit conditions), $r_{y_A F_{ext}}^{host+TMD}(\omega)$ showed two distinct peaks as can be seen in an electrically uncoupled system. However, these two distinctive peaks in the FRF diminished gradually as the resistance are increased. For the 40 kΩ resistor, both peaks almost diminished and the response of the host structure exhibited a flat plateau (thick blue line), the same way as portrayed by optimum benchmark response $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$. As the resistance was increased beyond the optimum value of 40 kΩ, the system de-tuned itself and a single peak in the response plot appeared. This might be due to the fact that for resistor values higher than optimum, the system behaved like an over-damped system. Moreover, at very high loads (i.e. 500 kΩ), the resonance frequency of the piezoelectric TVA/EH system changes about 6% from short-circuit to open-circuit conditions (5), which is another cause of de-tuning of the TVA.

Figure 6.7 shows the comparison of three different optimum receptance plots for the single-circuit parallel R-L-C circuit: (i) the experimental optimum result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$, (ii) the simulated result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$ with the R-L-C values computed the optimisation technique of Sect. 5.3; and (iii) benchmark optimum $\left\{r_{y_A F_{ext}}^{host+TMD}(\omega)\right\}_{uncoup,opt}$. It can be seen in Fig. 6.7 that the optimised curve

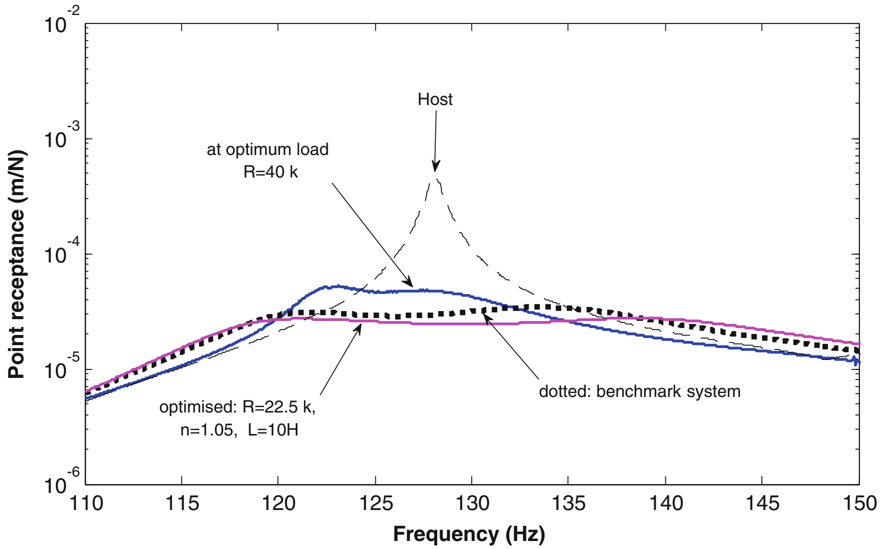

Fig. 6.7 Single-circuit parallel R-L-C; comparisons of optimum FRFs, experimental and theoretical

generated by the MATLAB optimisation toolbox matches closely the benchmark response. The experimental optimisation response of the host is reasonably close to the benchmark response, although it is not as well matched to it as the simulated optimised curve. The difference between the experimental and simulated optimal FRFs is mainly attributed to the fact that the R-L-C values are somewhat different for the two cases: for the experimental optimal R = 40 kΩ, L = 15.8 H, C = 0, whereas for the simulated optimal R = 22.5 kΩ, L = 10 H, C = $1.05C_p$. The reason for this is that the optimisation strategy was very different for the two cases. In the experimental case, it was a manual tuning with the external capacitance constrained to be zero. With the simulated case, there were fewer constraints applied to the R-L-C parameters, particularly the capacitance.

As can be seen in Fig. 6.8, the agreement between the experimental and simulated FRFs $r^{host+TMD}_{y_A F_{ext}}(\omega)$ is much improved if they pertain to the *same* R-L-C parameters (R = 40 kΩ, L = 15.8 H, C = 0). The discrepancies between the two plots in this case are attributed to the limitations of the theoretical modelling, particularly any unaccounted internal resistances of the equipment used and non-symmetry (electrical/mechanical) of the two bimorphs (apart from manufacturing tolerances errors and clamping errors, etc.).

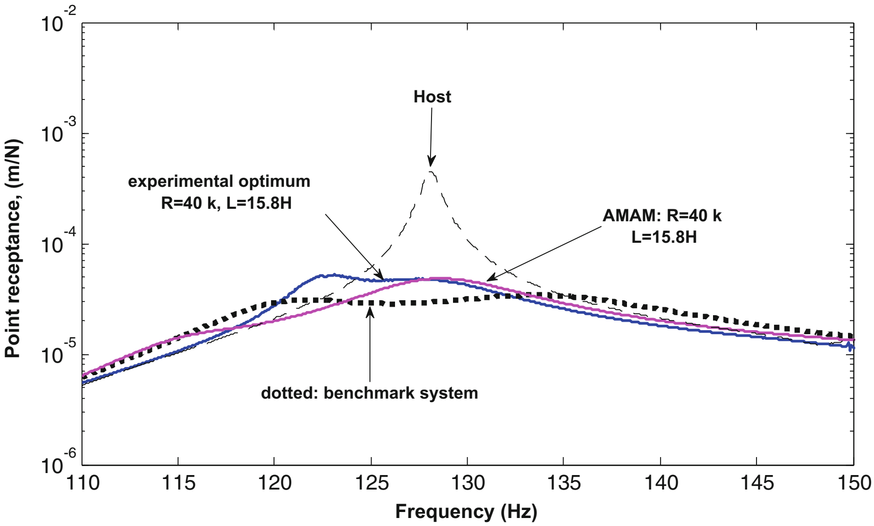

Fig. 6.8 Single-circuit parallel R-L-C; comparison of theory with experiment for the same parameters

6.2.2 FRFs: Single-Circuit Configuration, Parallel C, Series R-L

The circuit diagram for this case is shown in Fig. 5.11. Being a single-circuit configuration, like the previous case, the system has an increased internal capacitance (relative to the double-circuit configurations). For this case, like the previous one, it was found that it was possible to dispense with the external capacitor, with optimal conditions being achieved using the same inductance value of 15.8 H. However, in this case, the inductor is in series with the resistor. As mentioned in Chap. 5, the advantage of this is that it increases the net impedance of the system. Hence, the system attains tuned conditions at much lower values of *R* than previous parallel R-L arrangement i.e. the series R-L configuration is suitable for low-load applications and parallel R-L configuration is suitable for high-load (resistor) applications.

Figure 6.9 shows the receptances $r_{y_A F_{ext}}^{host+TMD}(\omega)$, for L = 15.8 H, no external capacitor and four values of resistance: R = 500 kΩ; R = 100 kΩ; R = 50 kΩ; and R = 2.1 kΩ. It can be seen that, for the present series R-L case, the electromechanical TMD is optimally tuned at a much lower resistance of 2.1 kΩ compared to 40 kΩ in the parallel R-L case.

Figure 6.10 shows the comparison of three different optimum receptance plots for the single-circuit parallel C, series R-L combination: (i) the experimental optimum result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$, (ii) the simulated result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$ with the R-L-C values computed using the optimisation technique of Sect. 5.3; and

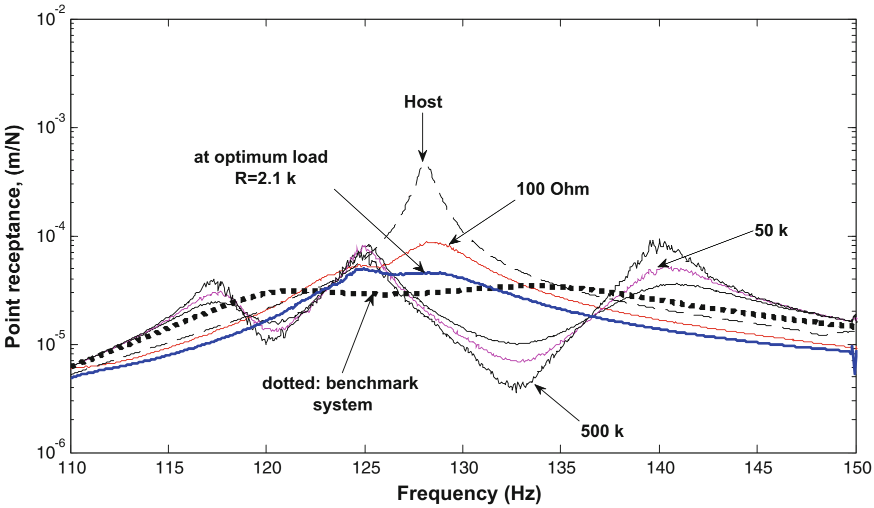

Fig. 6.9 Experimental results for single-circuit parallel C and series R-L

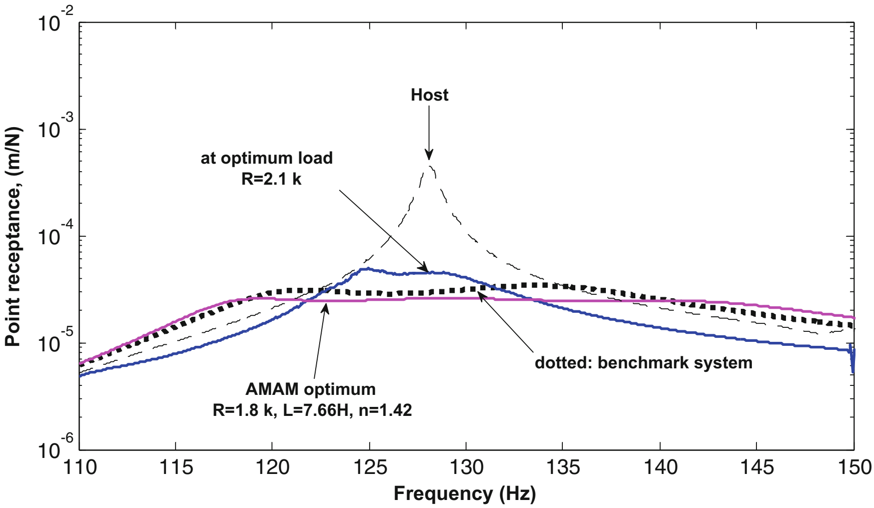

Fig. 6.10 Single-circuit parallel C and series R-L; comparisons of optimum FRFs, experimental and theoretical

(iii) benchmark optimum $\left\{ r_{y_A F_{ext}}^{host+TMD}(\omega) \right\}_{uncoup,opt}$. Both experimental and simulated optimal curves are reasonably well matched with the benchmark response. As discussed previously, the mismatch between the experimental and simulated optimal FRFs is mainly attributed to the fact that the R-L-C values are somewhat

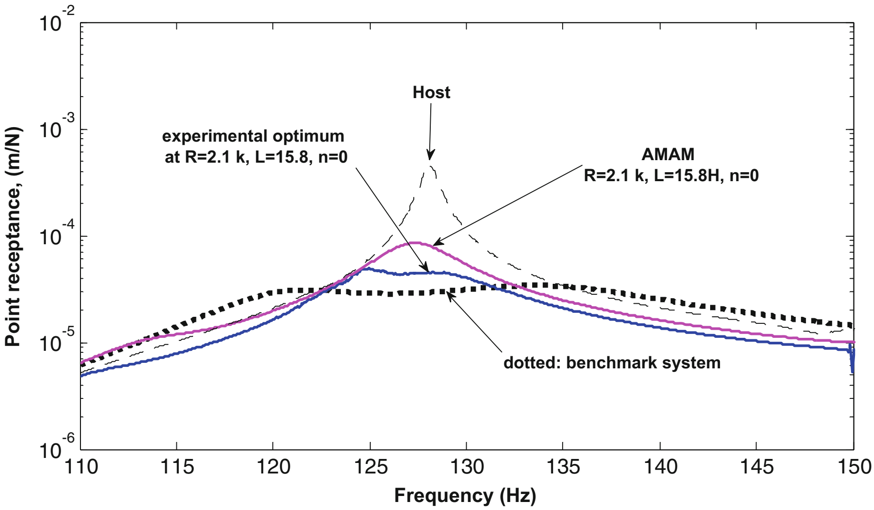

Fig. 6.11 Single-circuit parallel C and series R-L; comparison of theory with experiment for the same parameters

different for the two cases due to the different optimisation strategies used: for the experimental optimal R = 2.1 kΩ, L = 15.8 H, C = 0, whereas for the simulated optimal R = 1.8 kΩ, L = 7.66 H, C = 1.42C_p.

As before, Fig. 6.11 shows that the agreement between the experimental and simulated FRFs $r_{y_A F_{ext}}^{host+TMD}(\omega)$ is much improved if they pertain to the *same* R-L-C parameters (R = 2.1 kΩ, L = 15.8 H, C = 0).

6.2.3 FRFs: Double-Circuit Configuration, Parallel R-L-C

In this arrangement, each bimorph of the electromechanical TMD is connected across a separate identical circuit consisting of R-L-C in parallel as shown in Fig. 5.8. Figure 6.12 shows the receptances $r_{y_A F_{ext}}^{host+TMD}(\omega)$, for an external capacitor of 100 nF ($n = 1.3$), an inductor of 10 H and four values of resistance: R = 1000 kΩ; R = 50 kΩ; R = 25 kΩ; and R = 1 kΩ. The FRFs evolve in the same way with increase in resistance as was observed in the previous two sections. Optimal tuning was observed at around 50 kΩ, which is comparable to the 50 kΩ observed for the corresponding single-circuit case.

Figure 6.13 shows the comparison of three different optimum receptance plots for the double-circuit parallel R-L-C circuit: (i) the experimental optimum result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$, (ii) the simulated result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$ with the R-L-C values computed using the optimisation technique of Sect. 5.3 and (iii) benchmark

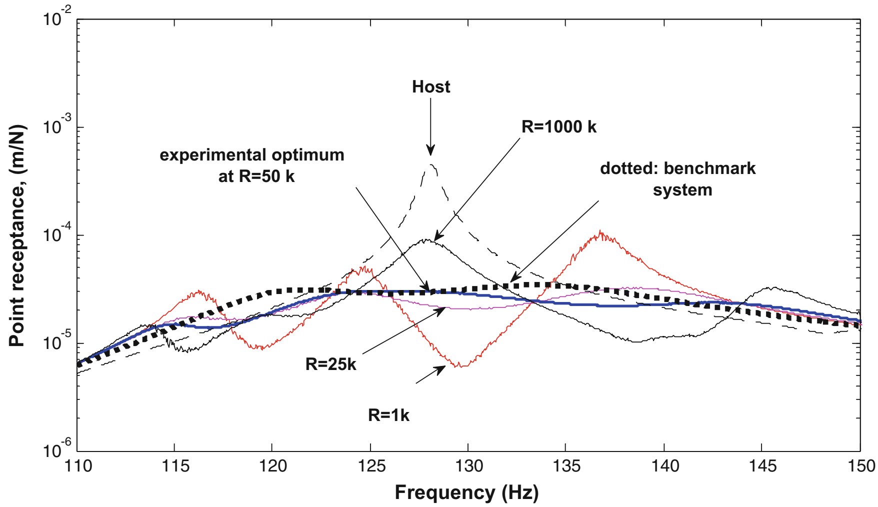

Fig. 6.12 Experimental results for double-circuit parallel R-L-C

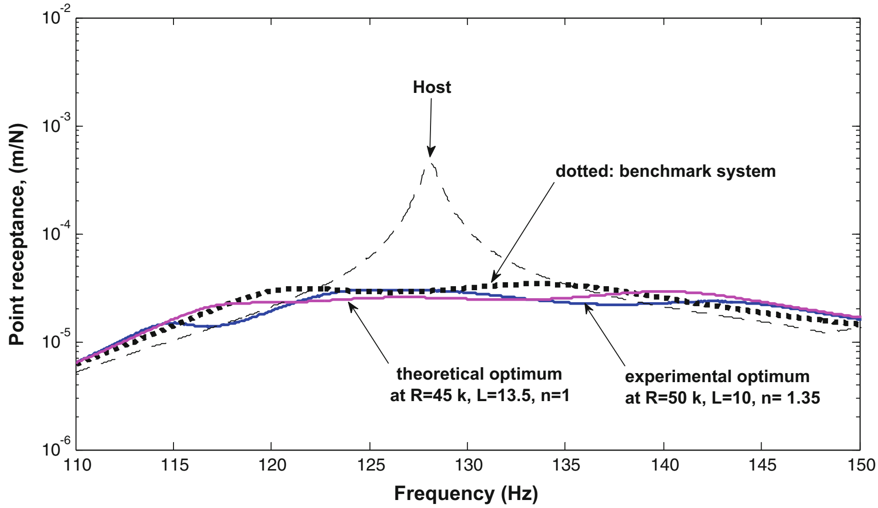

Fig. 6.13 Double-circuit parallel R-L-C; comparisons of optimum FRFs, experimental, theoretical and benchmark

optimum $\left\{ r_{y_A F_{ext}}^{host+TMD}(\omega) \right\}_{uncoup,opt}$. Both experimental and simulated optimal curves match very well with the benchmark response and the match between the simulated and experimental optimal curves is much better than for the previous single-circuit cases. The reason for this is that the experimental optimisation process

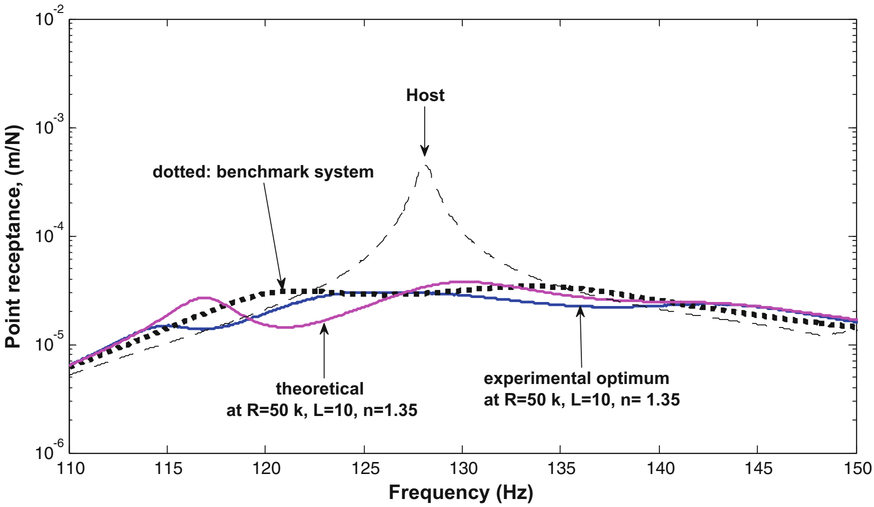

Fig. 6.14 Double-circuit parallel R-L-C; comparison of theory with experiment for the same parameters

was less constrained since the external capacitor was not forced to be zero (i.e. all three parameters R-L-C were adjusted, just as in the optimisation procedure used for the simulations). In fact, the R-L-C values were found to be quite close for the two cases: for the experimental optimal R = 50 kΩ, L = 10 H, C = 1.35C_p, whereas for the simulated optimal R = 45 kΩ, L = 13.5 H, C = 1C_p.

Figure 6.14 shows the experimental and simulated FRFs $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for the *same* R-L-C parameters (R = 50 kΩ, L = 10 H, C = 1.35C_p). As in the previous cases, agreement between theory and experiment is quite good.

6.2.4 FRFs: Double-Circuit Configuration, Parallel C, Series R-L

In this arrangement, each bimorph of the electromechanical TMD is connected across a separate identical circuit consisting of C in parallel with the series combination of R and L as shown in Fig. 5.9. As discussed in Sect. 6.3.2 for the single-circuit version, the R-L series combination increases the net impedance of the system, and hence, the system attains tuned conditions at much lower values of *R* than the parallel R-L arrangement. For this case, the experimental optimal response was obtained at a much lower load of 2.5 kΩ than the corresponding double-circuit parallel R-L-C case described in the previous section.

Figure 6.15 shows the comparison of three different optimum receptance plots for the double-circuit parallel C, series R-L circuit: (i) the experimental optimum

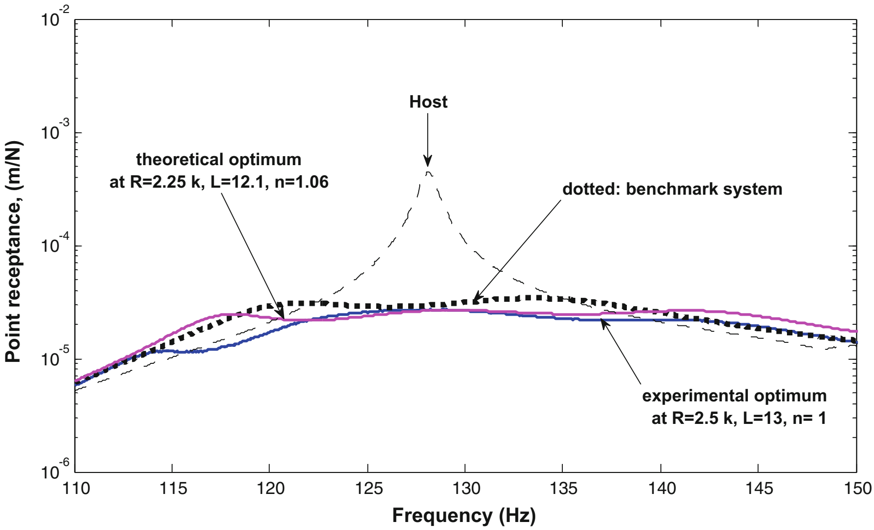

Fig. 6.15 Double-circuit parallel C and series R-L; comparisons of optimum FRFs, experimental, theoretical and benchmark

result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$, (ii) the simulated result for $r_{y_A F_{ext}}^{host+TMD}(\omega)$ with the R-L-C values computed using the optimisation technique of Sect. 5.3; and (iii) benchmark optimum $\left\{ r_{y_A F_{ext}}^{host+TMD}(\omega) \right\}_{uncoup,opt}$. Both experimental and simulated optimal curves match very well with the benchmark response and, as in the previous double-circuit case, the match between the experimental and simulated optimal curves is much better than the single-circuit cases, for the same reasons given in the previous section. The optimal R-L-C values were found to be quite close for the two cases: for the experimental optimal R = 2.5 kΩ, L = 13 H, C = $1C_p$, whereas for the simulated optimal R = 2.25 kΩ, L = 12.1 H, C = $1.06C_p$.

Figure 6.16 above shows the experimental and simulated FRFs $r_{y_A F_{ext}}^{host+TMD}(\omega)$ for the *same* R-L-C parameters (R = 2.5 kΩ, L = 13 H, C = $1C_p$). As in the previous cases, agreement between theory and experiment is quite good.

6.2.5 Summary of Experimental Optimal Results

Table 6.2 compares the experimentally observed performance of the four-circuit configurations. The performance indicator is the 'attenuation' which is defined as the ratio of the maximum peak of $\left| r_{y_A F_{ext}}^{host}(\omega) \right|$ to the maximum peak of $\left| r_{y_A F_{ext}}^{host+TMD}(\omega) \right|$ over the frequency range of interest. It is seen that the attenuation is

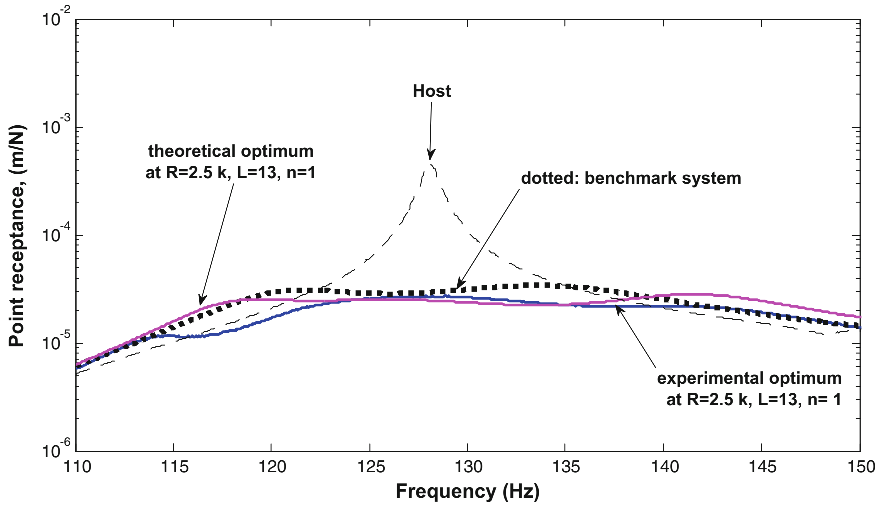

Fig. 6.16 Comparison of theory with experiment for the same parameters

Table 6.2 Vibration attenuation comparison of four cases

Circuit Type	R-L configuration	Attenuation achieved (vibration reduction factor)
Double circuit	Parallel R-L	15.3
	Series R-L	15.15
Single circuit	Parallel R-L	10.1
	Series R-L	10

at least 10, which is quite impressive, given that the effective mass of the TMD is less than 2% of the equivalent mass of the host structure for the vibration mode to be attenuated. In the tests performed, it was observed that the attenuation achieved by the double-circuit configurations with an external capacitor is significantly higher (around 15) than that achieved by the single-circuit configurations without an external capacitor (around 10). However, this comes at twice the cost, weight and space requirements.

6.3 Conclusions

The concept of the dual-function EH/TMD device, or electromechanical TMD, developed in the previous chapter, was experimentally validated in this chapter. The prototype was formed from two symmetric bimorph beams which were suitably shunted across different R-L-C circuits. The optimised damping of this device was supplied by the piezoelectric energy harvesting mechanism of the bimorphs. The device could be used to attenuate a particular mode of a generic structure. However, in this study, for illustrative purposes (only), the host structure was taken to be a free-free beam with an attachment block in the middle and the TMD was targeted at dampening its first flexural mode.

The performance of the device was evaluated by the same four-circuit configurations of the previous chapter, against the benchmark performance of an equivalent optimally damped mechanical system. As predicted in the previous chapter, the experiments showed that the benchmark performance was achievable through correct tuning for each of the four-circuit configurations. The effective mass of the TMD was less than 2% of the equivalent modal mass of the host structure, and experiments showed that the host structure vibration was attenuated by at least a factor of 10. It was also observed from the tests conducted that the vibration attenuation achieved by the double-circuit configurations with an external capacitor was significantly higher than that achieved by the single-circuit configurations without an external capacitor. However, this came at twice the cost, weight and space requirements. The experimental results agreed quite well the simulated results, thereby validating the theory presented in the previous chapter.

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Chapter 7
Example of Vibration Suppression of Electronic Box Using Dual Function EH/TVA

7.1 Applying the Proposed EH/TVA Theory

Having discussed the working of mechanical and electrical TVAs and their respective limitations in the previous chapters, the model of an 'electromechanical' TVA is presented here [1]. If the damping element of the mechanical TVA shown in Fig. 5.1(b), may be changed with the electrical damping generated due to energy harvesting, then the TVA will be called as an 'electromechanical TVA' [2, 3]. It is essential to note that, unlike in mechanical TVA, the damping level in the electromechanical TVA can simply be controlled by adjusting the value of electrical load attached to the energy harvesting circuit [1]. This chapter briefly presents a mathematical model and analysis of such dual function energy harvesting/tuned vibration absorber or EH/TVA beam device, in which suitably shunted piezoelectric beams are used as a TVA to attenuate the vibration of an electronic box as shown in Fig. 7.1 [1]. Thus, the advantages of mechanical and electrical TVAs are suitably combined in the proposed 'electromechanical' TVA.

7.2 Methodology

As mentioned in Chaps. 5 and 6 that the objective of this study is to suppress the frequency response function (FRF) of an electronic box at one of its natural/ resonance frequency against the external excitation 'F_{ext}' over the range of troublesome excitation frequencies ω, in the surrounding of the target natural frequency. The expressions for the FRFs of the electronic box, with and without the proposed TVA attached, are determined by following the theory mentioned in reference [2, 3]. Additionally, the classical theory of Den Hartog [4] is adapted in order to obtain the optimal level of hypothetical viscous damping that will be used as the benchmark for the performance of the electromechanical TVA [1, 5].

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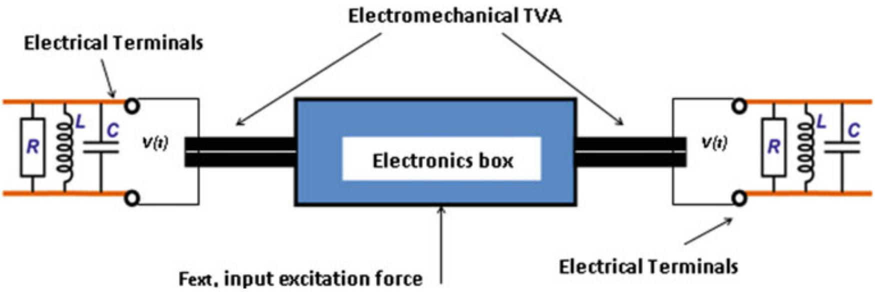

Fig. 7.1 Working of the dual function 'electromechanical' TVA energy harvester [1]

Table 7.1 Modal parameters of electronic box assembly

$\Omega_1/(2\pi)$ (Hz)	0	$M_A^{(1)} = 1/\left\{\widehat{\varphi}_A^{(1)}\right\}^2$ (gram)	276
$\Omega_2/(2\pi)$ (Hz)	127.65	$M_A^{(2)} = 1/\left\{\widehat{\varphi}_A^{(1)}\right\}^2$ (gram)	470

As illustrated earlier, in order to design a TVA to suppress the vibration of any structure, the only information it needs is the target frequency and the modal mass of the structure for the required degree of freedom (DOF) [1]. For this case, the target resonance frequency and the modal mass, at that resonance mode, of the electronic box can be calculated using normal mode analysis type available in any structural analysis (FEA) software. Having determined the target frequency and the modal mass of the host structure using FEA software, the effective mass of the cantilevered piezoelectric beam or electromechanical TVA can be determined by following the technique presented in Chaps. 5 and 6, i.e. using classical mechanical theory. The proposed theory is then validated by considering the example of electronic box whose parameters are approximately similar as was used in previous chapters (Tables 5.1 and 5.2) and as given here in Tables 7.1 and 7.2.

Table 7.2 Parameters of either beam of the electromechanical TVA [1]

Property	Units	Value
Young's Modulus of the Piezoelectric, Y_p	GPa	66
Young's Modulus of the shim, Y_{sh}	GPa	72
Density of the piezoelectric material	Kg/m^3	7800
Density of the shim material	Kg/m^3	2700
Piezoelectric constant, d_{31}	pm/volt	−190
Relative dielectric constant (at constant stress)		1800
Overhanging length of the beam, l	mm	58.75
Width of the beam, b	mm	25
Thickness of each piezoelectric layer, h_p (upper and lower layers)	mm	0.267
Thickness of the shim (substrate), h_{sh}	mm	0.285

7.3 Result and Analysis

For this electromechanical TVA, the effective mass ratio calculated as $\mu = 1.9\%$ which is quite small as compared to the mass ratio of the classical mechanical TVAs (typically 10–20%), demonstrating the compactness of the proposed TVA. The optimal damping ratio $\zeta_{a_{opt}}(= 8.2\%)$ is calculated using Eq. (5.3). Figure 7.2 shows different curves to illustrate the effectiveness of the proposed electromechanical TVA. The black, thick solid dotted curve in Fig. 7.2 shows the equivalent lumped parameter TVA receptance having optimal parameters, whereas the thin solid line shows the receptance of the electronic box with the electromechanical TVA.

The remarkable conformity between the thick, black dotted (benchmark mechanical TVA) and the thin, red solid lines (electromechanical TVA) validate the theory of the proposed 'electromechanical' absorber as shown in Fig. 7.2 [1]. The black solid line demonstrates the receptance of the electronic box without the TVA attached, and it can be seen that the response of the electronic box prior to and after the TVA has been suppressed considerably [1]. It is worth mentioning that the proposed TVA has extremely small mass ratio 'μ', less than 2%, showing its smallness and agility.

The inherent viscous damping present in the system is 1%, whereas the optimum damping required to tune the electromechanical TVA, as calculated using Eq. (5.3) is $\zeta_{a_{opt}} = 8.2\%$. This means that the remainder of 7.2% damping is provided by the energy harvesting effect, i.e. conversion of mechanical vibration energy into electrical energy using appropriate electrical circuitry parameters such as resistor, capacitor and inductor. The damping provided by the energy harvesting effect can be termed as electrical damping, and it is relatively easier to control the electrical

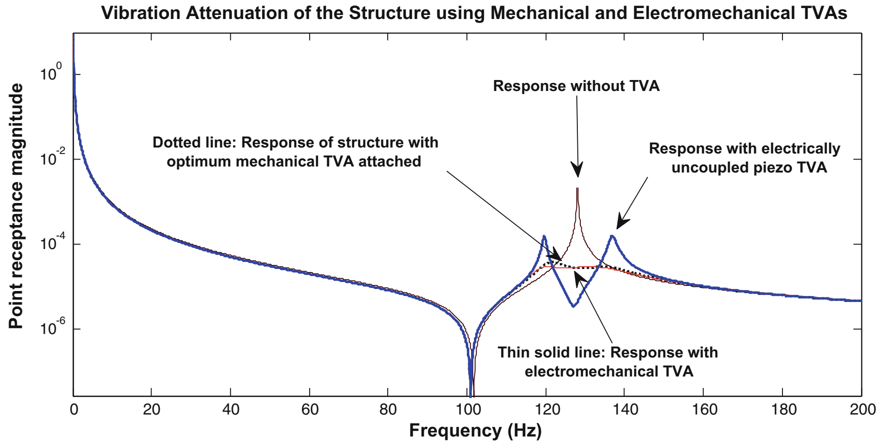

Fig. 7.2 FRF of an electronic box with and without 'mechanical' and 'electromechanical' TVA at the point of attachment [1]

damping of the system than the conventional viscous damping. The thick solid blue line present in Fig. 7.2 shows the response of the 'electromechanical' TVA without the electrical damping effects [1]. After appropriately selecting the values of the circuit components, such as resistor, capacitor and inductor, according to the technique derived in Chap. 5, the response of the system at the target mode suppresses ideally and gets excellent match with the classical mechanical TVA. This shows the importance of using the accurate values of resistor, inductor and the capacitor in the attached circuit to precisely tune the electromechanical TVA of Fig. 7.1 [1].

As presented in Chap. 5 and in [2], there can be different electronic configurations of the R-L-C circuits to tune the electromechanical TVA to generate the required amount of damping. However, for the example presented, a symmetric parallel resistor, inductor and capacitor arrangement is used on both sides of the piezoelectric beams as shown in Fig. 7.1. The optimum values of the attached R-L-C components were calculated using a Matlab optimisation program which is attached as Appendix A(b) of the Book. For this study, the resistor with value of 44.8 KΩ, inductor of value 13.5 H and the external capacitor of value 76 nF were used to generate the optimum amount of damping, which was needed to tune the electromechanical TVA to suppress the modal response of the electronic box.

It is worth mentioning that the above stated optimum values, of R-L-C, are not unique, and these can vary if the value of any of the component in the R-L-C combination changes [1]. In that case, the MATLAb optimisation program, provided in the Appendix-A, will produce entirely a new set of values of the R-L-C circuit. The results illustrated in Fig. 7.2 clearly verified that the electromechanical TVA is fully capable of suppressing the response of the target mode in the same way that closely mimics the benchmark response (shown by the black solid dotted line) of the classical TVA. In addition to achieve the optimum vibration attenuation, the proposed 'electromechanical' TVA is also generating useful electrical energy which can be conditioned and stored by adding the appropriate circuitry and storage device in the model [1]. However, the scope of this nonlinear analysis is beyond the scope of the present work.

7.4 Summary

The analytical example presented in this chapter validates the theoretical model derived, in Chap. 5 of this book, of a dual function piezoelectric energy harvester/tuned vibration absorber (i.e. electromechanical TVA) which can suppress vibration of the target mode and also harvest vibration energy at the same time. The work effectively proves the compactness, agility and performance of the proposed dual function 'electromechanical' TVA. It was shown that the vibration response of an electronic box was suppressed over a range of troublesome excitation frequencies significantly, and the system exhibited to be quite stable against input excitations.

The proposed dual function energy harvesting/TVA device consists of two symmetric beams which were properly shunted in order to suppress the primary

vibration mode of the host structure. The optimum amount of damping needed to tune the TVA was provided by the piezoelectric energy harvesting effect by selecting the optimised values of the attached resistor, capacitor and inductor components of the R-L-C circuit. It is important to note that the proposed electromechanical TVA shared the benefits of the classical, mechanical TVA and its electrical analogue (i.e. a shunted piezoelectric patch bonded directly to the host structure), reducing their relative disadvantages and combining their advantages.

The proposed compact and flexible 'electromechanical' TVA can have enormous applications in dynamic industrial and defence systems. Another advantage of using the proposed 'electromechanical' TVA is that the electrical energy produced by these TVA is very small (i.e. in the range of milliwatts to microwatts) but still it has the potential to power various modern low-powered electronic devices, e.g. the energy consumption of many wireless electronic nodes is in micro- to milliwatts.

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Chapter 8
Summary and Future Research

8.1 Summary

The studies presented in this book are intended to offer a deeper vision into the electromechanically coupled behaviour of PVEH systems and their potential application to vibration control. The outcomes of this study are expected to enhance the modelling of PVEH systems and also contribute towards their application to vibration control. A summary of the major findings of this study is listed below. The reader is referred to the detailed conclusions in the individual chapters for more information.

1. The first part of the book used the existing closed form distributed parameter AMAM as its theoretical basis but considerably contributed towards the knowledge of PVEH through a theoretical and experimental analysis. It was shown that the load that gave maximal resonant power was much higher than the load that induced a minimum response of the tip of a base-excited PVEH cantilever. Moreover, the study presented graphs showing the theoretical and experimental variations with electrical load of the resonance frequency, resonant voltage amplitude, resonant power and resonant deflection amplitude. These graphs provided a deeper insight into the electromechanical interaction within the harvester. Moreover, FRFs using Nyquist plots was presented. The Nyquist plots allowed for a more thorough validation of the theory than FRF magnitude plots. The centre of the Nyquist circle was shown to shift in a more noticeable way with the electrical load than in the FRF magnitude plots.
2. In the second part of the research, a mathematical modelling technique based on the dynamic stiffness matrix (DSM) method was developed to model piezo-electric beams. The method was based on the exact solution of the wave

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equation and so obviated the need for modal transformation as required in AMAM. For the same reason, DSM requires less elements than the finite element method for an assembly of uniform-section beams [3], offering a more accurate solution for high-frequency applications. In contrast to AMAM, the DSM readily lends itself to the modelling of beams with different boundary conditions or assemblies of beams of different cross sections.

3. Analytical investigations revealed that AMAM converged to DSM when a sufficiency of modes was used in the AMAM.
4. A thorough investigation of damping, and the damping related assumptions, in PVEH beam analysis was presented in the first two parts of the research. Nyquist plots were shown to be useful for the identification of the mechanical modal damping of the PVEH device. It was also demonstrated that the existence of ambient damping needs to be considered if the PVEH beam's performance at the higher modes is to be accurately quantified.
5. Analytical investigations using DSM revealed that a significant increase in the power output from a base-excited PVEH cantilever could be achieved through the application of a tip rotational restraint and the use of segmented electrodes.
6. The analytical investigations using DSM revealed the neutralising effects of a tuned harvester beam on the vibration at its base for different electrical loads. The findings suggested the use of a piezoelectric beam shunted by variable capacitance for the dual function of adaptive vibration neutralisation/energy harvesting. The vibration neutralizer is one type of tuned vibration absorber that is designed to suppress harmonic vibration at a particular excitation frequency.
7. The final part of the research extended the above concept to the other type of tuned vibration absorber—the tuned mass damper (TMD)—which suppresses a particular vibration mode of a generic host structure over a broadband of excitation frequencies. In-depth theoretical and experimental investigations were presented to validate the concept of the dual EH/TMD beam device or 'electromechanical' TMD. This device comprised a pair of bimorphs shunted by resistor–capacitor–inductor circuitry. The optimal damping required by this TMD was generated by the PVEH effect of the bimorphs. The results demonstrated that the ideal degree of vibration attenuation could be achieved by the proposed device through appropriate tuning of the circuitry. It was shown that vibration reduction factors of 10 or more were achievable by a EH/TMD beam device whose effective mass was less than 2% of the equivalent modal mass of the host structure. The EH effect was thus shown to provide a far easier way of controlling/adjusting the TMD damping compared to conventionally damped

TMDs. The proposed dual EH/TMD beam device combines the advantages of the classical (mechanical) TMD and the electrical vibration absorber, presenting the prospect of a functionally more readily adaptable class of 'electromechanical' tuned vibration absorbers.

8.2 Future Research

The research performed in this study has effectively contributed to the modelling of PVEH systems and its application to vibration control. The following suggestions are made for developing the work in this book:

- It is important to note that the research in this book assumed a linear electrical load. Hence, the nonlinear elements used in AC-DC rectification required for energy storage were not accommodated. Hence, there is considerable scope for developing the modelling of this book to accommodate energy storage devices.
- Further investigation can involve the development of more efficient energy harvesting and management circuitry for the transfer of generated energy from the piezoelectric layers to the energy storage device or electrical load. Such electronics can be designed to be directly embedded in the energy harvesting device.
- The developed EH/TMD beam device could be made adaptive to cope with possible variations in the modal parameters of the host structure, which would result in de-tuning of the device, and consequently, non-optimal performance (these could occur due to environmental or operational fluctuations). A self-sufficient control system could be explored, with the ability to adapt the circuit parameters (R-L-C). A suitably designed microcontroller can serve to retune the EH/TVA beam by varying the effective mass or stiffness (e.g. as in Fig. 2.6).
- Both the EH/TMD beam device (Chaps. 5 and 6) and the EH/vibration neutraliser beam device (Chap. 4) can potentially be developed to accommodate an energy storage device. This would involve nonlinear analysis which is needed due to the addition of AC-DC rectification elements in the circuit.
- The theoretical analysis presented for a bimorph showed that a significant increase in the power generated can be achieved for a given working frequency through the application of a tip rotational restraint, the use of segmented electrodes and a resized tip mass. Further theoretical and experimental analysis in this area is recommended.

References

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Appendix A-MatLab Program Codes

MATLAB Code for Voltage, Current and Power FRFs of a Distributed Parameter PVEH (Chapter 3)

```
%Program code of the model presented in Chap. 3
%PVEH system using distributed parameter model
L=60e-3; %Length of the overhung piezobeam
b= 25e-3;%Width of the piezobeam
hs= 0.3e-3;%Thickness of the shim
hp = 0.267e-3; %Thickness of the piezo
Ys = 7.2e10; sm = 1/Ys; %Young's Modulus and compliant of the shim
Yp = 6.2e10; s11 = 1/Yp; %Young's Modulus and compliant of piezo
rho_s = 2700; %Density of shim
rho_p = 7800; %Density of piezo
d31 = -190e-12; %-190e-12, Piezoconstant;
zeta = 0.008; damping ratio
yc = (hp+hs)/2; %Location of neutral axis
w=0:0.5:2000; %Excitation frequency range in Radians
w_hz = w./2/pi; %Excitation frequency in Hz
mass_beampiezo = 2*rho_p*L*b*hp; mass_beamshim = rho_s*L*b*hs;
mass_beam = mass_beampiezo + mass_beamshim; %Total mass of EH beam
mass_dist = mass_beam/L; % Distributed mass per length
permit_cons_elect = 8.854187817e-12 * 3800; %Permitivity in free space

perm_cons_strain = (permit_cons_elect - d31^2*Yp); %Permitivity at constant
strain
R = 1000e3; % Resistance value in Ohms

% Mechanical modal constants for first mode of clamp-free beam, %Equations
(3.8), (3.9)
Lamda = 1.87510407; % Standard value for cantilever beam (Mode shape books)
Sigma = 0.734095514; % % Standard value for cantilever beam
phi_r = ((cosh(Lamda)-cos(Lamda))-Sigma*(sinh(Lamda)-sin(Lamda)))/sqrt
(mass_beam); %Part of Equation %(3.8), terms within brackets
```


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```
trans_constant_1 = 2*Sigma/Lamda; %%%Gamma translation
trans_constant_2 = sqrt(L/mass_dist);
trans_constant = trans_constant_1*trans_constant_2;
phi_deriv1 = (sinh(Lamda)+ sin(Lamda))-Sigma*(cosh(Lamda)-cos(Lamda));
phi_derivative = Lamda*phi_deriv1/(sqrt(mass_beam)*L);

% Electromechanical constants in mechanical domain %

YI = b*(s11*hs^3 + 6*sm*hp*hs^2 + 12*sm*hs*hp^2 + 8*sm*hp^3)/
(12*sm*s11);% Equivalence of % Equation(3.3)
elect_const = -(Yp*d31*b*(hp + hs))/(2); % Equation (3.16)
Xr = elect_const * phi_derivative; %Equation (3.18)

% Calculating natural frequencies of the PV energy harvester
w_r = Lamda^2*sqrt(YI/(mass_beam*L^3)); %Equivalence of equation (3.11)
w_r_Hz = w_r/2/pi; %Natural frequency in Hz

% Voltage constant, V and Xr, electromechanical constants in mechanical
domain
% Capacitance of piezoelectric beam
Cp = perm_cons_strain * b * L / (hp); % Equation (3.24), 2 for series connection
modal_const = -d31*Yp*yc*b*phi_derivative; %
% Term-by-term voltage FRF calculation for single-mode expression, Equation
(3.31)
Volt_denom1 = (w_r^2 - w.^2) + (j*2*w.*w_r*zeta);
Volt_nume = j*2*w.*R*mass_dist*trans_constant*modal_const;
Volt_denom11 = (j*2*w.*R*Xr*modal_const);
Volt_denom22 = (2+j*w.*R*Cp);
Volt_denom33 = Volt_denom1.*Volt_denom22;
Volt_denom44 = Volt_denom22.*Volt_denom1;
Volt_denom_final = (Volt_denom44 + Volt_denom11);
%% Voltage FRF as per equation (3.31), for per "g", multiply by 9.81
VOLTAGE_FRF = (Volt_nume./Volt_denom_final);
VOLTAGE_FRF_abs = abs(VOLTAGE_FRF);
plot(w_hz,VOLTAGE_FRF_abs,'k')
%semilogy(w_hz,VOLTAGE_FRF,'r')% for semilog "y" axis
title('R = … ohm, Voltage')
xlabel('Frequency, Hz')
ylabel('Volt')
%axis([0 200 0 2])% Specifying the range on the axis if needed
figure;
%hold on
% Calculating CURRENT (mA)FRFs for PVEH System %
Current_FRF = VOLTAGE_FRF./R*1e3;
title = ('CURRENT FRFs')
xlabel('Frequency Hz')
ylabel('Current')
```

```
semilogy(w_hz,Current_FRF)
figure;
%hold on
%Calculating POWER FRFs for PVEH system normalised by g of acceleration
Power = (VOLTAGE_FRF.^2./R)*1000; % milliWatts
plot(w_hz,Power,'–')
title('Power')
xlabel('Frequency, Hz')
ylabel('miliwatt')
% It is important to note that the program is valid for a particular case based on equations of Chap. 3. It can % be different for different piezo materials, input frequencies, resonance frequencies, connected % resistor % value, damping value and other piezoelectric constants.
```

% Ch-3, Program 2: Example Nyquist Plot - Uncoupled

```
clear all;
w = 1:1:500*2*pi; % define frequency range
wr = 121.1*2*pi; % natural frequency
zeta = 0.01; %damping
L= 60e-3; % overhung length of the bimorph beam
b = 25e-3;%width of the bimorph beam
hs = 0.3e-3; %shim thickness;
hp = 0.267e-3; %piezo thickness
Ys = 7.2e10; sm = 1/Ys; % Young’s modulii of shim and piezo
Yp = 6.2e10; s11 = 1/Yp; %compliances
rho_s = 2700; %density of shim
rho_p = 7800; %density of piezo
d31 = -320e-12; %Electromechanical coupling coefficient
yc = (hp+hs)/2; %neutral axis
w_hz = w./2/pi; % excitation frequency in hertz
mass_beampiezo = 2*rho_p*L*b*hp;
mass_beamshim = rho_s*L*b*hs; %mass of %piezos & shim calculation
mass_beam = mass_beampiezo + mass_beamshim; %total mass
mass_dist = mass_beam/L; %distributed mass per lenght
permit_cons_elect = 8.854187817e-12 * 3800;
perm_cons_strain = (permit_cons_elect - d31^2*Yp); %as per theory
R = 1e3; %Resistance load of 1k ohm
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
Lamda = 1.87510407; %standard value for cantilever
Sigma = 0.734095514; %standard value for cantilever
phi_r = ((cosh(Lamda)-cos(Lamda))-Sigma*(sinh(Lamda)-sin(Lamda)))/sqrt(mass_beam) %calculation of mode shape
trans_constant_1 = 2*Sigma/Lamda; %%%Gamma translation
trans_constant_2 = sqrt(L/mass_dist);
trans_constant = trans_constant_1*trans_constant_2;
```

```
phi_deriv1 = (sinh(Lamda)+ sin(Lamda))-Sigma*(cosh(Lamda)-cos(Lamda));
phi_derivative = Lamda*phi_deriv1/(sqrt(mass_beam)*L);

%%%%%%%%%% ELECTROMECHANICAL CONSTANTS IN MECHANICAL DOMAIN %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

YI = b*(s11*hs^3 + 6*sm*hp*hs^2 + 12*sm*hs*hp^2 + 8*sm*hp^3)/(12*sm*s11); %stiffness calculation
elect_const = -(Yp*d31*b*(hp + hs))/(2); %vita or theta
Xr = elect_const * phi_derivative; %as per theory

% CALCULATION of NATURAL FREQUENCIES
w_r = Lamda^2*sqrt(YI/(mass_beam*L^3));
w_r_Hz = w_r/2/pi
Cp = perm_cons_strain * b * L / (hp); % 2 for series connection

modal_const = -d31*Yp*yc*b*phi_derivative;
c1=w*mass_dist*R*modal_const*trans_constant;
denom1 = (w_r^2 - w.^2).^2 + (2*w.*w_r*zeta).^2;

%%Real and Imaginary parts of FRF for Nyquist plot
Re_nume = (2*w.*w_r*zeta).*w;
Im_nume = w.*(w_r^2 - w.^2);
Re = (Re_nume./denom1)*9.81;
Im = (Im_nume./denom1)*9.81;
plot(Re,Im)
```

MATLAB Code for Energy Harvester Tuned Vibration Absorbers (Chaps. 5–6)

Step-1: Determination of resonance and other tuning parameters of the host and TVA as per equations of Chap. 5. The host structure is a free-free beam and TVA is a clamp-free beam, so care shall be made while selecting modal equations.

```
% See Equations from (5.1) to (5.10)
clear all; clc;
rho = 2720; % Density of host beam (Aluminium free-free beam)
Ys = 6.7e10; % Young's Modulus of host beam
l = 361.5e-3; % Length of host structure beam
b = 51e-3; % width
h = 3.5e-3; % height of host beam
I = b*h^3/12; % Inertia of host beam
mhost = rho*b*h*l; % mass of host beam
mhostunit = rho*b*h; % mass of host beam per unit length
wa = 121.1*2*pi; % natural frequency of the piezo absorber
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clamp_mass = 2*50*45*3.5e-9*2700; % mass of the clamp located in the
middle of host beam

screws = 12e-3; % mass of the screws used to clamp piezo beams, etc.

% As per theory, 40% mass of the overhung piezo cantlivers is redundant
% calculating mass of 4 piezo layers and two shim layers (2 x EH beams)
% 60 % of overhung cantilever mass is effective
%Thus, 40% overhung + piezo beam under clamp is redundant mass
% clamp mass + screws mass are redundant & simply add to host

tva_reduntant1 = 0.4*(4*.267e-3*60e-3*25e
-3*7800+ 2*0.3e-3*25e-3*60e-3*2700);
% tva_redundant1 = redundant overhung piezo portion of 2 beams
tva_reduntant2 = (4*.267e-3*12.2e-3*25e-3*7800+ 2*0.3e-3*25e
-3*12.2e-3*2700);
% tva_redundant2 = under clamp piezo portion of 2 beams
tva_total_redun = tva_reduntant1 + tva_reduntant2 + clamp_mass + screws
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
tva_effect = 0.6*(4*.267e-3*60e-3*25e-3*7800+ 2*0.3e-3*25e-3*60e-3*2700)
mtotal = rho*b*h*l + tva_total_redun; % total mass of host structure with
redundancy
lambdar = 4.73004; % standard for free-free beam for mode-1
w0=(lambdar.^2)*sqrt(Ys*I/(mhostunit*l^4)); % first resonance frequency of
host structure
w0_hz_host = w0/2/pi; % host structure resonance frequency in hz
mu=tva_effect/mtotal; % mass ratio, Equation (5.2)
mhost_freq = (1+mu)*wa/2/pi; % Equation (5.1)
% w0 frequencies
wh = w0/(sqrt(1+tva_total_redun/mhost))
wh_hz=wh/2/pi
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% %%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%
```

Part 2:

MATLAB program of the ‘proposed electromechanical TVA’ is presented in Chaps. 5 and 6. The details of the program are as below:

a. The program consists of one main ‘.m’ file, one MATLAB data file and three custom-made MATLAB ‘function files’. Upon execution of the main ‘.m’ file, the associated functions and the data files are loaded and the main program calculates optimal R-L-C circuit parameters for which the TVA suppresses the target mode of the host structure optimally. The codes of all these five files are provided below. The users are strongly recommended to understand the main features of the theory presented in Chaps. 5 and 6 in order to fully benefit from these codes.

b. The program, upon execution, prompts for information about the type of piezoelectric circuit, R-L-C arrangement (out of the four circuit configurations), etc., which users has to enter according to their system.
c. It is a convenient practice to start with shorter vectors (start from 1 or 2 values) of resistors ‘RR’ and capacitors ‘nn’ values to simplify the optimisation process and to understand the model.
d. Keep all the MATLAB files in one folder/directory. A thorough understanding of the theory and the optimization procedure mentioned in Chap. 5 is mandatory to run the below simulation.

Main executable ‘.m’ file:

```
clear;
%optfact=0.5;
optfact=0.1;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
filename='exp_free_free_beam_data_file'; %%%this file should contain M0, w0,
M0d, w0d, b, hp, hs, l, mtva, mu, rhop, rhos, w0, wa, eps33_S, d31
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
piezo_ser_or_par=input('Are piezo layers of bimorph TVA connected in series
(1) or parallel (2)? ');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
interconnected=input('Are the two TVA cantilevers connected across the same
circuit? yes (1), no (2): ');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
seriesinductor=input('Is inductor in series (1) or in parallel (2) with resistor? ');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
zeta1=input('input equivalent viscous damping ratio of bimorph beam at its
tuned mode ');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%
if piezo_ser_or_par==1
aa=2;ff=1;
end;
```

```
if piezo_ser_or_par==2
aa=1;ff=2;
end;
if interconnected==1
ff=2*ff;
end;

lambdar1=1.87510;
RR=1-16*(1.87510/4.73004)^4;
fun1='recTMD_eh_bimorph_parallelCvb_f_f_b_exp';
fun2='rechostplusTMD_eh_bimorph_parallelCvb_for_optL_f_f_b_exp';
fun3='rechostplusTMD_eh_bimorph_parallelCvb_for_optv2_f_f_b_exp';

Rvec=[1 1e2 1e3 1e4 2.5e4 5e4 1e5 2e5 1e6]; % resistor vector
nnvec=[0 0.5 1 2 3 5 10]; % capacitor vector
eval(['run ' filename]);
maxonly=input('optimise electrical parameters based on greatest peak only? yes
(1), no(2): ');
nmodes=300; % number of modes to include
zeta1opt=sqrt(3*mu/(8*(1+mu)^3)); % optimal damping, Equation (5.3)
disp(['resonance to be damped is ' num2str(w0/(2*pi)) 'Hz: ']);
flimits=input('Enter lower and upper frequencies for consideration (in Hz)
([lower upper]): ');% e.g. [1 300] hz, start end
fvec=flimits(1):0.1:flimits(2); % frequency vector in hz
wvec=2*pi*fvec; % frequency vector in radians
Cfactuncoup=0;% at uncoupled conditions, value is zero
if seriesinductor==1
Lnoinductor=0;
else
Lnoinductor=inf;
end;
daopt = feval(fun1,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,0.1,Cfactuncoup,
Lnoinductor,wvec,zeta1opt,nmodes,aa,ff,0,seriesinductor);
daopt_Rverysmall=feval(fun1,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,0.1,
Cfactuncoup,Lnoinductor,wvec,zeta1opt,nmodes,aa,ff,1,seriesinductor);%
receptance of the optimal absorber
dhost=(1/M0)./(w0^2-wvec.^2); % receptance host beam only
dhostwithclampnopiezo=(1/M0dorig)./(w0dorig^2-wvec.^2); % receptance host
with clamp, no piezos
dhostwithclamp=(1/M0d)./(w0d^2-wvec.^2);
dhostwithclampTVAopt=dhostwithclamp./(1+dhostwithclamp./daopt);% recep-
tance of host with clamp and TVA optimal
meff=RR*mtva;mred=(1-RR)*mtva;% Equation 5.11(b-c)
ktva=meff*wa^2; % Equation 5.11(a) ctva=2*zeta1opt*meff*wa;
daopt_2dof=-meff*wvec.^2+ktva+j*wvec*ctva;% Equivalent 2DOF system of
Fig 5.3(b)
```

```
daopt_2dof=daopt_2dof./(-meff*(ktva+j*wvec*ctva).*wvec.
^2-mred*daopt_2dof.*wvec.^2);
dhostwithclampTVAopt_2dof=dhostwithclamp./
(1+dhostwithclamp./daopt_2dof);
figure;
subplot(2,1,1),semilogy(fvec,abs(daopt),'k-');hold on;subplot(2,1,1),semilogy
(fvec,abs(daopt_2dof),'k:');subplot(2,1,1),semilogy(fvec,abs
(daopt_Rverysmall),'r-');
title(['Short circuit TMD point receptance with optimal damping: exact beam
model ' int2str(nmodes) ' clamped-free modes (solid); 2-dof approx (dotted)']);
subplot(2,1,2),semilogy(fvec,abs(dhost),':');hold on;subplot(2,1,2),semilogy
(fvec,abs(dhostwithclampnopiezo),'-.');subplot(2,1,2),semilogy(fvec,abs
(dhostwithclampTVAopt),'k-');subplot(2,1,2),semilogy(fvec,abs
(dhostwithclampTVAopt_2dof),'k–');
title('Point receptance of host: without TMD (dotted); without TMD but with
clamp (dash-dot); short-circuited optimal TMD, beam model (solid);
short-circuited optimal TMD 2-dof model (dashed)');
xlabel('frequency (Hz)');
ylabel('point receptance magnitude');
Cp=eps33_S*b*l/hp;% Capacitance of piezo layers
dauncoupled=feval(fun1,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,0.1,Cfactuncoup,
Lnoinductor,wvec,zeta1,nmodes,aa,ff,0,seriesinductor);
dhostwithclampTVAuncoupled=dhostwithclamp./
(1+dhostwithclamp./dauncoupled);
%Host receptance when no electrical circuit is included
ttt=find((wvec>0.75*wa)&(wvec<1.25*wa));
wvecconsidered=wvec(ttt);
absdhostwithclampTVAoptred=abs(dhostwithclampTVAopt(ttt));
dhostwithclampred=dhostwithclamp(ttt);
for gg=1:length(Rvec) % resistor vector
R=Rvec(gg);
figure;
subplot(2,1,1),semilogy(fvec,abs(dhost),':');hold on;
subplot(2,1,1),semilogy(fvec,abs(dhostwithclampnopiezo),'-.')
subplot(2,1,1),semilogy(fvec,abs(dhostwithclampTVAopt),'k:','LineWidth',1.5);
xlabel('frequency (Hz)');
ylabel('point receptance magnitude');
col='brmkcgy';% different clours
for hh=1:length(nnvec);% loop for capacitor vector
nn=nnvec(hh);
da = feval(fun1,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,R,nn,Lnoinductor,wvec,ze-
ta1,nmodes,aa,ff,1,seriesinductor);
dhostwithclampTVA=dhostwithclamp./(1+dhostwithclamp./da);
subplot(2,1,1),semilogy(fvec,abs(dhostwithclampTVA),col(hh));
end;
```

```
subplot(2,1,1),semilogy(fvec,abs(dhostwithclampTVAuncoupled),'b-
','LineWidth',1.5);
title(['Resistance of ' num2str(R) ' ohms, no inductor: thick solid line represents
electrically uncoupled cond.']);
axis([min(fvec) max(fvec) 0.99*min(abs(dhost)) 1.01*max(abs(dhost))]);
subplot(2,1,2),semilogy(fvec,abs(dhost),':');hold on;
subplot(2,1,2),semilogy(fvec,abs(dhostwithclampnopiezo),'-.')
subplot(2,1,2),semilogy(fvec,abs(dhostwithclampTVAopt),'k:','LineWidth',1.5);
xlabel('frequency (Hz)');
ylabel('point receptance magnitude');
if maxonly==1
absdhostwithclampTVAoptred=max(absdhostwithclampTVAoptred);
end;
for hh=1:length(nnvec); % for capacitors
nn=nnvec(hh);
Lapprox=1/(Cp*(ff/aa+nn)*wa^2);%bimorph parallel C
% Initially, approximate values of Inductor
inputtt=nn;
% “fgoalattain” Optimization toolbox function
L=fgoalattain(fun2,Lapprox,optfact*absdhostwithclampTVAoptred(:),
optfact*absdhostwithclampTVAoptred(:),[],[],[],[],0,Lapprox*100,[],[],R,in-
puttt,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,wvecconsidered,zeta1,nmodes,aa,ff,
dhostwithclampred,maxonly,seriesinductor);
da = feval(fun1,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,R,nn,L,wvec,zeta1,nmodes,
aa,ff,1,seriesinductor);
dhostwithclampTVA=dhostwithclamp./(1+dhostwithclamp./da);
subplot(2,1,2),semilogy(fvec,abs(dhostwithclampTVA),col(hh));
Lmat(hh,gg)=L;
Lfactormat(hh,gg)=L/Lapprox;
dhostwithclampTVAmatwithL(hh,:,gg)=dhostwithclampTVA;
end;
subplot(2,1,2),semilogy(fvec,abs(dhostwithclampTVAuncoupled),'b-
','LineWidth',1.5);
title(['Resistance of ' num2str(R) ' ohms, with inductor: thick solid line represents
electrically uncoupled cond.']);
axis([min(fvec) max(fvec) 0.99*min(abs(dhost)) 1.01*max(abs(dhost))]);
end;
Rvec
col(1:length(nnvec))
optimise=input('optimise for a chosen resistance R and inductance L? yes(1), no
(2): ');
if optimise
ggsel=input('id number of chosen resistance ');
hhsel=input('id number of chosen inductance ');
Rsel=Rvec(ggsel)
nnsel=nnvec(hhsel)
```

```
Lsel=Lmat(hhsel,ggsel)
elecparas = fgoalattain(fun3,[0.9*Rsel;nnsel;Lsel],
optfact*absdhostwithclampTVAoptred(:),
optfact*absdhostwithclampTVAoptred(:),[],[],[],[],[0;0;0],[1e6;inf;Lsel],[],[],l,
hp,hs,b,mtva,Yp,Ys,eps33_S,d31,wvecconsidered,zeta1,nmodes,aa,ff,dhost-
withclampred,maxonly,seriesinductor);
Rselelopt=elecparas(1)
nnselelopt=elecparas(2)
Lselelopt=elecparas(3)
daselelopt = feval(fun1,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,Rselelopt,nnselelopt,
Lselelopt,wvec,zeta1,nmodes,aa,ff,1,seriesinductor);
dhostwithclampTVAelopt=dhostwithclamp./(1+dhostwithclamp./daselelopt);
figure;
semilogy(fvec,abs(dhostwithclampTVAelopt),'m-','LineWidth',1.5);
hold on;
semilogy(fvec,abs(dhost),':');
semilogy(fvec,abs(dhostwithclampnopiezo),'-.')
semilogy(fvec,abs(dhostwithclampTVAopt),'k:','LineWidth',1.5);
end;
%%%%%%%%%%%%%%%%%%%%%% MAIN Program File Ends her %%
%%%%%%%%%%%%
```

Data File: 'exp_free_free_beam_data_file' for free-free beam host structure

```
M0d=0.460+0.003;%%%corrected for clamped part of piezo
w0d=128.1*2*pi;
w0d=sqrt(0.460*w0d^2/M0d);%%%corrected for clamped part of piezo
M0dorig=0.460;%%without clamped part of piezo
w0dorig=128.1*2*pi;%%%without clamped part of piezo
mtva=15e-3;
mu=RR*mtva/((1-RR)*mtva+M0d)
l=58.75e-3;
M0=0.146;
w0=135.2*2*pi;
eps33_S=1.1196e-8;
d31=-320e-12;
b=25e-3;
hp=0.267e-3;hs=0.3e-3;
Yp=62e9;Ys=72e9;
%rhop=7800;rhos=2700;mtva=2*(2*rhop*b*hp+rhos*b*hs)*l;
lambdar=1.87510;
%m=(2*rhop*b*hp+rhos*b*hs);mb=m*l;
mb=mtva/2;m=mb/l;
YI=b*((Ys/12)*hs^3+(Yp/6)*hp^3+(Yp/2)*hp*(hp+hs)^2);
wa=(lambdar^2)*sqrt(YI/(m*l^4));
sqrtterm=sqrt(1+(1-RR)*mtva/M0d);
%wa=125.6*2*pi;
```

Associated Custom-made Function Files
%fun1='recTMD_eh_bimorph_parallelCvb_f_f_b_exp';

```
function Rec = recTMD_eh_bimorph_parallelCvb_f_f_b_exp(l,hp,hs,b,mtva,
Yp,Ys,eps33_S,d31,R,nn,L,wvec,zeta1,nmodes,aa,ff,coupling,seriesinductor)
lambdar=[1.87510 4.69409 7.85476 10.9955 14.1372 (2*(6:nmodes+6)-1)
*pi/2];
lambdar=lambdar(1:nmodes);
%m=(2*rhop*b*hp+rhos*b*hs);mb=m*l;
mb=mtva/2;m=mb/l;
sigmar=[0.7340955 1.0184664 0.9992245 1.0000336 0.9999986];
if nmodes>5
sigmar=[sigmar ones(1,nmodes-5)];
end;
dphidr_x_l=(sinh(lambdar(1:5))-sigmar(1:5).*(cosh(lambdar(1:5)))+sin(lambdar
(1:5))-sigmar(1:5).*(-cos(lambdar(1:5)))).*lambdar(1:5)/(l*sqrt(mb));
if nmodes>5
dphidr_x_l=[dphidr_x_l 2*lambdar(6:nmodes).*(sin(lambdar(6:nmodes)))./
(l*sqrt(mb))];
end;
gammar_w=2*l*sigmar./(sqrt(mb)*lambdar);
hpc=(hs+hp)/2;
Phir=-dphidr_x_l*d31*Yp*hpc*hp*aa/(eps33_S*l);
YI=b*((Ys/12)*hs^3+(Yp/6)*hp^3+(Yp/2)*hp*(hp+hs)^2);
wr=(lambdar.^2)*sqrt(YI/(m*l^4));
wrsqr=wr.^2;
csI=zeta1*2*YI/wr(1);
zetar=csI*wr/(2*YI);
Cp=eps33_S*b*l/hp;
ttheta=-(hp^2+hp*hs)*d31*Yp*b/(aa*hp);
if coupling
ksir=ttheta*dphidr_x_l;
else
ksir=zeros(length(wr),1);
end;
wsqrvec=wvec.^2;
wrsqr=wrsqr(:);
wr=wr(:);
zetar=zetar(:);
Phir=Phir(:).';
gammar_w=gammar_w(:);
ksir=ksir(:);
phirddd_0=-2*sigmar.*((lambdar/l).^3)/sqrt(mb);
phirddd_0=phirddd_0(:);
zetarwr=zetar.*wr;
```

```
Reccmat=wrsqr(:,ones(1,length(wvec)))-wsqrvec(ones(1,length(wr)),:)
+j*2*zetarwr(:,ones(1,length(wvec))).*wvec(ones(1,length(wr)),:);
Reccmat=1./Reccmat;
S1=Phir*(gammar_w(:,ones(1,length(wvec))).*Reccmat);
S2=Phir*(ksir(:,ones(1,length(wvec))).*Reccmat);
if seriesinductor==1
T1=(-wsqrvec*L+j*wvec*R)*ff/aa;
T2=1/Cp-wsqrvec*L*(nn+ff/aa)+j*wvec*R*(nn+ff/aa);
else
T1=-wsqrvec*ff/aa;
T2=1/(L*Cp)-wsqrvec*(nn+ff/aa)+j*wvec*1/(R*Cp);
end;
U1=T1.*S1./(T2+T1.*S2);
gammar_w_mod=gammar_w(:,ones(1,length(wvec)))-ksir(:,ones(1,length
(wvec))).*U1(ones(1,length(wr)),:);
S3=2*YI*m*wsqrvec.*sum(gammar_w_mod.*(phirddd_0(:,ones(1,length
(wvec))).*Reccmat),1);
S3=S3.*(1+j*wvec*csI/YI);%%%%%%
Rec=1./S3;
```

fun2='rechostplusTMD_eh_bimorph_parallelCvb_for_optL_f_f_b_exp';

```
function absRec = rechostplusTMD_eh_bimorph_parallelCvb_for_optL_f_f_b_
exp(L,R,nn,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,wvec,zeta1,nmodes,aa,ff,
dhostred,maxonly,seriesinductor)
lambdar=[1.87510 4.69409 7.85476 10.9955 14.1372 (2*(6:nmodes+6)-1)
*pi/2];
lambdar=lambdar(1:nmodes);
%m=(2*rhop*b*hp+rhos*b*hs);mb=m*l;
mb=mtva/2;m=mb/l;
sigmar=[0.7340955 1.0184664 0.9992245 1.0000336 0.9999986];
if nmodes>5
sigmar=[sigmar ones(1,nmodes-5)];
end;
dphidr_x_l=(sinh(lambdar(1:5))-sigmar(1:5).*(cosh(lambdar(1:5)))+sin(lambdar
(1:5))-sigmar(1:5).*(-cos(lambdar(1:5)))).*lambdar(1:5)/(l*sqrt(mb));
if nmodes>5
dphidr_x_l=[dphidr_x_l 2*lambdar(6:nmodes).*(sin(lambdar(6:nmodes)))./
(l*sqrt(mb))];
end;
gammar_w=2*l*sigmar./(sqrt(mb)*lambdar);
hpc=(hs+hp)/2;
Phir=-dphidr_x_l*d31*Yp*hpc*hp*aa/(eps33_S*l);
YI=b*((Ys/12)*hs^3+(Yp/6)*hp^3+(Yp/2)*hp*(hp+hs)^2);
wr=(lambdar.^2)*sqrt(YI/(m*l^4));
wrsqr=wr.^2;
```

```
csI=zeta1*2*YI/wr(1);
zetar=csI*wr/(2*YI);
Cp=eps33_S*b*l/hp;
ttheta=-(hp^2+hp*hs)*d31*Yp*b/(aa*hp);
ksir=ttheta*dphidr_x_l;
wsqrvec=wvec.^2;
wrsqr=wrsqr(:);
wr=wr(:);
zetar=zetar(:);
Phir=Phir(:).';
gammar_w=gammar_w(:);
ksir=ksir(:);
phirddd_0=-2*sigmar.*((lambdar/l).^3)/sqrt(mb);
phirddd_0=phirddd_0(:);
zetarwr=zetar.*wr;
Reccmat=wrsqr(:,ones(1,length(wvec)))-wsqrvec(ones(1,length(wr)),:)
+j*2*zetarwr(:,ones(1,length(wvec))).*wvec(ones(1,length(wr)),:);
Reccmat=1./Reccmat;
S1=Phir*(gammar_w(:,ones(1,length(wvec))).*Reccmat);
S2=Phir*(ksir(:,ones(1,length(wvec))).*Reccmat);
if seriesinductor==1
T1=(-wsqrvec*L+j*wvec*R)*ff/aa;
T2=1/Cp-wsqrvec*L*(nn+ff/aa)+j*wvec*R*(nn+ff/aa);
else
T1=-wsqrvec*ff/aa;
T2=1/(L*Cp)-wsqrvec*(nn+ff/aa)+j*wvec*1/(R*Cp);
end;
U1=T1.*S1./(T2+T1.*S2);
gammar_w_mod=gammar_w(:,ones(1,length(wvec)))-ksir(:,ones(1,length
(wvec))).*U1(ones(1,length(wr)),:);
S3=2*YI*m*wsqrvec.*sum(gammar_w_mod.*(phirddd_0(:,ones(1,length
(wvec))).*Reccmat),1);
S3=S3.*(1+j*wvec*csI/YI);%%%%%%
Rec=1./S3;
dhostred=dhostred(:).';
Rec=dhostred./(1+dhostred./Rec);
absRec=abs(Rec(:));
if maxonly==1
absRec=max(absRec);
end;
```

fun3='rechostplusTMD_eh_bimorph_parallelCvb_for_optv2_f_f_b_exp';

```
function absRec = rechostplusTMD_eh_bimorph_parallelCvb_for_optv2_f_f_
b_exp(elecparas,l,hp,hs,b,mtva,Yp,Ys,eps33_S,d31,wvec,zeta1,nmodes,aa,ff,
dhostred,maxonly,seriesinductor)
```

```
R=elecparas(1);
nn=elecparas(2);
L=elecparas(3);
lambdar=[1.87510 4.69409 7.85476 10.9955 14.1372 (2*(6:nmodes+6)-1)
*pi/2];
lambdar=lambdar(1:nmodes);
%m=(2*rhop*b*hp+rhos*b*hs);mb=m*l;
mb=mtva/2;m=mb/l;
sigmar=[0.7340955 1.0184664 0.9992245 1.0000336 0.9999986];
if nmodes>5
sigmar=[sigmar ones(1,nmodes-5)];
end;
dphidr_x_l=(sinh(lambdar(1:5))-sigmar(1:5).*(cosh(lambdar(1:5)))+sin(lambdar
(1:5))-sigmar(1:5).*(-cos(lambdar(1:5)))).*lambdar(1:5)/(l*sqrt(mb));
if nmodes>5
dphidr_x_l=[dphidr_x_l 2*lambdar(6:nmodes).*(sin(lambdar(6:nmodes)))./
(l*sqrt(mb))];
end;
gammar_w=2*l*sigmar./(sqrt(mb)*lambdar);
hpc=(hs+hp)/2;
Phir=-dphidr_x_l*d31*Yp*hpc*hp*aa/(eps33_S*l);
YI=b*((Ys/12)*hs^3+(Yp/6)*hp^3+(Yp/2)*hp*(hp+hs)^2);
wr=(lambdar.^2)*sqrt(YI/(m*l^4));
wrsqr=wr.^2;
csI=zeta1*2*YI/wr(1);
zetar=csI*wr/(2*YI);
Cp=eps33_S*b*l/hp;
ttheta=-(hp^2+hp*hs)*d31*Yp*b/(aa*hp);
ksir=ttheta*dphidr_x_l;
wsqrvec=wvec.^2;
wrsqr=wrsqr(:);
wr=wr(:);
zetar=zetar(:);
Phir=Phir(:).';
gammar_w=gammar_w(:);
ksir=ksir(:);
phirddd_0=-2*sigmar.*((lambdar/l).^3)/sqrt(mb);
phirddd_0=phirddd_0(:);
zetarwr=zetar.*wr;
Reccmat=wrsqr(:,ones(1,length(wvec)))-wsqrvec(ones(1,length(wr)),:)
+j*2*zetarwr(:,ones(1,length(wvec))).*wvec(ones(1,length(wr)),:);
Reccmat=1./Reccmat;
S1=Phir*(gammar_w(:,ones(1,length(wvec))).*Reccmat);
S2=Phir*(ksir(:,ones(1,length(wvec))).*Reccmat);
if seriesinductor==1
```

```
T1=(-wsqrvec*L+j*wvec*R)*ff/aa;
T2=1/Cp-wsqrvec*L*(nn+ff/aa)+j*wvec*R*(nn+ff/aa);
else
T1=-wsqrvec*ff/aa;
T2=1/(L*Cp)-wsqrvec*(nn+ff/aa)+j*wvec*1/(R*Cp);
end;
U1=T1.*S1./(T2+T1.*S2);
gammar_w_mod=gammar_w(:,ones(1,length(wvec)))-ksir(:,ones(1,length
(wvec))).*U1(ones(1,length(wr)),:);
S3=2*YI*m*wsqrvec.*sum(gammar_w_mod.*(phirddd_0(:,ones(1,length
(wvec))).*Reccmat),1);
S3=S3.*(1+j*wvec*csI/YI);%%%%%%
Rec=1./S3;
dhostred=dhostred(:).';
Rec=dhostred./(1+dhostred./Rec);
absRec=abs(Rec(:));
if maxonly==1
absRec=max(absRec);
end;
```

Appendix B

Mathematical Derivation Constants

The frequency that gives maximum modulus of $\beta(\omega)|_r$ in Eq. (3.32) (i.e. the resonance frequency) satisfies the following cubic in ω^2:

$$B_a(\omega^2)^3 + B_b(\omega^2)^2 + B_c\omega^2 + B_d = 0 \tag{B.1}$$

where

$$B_a = 2K_d^2 K_a^2 \tag{B.2}$$

$$B_b = -2K_d^2 K_a K_c + K_d^2 K_a^2 K_b^2 + 4K_d^2 + 3K_e^2 K_a^2 - 2K_d^2 K_a^2 \omega_r^2 \tag{B.3}$$

$$B_c = 8K_e^2 + 2K_e^2 K_a^2 K_b^2 - 4K_e^2 K_a K_c - 4K_e^2 K_a^2 \omega_r^2 \tag{B.4}$$

$$B_d = 4K_e^2 K_b^2 - 8K_e^2 \omega_r^2 + K_e^2 K_c^2 + 4K_e^2 K_b K_c + 2K_e^2 K_a K_c \omega_r^2 + K_e^2 K_a^2 \omega_r^4 - 4K_d^2 \omega_r^4 \tag{B.5}$$

$$K_a = RC_p,\ K_b = 2\xi_r\omega_r,\ K_c = 2R\alpha_r\chi_r,\ K_e = RC_p m\gamma_r^u\phi_r(l),\ K_e = 2m\gamma_r^u\phi_r(l) \tag{B.6a–d}$$

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Appendix C

Testing Laboratory Equipment

A brief introduction of the vibration testing and data acquisition system used in the experimental study is as follows:

- A pc-controlled data acquisition system (LMS Scadas 5 with LMS Test.Lab Rev 7A software) was used to control the input excitation as well as used to record the response and to generate the frequency response functions.
- A PCB 352C22 accelerometer of sensitivity 9.08 mV/g and resolution of 0.002g rms was used to measure the acceleration of the host structure.
- A PCB force sensor of model 208 B01, sensitivity 114.11 mv/N, maximum static force sensing capability 0.27 kN in tension and compression, frequency limit 0.01 Hz–36 kHz was used to record the input excitation force.
- A four-channel PCB 442B104 ICP sensor signal conditioner.
- Amplifier.
- An electrodynamic shaker manufactured by Ling Dynamic Systems.
- A handheld Agilent U1732A LCR (inductor, capacitor and resistor) metre with test frequency settings of 100 Hz, 120 Hz, 1 kHz and 10 kHz.

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